

## LIST OF DESIGNATED REFERENCE AND EQUIVALENT METHODS

**Issue Date: June 27, 2013**  
([www.epa.gov/ttn/amtic/criteria.html](http://www.epa.gov/ttn/amtic/criteria.html))

These methods for measuring ambient concentrations of specified air pollutants have been designated as "reference methods" or "equivalent methods" in accordance with Title 40, Part 53 of the Code of Federal Regulations (40 CFR Part 53). Subject to any limitations (e.g., operating range or temperature range) specified in the applicable designation, each method is acceptable for use in state or local air quality surveillance systems under 40 CFR Part 58 unless the applicable designation is subsequently canceled. Automated methods for pollutants other than PM<sub>10</sub> are acceptable for use only at shelter temperatures between 20°C and 30°C and line voltages between 105 and 125 volts unless wider limits are specified in the method description.

Prospective users of the methods listed should note (1) that each method must be used in strict accordance with its associated operation or instruction manual and with applicable quality assurance procedures, and (2) that modification of a method by its vendor or user may cause the pertinent designation to be inapplicable to the method as modified. (See Section 2.8 of Appendix C, 40 CFR Part 58 for approval of modifications to any of these methods by users.)

Further information concerning particular designations may be found in the *Federal Register* notice cited for each method or by writing to the National Exposure Research Laboratory, Human Exposure and Atmospheric Sciences Division (MD-D205-03), U.S. Environmental Protection Agency, Research Triangle Park, North Carolina 27711. Technical information concerning the methods should be obtained by contacting the source listed for each method. Source addresses are listed at the end of the listing of methods, except for the addresses for lead method sources, which are given with the method. New analyzers or PM<sub>10</sub> samplers sold as reference or equivalent methods must carry a label or sticker identifying them as designated methods. For analyzers or PM<sub>10</sub> or samplers sold prior to the designation of a method with the same or similar model number, the model number does not necessarily identify an analyzer or sampler as a designated method. Consult the manufacturer or seller to determine if a previously sold analyzer or sampler can be considered a designated method or if it can be upgraded to designation status. Analyzer users who experience operational or other difficulties with a designated analyzer or sampler and are unable to resolve the problem directly with the instrument manufacturer may contact EPA (preferably in writing) at the above address for assistance.

This list will be revised as necessary to reflect any new designations or any cancellation of a designation currently in effect. The most current revision of the list will be available for inspection at EPA's Regional Offices, and copies may be obtained at the Internet site identified above or by writing to the National Exposure Research Laboratory at the address specified above.

**Most Recent Designations**

**Particulate Matter - TSP*****Reference Method for TSP*****Manual Reference Method: 40 CFR Part 50, Appendix B**

Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High-Volume Method)

***Federal Register: Vol. 47, page 54912, 12/06/82 and Vol. 48, page 17355, 04/22/1983***

**Particulate Matter – PM<sub>10</sub>****Andersen Model RAAS10-100 PM<sub>10</sub> Single Channel PM<sub>10</sub> Sampler****Manual Reference Method: RFPS-0699-130**

“Andersen Instruments, Incorporated Model RAAS10-100 Single Channel Reference Method PM<sub>10</sub> Sampler,” with RAAS-10 PM<sub>10</sub> inlet or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, configured as a PM<sub>10</sub> reference method, and operated for 24-hour continuous sample periods at a flow rate of 16.67 liters/ minute, and in accordance with the Model RAAS105-100 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix J or Appendix M.

**Federal Register: Vol. 64, page 33481, 06/23/1999**

**Andersen Model RAAS10-200 PM<sub>10</sub> Single Channel PM<sub>10</sub> Audit Sampler****Manual Reference Method: RFPS-0699-131**

“Andersen Instruments, Incorporated Model RAAS10-200 Single Channel Reference Method PM<sub>10</sub> Audit Sampler,” with RAAS-10 PM<sub>10</sub> inlet or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, configured as a PM<sub>10</sub> reference method, and operated for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the Model RAAS105-200 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix J or Appendix M.

**Federal Register: Vol. 64, page 33481, 06/23/1999**

**Andersen Model RAAS10-300 PM<sub>10</sub> Multi Channel PM<sub>10</sub> Sampler****Manual Reference Method: RFPS-0699-132**

“Andersen Instruments, Incorporated Model RAAS10-300 Multi Channel Sequential Reference Method PM<sub>10</sub> Sampler,” with RAAS-10 PM<sub>10</sub> inlet or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, configured as a PM<sub>10</sub> reference method, and operated for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the Model RAAS105-300 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix J or Appendix M.

**Federal Register: Vol. 64, page 33481, 06/23/1999**

**BGI Incorporated Model PQ100 Air Sampler****Manual Reference Method: RFPS-1298-124**

“BGI Incorporated Model PQ100 Air Sampler,” with BGI 16.7 Inlet Kit or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, configured as a PM<sub>10</sub> reference method, for 24-hour continuous sample periods at a flow rate of 16.7 liters/minute, with original firmware Version 5.X and lower or new firmware version 6.0 and higher, operated in accordance with the original Model PQ100 Instruction Manual or manual revision Version 7.0, as appropriate, and with the requirements specified in 40 CFR Part 50, Appendix J or Appendix M, using either the original or the newer PQ200-type filter cassettes, and with or without the optional Solar Panel Power Supply.

**Federal Register: Vol. 63, page 69625, 12/17/1998**

**Latest modification: 01/2009**

**BGI Incorporated Model PQ200 Air Sampler****Manual Reference Method: RFPS-1298-125**

“BGI Incorporated Model PQ200 Air Sampler,” with “flat plate” PM<sub>10</sub> inlet or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, configured as a PM<sub>10</sub> reference method, and operated for 24-hour continuous sample periods in accordance with the Model PQ200 Instruction Manual and with the requirements specified in 40 CFR Part 50, Appendix J or Appendix M, and with or without the optional Solar Panel Power Supply.

**Federal Register: Vol. 63, page 69625, 12/17/1998**

**DKK-TOA Models FPM-222/222C, FPM223/223C, and DUB-222(S)/223(S) PM<sub>10</sub> Monitor****Automated Equivalent Method: EQPM-0905-156**

“DKK-TOA Models FPM-222, FPM-222C, FPM-223, FPM-223C, DUB-222(S), and DUB-223(S) Particulate Monitor,” for monitoring PM<sub>10</sub> in Ambient Air (beta attenuation monitor), configured for PM<sub>10</sub>, with Firmware Version DUB4-658355, Corrected Slope Factor (FACT SLOPE) set to 1.232, Corrected Zero Value (FACT ZERO) set to 1.8, and with or without any of the following options: Auto Check and Serial Recorder.

**Federal Register: Vol. 70, page 56684, 09/28/2005**

**Ecotech Model 3000 PM<sub>10</sub> High Volume Air Sampler****Manual Reference Method: RFPS-0706-162**

"Ecotech Pty. Ltd. Model 3000 PM<sub>10</sub> High Volume Air Sampler," configured with the Ecotech PM<sub>10</sub> Size-Selective Inlet (SSI)(P-ECO-HVS3000-02), with the flow rate set to 1.13 m<sup>3</sup>/min (67.8 m<sup>3</sup>/hour).

**Federal Register: Vol. 71, page 42089, 07/25/2006**

**Environnement S.A. Model MP101M PM<sub>10</sub> Monitor****Automated Equivalent Method: EQPM-0404-151**

"Environnement S. A. Model MP101M PM<sub>10</sub> Beta Gauge Monitor," configured with the louvered PM<sub>10</sub> inlet specified in 40 CFR 50 Appendix L or its flat-topped predecessor version and one of the three optional temperature-regulated sampling tubes (RST), and operated with a full scale measurement range of 0 - 0.500 mg/m<sup>3</sup> (0 - 500 :g/m<sup>3</sup>), with the sample flow rate set to 1.00 m<sup>3</sup>/h and flow regulation set to yes, the "norms selection" set to m<sup>3</sup> (actual volume), the "cycle" set to 24 hours, the "period" set to none, and the "counting time" set to 200 seconds.<sup>2</sup>

**Federal Register: Vol. 69, page 18569, 4/8/2004**

**Graseby Andersen/GMW Model 1200 High-Volume Air Sampler****Manual Reference Method: RFPS-1287-063**

"Sierra-Andersen or General Metal Works Model 1200 PM<sub>10</sub> High-Volume Air Sampler System," consisting of a Sierra-Andersen or General Metal Works Model 1200 PM<sub>10</sub> Size-Selective Inlet and any of the high-volume air samplers identified as SAUV-10H, SAUV-11H, GMW-IP-10, GMW-IP-10-70, GMW-IP-10-801, or GMW-IP-10-8000, which include the following components: Anodized aluminum high-volume shelter with either acrylonitrile butadiene styrene plastic filter holder and motor/blower housing or stainless steel filter holder and phenolic plastic motor/blower housing; 0.6 hp motor/blower; pressure transducer flow recorder; either an electronic mass flow controller or a volumetric flow controller; either a digital timer/programmer, seven-day mechanical timer, six-day timer/programmer, or solid-state timer/programmer; elapsed time indicator; and filter cartridge.

**Federal Register: Vol. 52, page 45684, 12/01/1987 and Vol. 53, page 1062, 01/15/1988**

**Graseby Andersen/GMW Model 321-B High-Volume Air Sampler****Manual Reference Method: RFPS-1287-064**

"Sierra-Andersen or General Metal Works Model 321-B PM<sub>10</sub> High-Volume Air Sampler System," consisting of a Sierra-Andersen or General Metal Works Model 321-B PM<sub>10</sub> Size-Selective Inlet and any of the high-volume air samplers identified as SAUV-10H, SAUV-11H, GMW-IP-10, GMW-IP-10-70, GMW-IP-10-801, or GMW-IP-10-8000, which include the following components: Anodized aluminum high-volume shelter with either acrylonitrile butadiene styrene plastic filter holder and motor/blower housing or stainless steel filter holder and phenolic plastic motor/blower housing; 0.6 hp motor/blower; pressure transducer flow recorder; either an electronic mass flow controller or a volumetric flow controller; either a digital timer/programmer, seven-day mechanical timer, six-day timer/programmer, or solid-state timer/programmer; elapsed time indicator; and filter cartridge.

**Federal Register: Vol. 52, page 45684, 12/01/1987 and Vol. 53, page 1062, 01/15/1988**

**Graseby Andersen/GMW Model 321-C High-Volume Air Sampler****Manual Reference Method: RFPS-1287-065**

"Sierra-Andersen or General Metal Works Model 321-C PM<sub>10</sub> High-Volume Air Sampler System," consisting of a Sierra-Andersen General Metal Works Model 321-C PM<sub>10</sub> or Size-Selective Inlet and any of the high-volume air samplers identified as SAUV-10H, SAUV-11H, GMW-IP-10, GMW-IP-10-70, GMW-IP-10-801, or GMW-IP-10-8000, which include the following components: Anodized aluminum high-volume shelter with either acrylonitrile butadiene styrene plastic filter holder and motor/blower housing or stainless steel filter holder and phenolic plastic motor/blower housing; 0.6 hp motor/blower; pressure transducer flow recorder; either an electronic mass flow controller or a volumetric flow controller; either a digital timer/programmer, seven-day mechanical timer, six-day timer/programmer, or solid-state timer/programmer; elapsed time indicator; and filter cartridge.

**Federal Register: Vol. 52, page 45684, 12/01/1987 and Vol. 53, page 1062, 01/15/1988**

**Graseby Andersen/GMW Models SA241 and SA241M Dichotomous Sampler****Manual Reference Method: RFPS-0789-073**

“Sierra-Andersen Models SA241 and SA241M or General Metal Works Models G241 and G241M PM<sub>10</sub> Dichotomous Samplers,” consisting of the following components: Sampling Module with SA246b or G246b 10 Φm inlet or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, 2.5 Φm virtual impactor assembly, 37 mm coarse and fine particulate filter holders, and tripod mount; Control Module with diaphragm vacuum pump, pneumatic constant flow controller, total and coarse flow rotameters and vacuum gauges, pressure switch (optional), 24-hour flow/event recorder, digital timer/programmer or 7-day skip timer, and elapsed time indicator.

**Federal Register: Vol. 54, page 31247, 07/27/1989**

**Graseby Andersen/GMW Model FH621-N Beta Monitor****Automated Equivalent Method: EQPM-0990-076**

“Andersen Instruments Model FH621-N PM<sub>10</sub> Beta Attenuation Monitor,” consisting of the following components: FH101 Vacuum Pump Assembly; FH102 Accessory Kit; FH107 Roof Flange Kit; FH125 Zero and Span PM<sub>10</sub> Mass Foil Calibration Kit; FH621 Beta Attenuation 19-inch Control Module; SA246b PM<sub>10</sub> Inlet (16.7 liter/min) or the louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19; operated for 24-hour average measurements, with an observing time of 60 minutes, the calibration factor set to 2400, a glass fiber filter tape, an automatic filter advance after each 24-hour sample period, and with or without either of the following options: FH0P1 Indoor Cabinet; FH0P2 Outdoor Shelter Assembly.

**Federal Register: Vol. 55, page 38387, 09/18/1990**

**Met One or Sibata Models BAM/GBAM 1020, BAM/GBAM 1020-1, Horiba APDA-371, or Ecotech Spirant BAM1000****Automated Equivalent Method: EQPM-0798-122**

“Met One Instruments or Sibata Scientific Technology Models BAM 1020, GBAM 1020, BAM 1020-1, GBAM 1020-1, Horiba APDA-371 PM<sub>10</sub> Beta Attenuation Monitor, or Ecotech Spirant BAM1000” including the BX-802 EPA PM<sub>10</sub> inlet (or alternative louvered PM<sub>10</sub> inlet meeting 40 CFR 50 Appendix L specifications), operated for 24-hour average measurements, with a filter change frequency of one hour, with glass fiber filter tape, and with or without any of the following options: BX-823, tube extension; BX-825, heater kit; BX-826, 230 VAC heater kit; BX-827 “Smart Heater” set for maintaining moisture between 35% and 45% and no ΔT control; BX-828, roof tripod; BX-902, exterior enclosure; BX-903, exterior enclosure with temperature control; BX-961, mass flow controller; BX-967, internal calibration device, BX-970 touch-screen display with USB interface. For software (firmware) versions V3.0 or higher, a user-selectable measurement time (COUNT TIME) of 4, 6, 8 or 10 minutes selected, along with appropriate sample time (BAM SAMPLE) setting of 50, 46, 42 or 38 minutes, respectively, to maintain a 60-minute measurement cycle. For software (firmware) versions V3.5 or higher, user-selectable option to sample under actual conditions (Flow Type: ACTUAL) and report under standard conditions (Reporting: STD), which requires the use of P/N BX-592 external temperature sensor or P/N BX-596 external temperature/barometric pressure sensor. The user may also sample under standard conditions (Flow Type: STD) and report under standard conditions (Reporting: STD) with any software/firmware 2.0 or higher. Instrument must be operated in accordance with the appropriate instrument manual.

**Federal Register: Vol. 63, page 41253, 08/03/1998**

**Latest modifications: 06/2009; 07/2010; 8/2010; 8/2012**

**Opsis Model SM200 PM<sub>10</sub> Monitor****Automated Equivalent Method: EQPM-0810-193**

“Opsis Model SM200 Monitor,” beta gauge semi-continuous ambient particulate monitor operated for 24 hours at a flow rate of 16.67 LPM between 5° and 40°C using 47 mm PTFE membrane filter media, in the mass measurement range of 0 to 60 mg, configured with a BGI Model SSI25 PM<sub>10</sub> inlet meeting criteria specified in 40 CFR 50 Appendix L, with a roof mounting kit, and with or without an inlet tube heater (as recommended based on site RH conditions), according to the SM200 User’s Guide.

**Federal Register: Vol. 75, page 51039, 08/18/2010**

**Oregon DEQ Medium Volume PM<sub>10</sub> Sampler****Manual Reference Method: RFPS-0389-071**

“Oregon DEQ Medium Volume PM<sub>10</sub> Sampler.” NOTE: This method is no longer commercially available.

**Federal Register: Vol. 54, page 12273, 03/24/1989**

**Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System or SWAM 5a Dual Channel Monitor****Automated Equivalent Method: EQPM-0912-205**

“Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System” or “SWAM 5a Dual Channel Monitor” configured for 1-hour measurements of PM<sub>10</sub> by beta attenuation on a single sampling line (Line A or B, but not both together), with the standard, louvered US EPA PM<sub>10</sub> size selective inlet specified in 40 CFR 50 Appendix L, using 47 mm glass fiber filters, at a sample flow set to 16.67 liters/min and software version 05-02.07.63 or later. Operated in accordance with the Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System Operation Manual.”

**Federal Register: Vol. 77, page 60985, 10/5/2012**

**Thermo Andersen Series FH 62 C14 Continuous PM<sub>10</sub> Monitor****Thermo Scientific Model 5014i Beta (5014i Beta), Continuous Ambient Particulate Monitor****Automated Equivalent Method: EQPM-1102-150**

“Thermo Andersen Series FH 62 C14 Continuous PM<sub>10</sub> Ambient Particulate Monitor and Thermo Scientific Model 5014i Beta (5014i Beta), Continuous Ambient Particulate Monitor,” operated for 24-hour average measurements, with the specified 10-micron EPA PM<sub>10</sub> inlet (or alternative louvered PM<sub>10</sub> inlet meeting 40 CFR 50 Appendix L specifications), inlet connector, sample tube with heater, roof flange kit, mass foil kit, pump kit, sample filter tape; with operational settings of 1000 L/h (16.67 L/min) sample flow rate, daily filter change, auto filter change at volumetric flow <950 L/h, auto filter change at mass >1500 micrograms, and factory default calculation mode settings operated with software version 1.07. Operated, calibrated and serviced according to the appropriate Operator Manual.

**Federal Register: Vol. 67, page 76174, 12/11/2002**

**Latest modifications: 07/2009; 12/2012**

**Thermo Scientific or Rupprecht & Patashnick Partisol<sup>®</sup> Model 2000 Air Sampler****Manual Reference Method: RFPS-0694-098**

“Thermo Scientific Partisol<sup>®</sup> 2000 Air Sampler” or “Rupprecht & Patashnick Partisol<sup>®</sup> Model 2000 Air Sampler,” consisting of a Hub Unit and 0, 1, 2, or 3 Satellite Units, with each sampling station used for PM<sub>10</sub> measurements equipped with a Rupprecht & Patashnick PM<sub>10</sub> inlet and operated for continuous 24-hour periods using the Basic, Manual, Time, Analog Input, or Serial Input programming modes, and with or without any of the following options: PM<sub>2.5</sub>-style filter cassette holder; louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19 in lieu of standard inlet; 57-002320 Stand for Hub or Satellite; 59-002542 Advanced EPROM; 10-001403 Large Pump (1/4 hp); 120 VAC. Hardware for Indoor Installation consists of: 51-002638-xxxx Temperature Sensor (Extended Length); 55-001289 Roof Flange (1 1/4"); 57-000604 Support Tripod for Inlet; 57-002526-0001 Sample Tube Extension (1 m); 57-002526-0002 Sample Tube Extension (2 m). Hardware for Outdoor Installation in Extreme Cold Environments consists of: 10-002645 Insulating Jacket for Hub Unit.

**Federal Register: Vol. 59, page 35338, 07/11/1994**

**Thermo Scientific Partisol<sup>®</sup> 2000-D Dichotomous Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2000i-D Dichotomous Air Sampler****Manual Equivalent Method: EQPS-0311-197**

“Thermo Scientific Partisol<sup>®</sup> 2000-D Dichotomous Air Sampler” or “Thermo Fisher Scientific Partisol<sup>®</sup> 2000i-D Dichotomous Air Sampler,” configured for dual-filter, single-event sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particles, operated with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate fine and coarse PM into two samples for collection on two separate filter membranes, for a 24-hour sampling period and in accordance with the Partisol<sup>®</sup> 2000-D or Partisol<sup>®</sup> 2000i-D instruction manual, as appropriate. Partisol<sup>®</sup> 2000i-D operated with firmware version 2.0 or greater.

**Federal Register: Vol. 76, page 15974, 03/22/2011**

**Latest modification: 06/2011**

***Thermo Scientific Partisol® 2000-FRM PM<sub>10</sub> Air Sampler or Thermo Fisher Scientific Partisol® 2000i PM<sub>10</sub> Air Sampler or Rupprecht and Patashnick Partisol®-FRM 2000 PM<sub>10</sub> Air Sampler***

**Manual Reference Method: RFPS-1298-126**

“Thermo Scientific Partisol® 2000-FRM PM<sub>10</sub> Air Sampler” or “Thermo Fisher Scientific Partisol® 2000i PM<sub>10</sub> Air Sampler” or “Rupprecht and Patashnick Partisol®-FRM 2000 PM<sub>10</sub> Air Sampler,” with PM<sub>10</sub> inlet or louvered inlet specified in 40 CFR 50, Appendix L, Figs. L-2 through L-19, configured as a PM<sub>10</sub> reference method with a U.S. EPA PM<sub>10</sub> inlet with straight downtube adapter and operated for 24-hour continuous sampling periods in accordance with the Partisol® 2000-FRM or Partisol® 2000i instruction manual, as appropriate, and with the requirements specified in 40 CFR Part 50, Appendix J or Appendix M. Model 2000i operated with firmware version 2.0 or greater.

**Federal Register: Vol. 63, page 69625, 12/17/1998**

**Latest modification: 06/ 2011**

***Thermo Scientific Partisol®-Plus 2025 PM<sub>10</sub> Sequential Air Sampler or Thermo Fisher Scientific Partisol® 2025i PM<sub>10</sub> Sequential Air Sampler or Rupprecht and Patashnick Partisol®-Plus 2025 PM<sub>10</sub> Sequential Air Sampler***

**Manual Reference Method: RFPS-1298-127**

“Thermo Scientific Partisol®-Plus 2025 Sequential Air Sampler” or “Thermo Fisher Scientific Partisol® 2025i Sequential Air Sampler” or “Rupprecht and Patashnick Partisol®-Plus 2025 PM<sub>10</sub> Sequential Air Sampler,” with PM<sub>10</sub> inlet or louvered inlet specified in 40 CFR 50, Appendix L, Figs. L-2 through L-19, configured as a PM<sub>10</sub> reference method and operated for 24-hour continuous sampling periods. Partisol®-Plus 2025 to be operated with any software version 1.003 through 1.5 and the Partisol® 2025i with firmware version 2.0 or greater, with the modified filter shuttle mechanism, in accordance with the Partisol®-Plus 2025 or Partisol® 2025i instruction manual, as appropriate, and with the requirements specified in 40 CFR Part 50, Appendix J or Appendix M.

**Federal Register: Vol. 63, page 69625, 12/17/1998**

**Last modified: 06/ 2011**

***Thermo Scientific Dichotomous Partisol®-Plus 2025-D Sequential Air Sampler or Thermo Fisher Scientific Dichotomous Partisol® 2025i-D Sequential Air Sampler***

**Manual Equivalent Method: EQPS-0311-198**

“Thermo Scientific Dichotomous Partisol®-Plus 2025-D Sequential Air Sampler” or “Thermo Fisher Scientific Dichotomous Partisol® 2025i-D Sequential Air Sampler,” configured for dual-filter sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particles, with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate the fine and coarse PM into two samples for collection on two separate filter membranes, and operated with the modified filter shuttle mechanism implemented May 31, 2008, and firmware version 1.500 or greater for the Partisol®-Plus 2025-D and version 2.0 or greater for the Partisol® 2025i-D, for 24-hour continuous sampling periods, in accordance with the Partisol®-Plus 2025-D or Partisol® 2025i-D instruction manual, as appropriate.

**Federal Register: Vol. 76, page 15975, 03/22/2011**

**Latest modification: 06/ 2011**

***Thermo Scientific TEOM® 1400AB/TEOM® 1405 Ambient Particulate Monitor or Rupprecht & Patashnick TEOM® Series 1400/1400a PM<sub>10</sub> Monitors***

**Automated Equivalent Method: EQPM-1090-079**

“Thermo Scientific TEOM® 1400AB [PM<sub>10</sub>] Ambient Particulate Monitor” or “Rupprecht & Patashnick TEOM® Series 1400 and Series 1400a PM-10 Monitors,” (including serial number prefixes 1400, 140A, 140AA, 140AB, 140AT, and 140UP, 1405A), consisting of the following components: TEOM® Sensor Unit; TEOM® Control Unit; Flow Splitter (3 liter/min sample flow); Teflon-Coated Glass Fiber Filter Cartridges; Rupprecht & Patashnick PM-10 Inlet (part number 57-00596), Sierra-Andersen Model 246b PM-10 Inlet (16.7 liter/min) or louvered inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19; operated for 24-hour average measurements, with the total mass averaging time set at 300 seconds, the mass rate/mass concentration averaging time set at 300 seconds, the gate time set at 2 seconds, and with or without any of the following

options: Tripod; Outdoor Enclosure; Automatic Cartridge Collection Unit (Series 1400a only); Flow Splitter Adapter (for 1 or 2 liter/min sample flow). Thermo Scientific TEOM® 1405 Ambient Particulate Monitor with combined sensor and control units and redesigned mass transducer and user interface, operated in accordance with the Thermo Scientific TEOM® 1405 instrument manual.

**Federal Register: Vol. 55, page 43406, 10/29/1990**

**Latest modification: 12/2008**



***Tisch Environmental Model TE-6070 PM<sub>10</sub> High-Volume Air Sampler or New Star Environmental Model NS-6070 PM<sub>10</sub> High-Volume Air Sampler*****Manual Reference Method: RFPS-0202-141**

"Tisch Environmental Model TE-6070 or New Star Environmental Model NS-6070 PM<sub>10</sub> High-Volume Air Sampler," consisting of a TE-6001 PM<sub>10</sub> size-selective inlet, 8" x 10" filter holder, aluminum outdoor shelter, mass flow controller or volumetric flow controller with brush or brushless motor, 7-day mechanical off/on-elapsed timer or 11-day digital off/on-elapsed timer, and any of the high volume sampler variants identified as TE-6070-BL or NS-6070-BL, TE-6070D or NS-6070D, TE-6070D-BL or NS-6070-BL, TE-6070V or NS-6070V, TE-6070V-BL or NS-6070V-BL, TE-6070-DV or NS-6070-DV, or TE-6070DV-BL or NS-6070DV-BL, with or without the optional stainless steel filter media holder/filter cartridge or continuous flow/pressure recorder.

**Federal Register: Vol. 67, page 15566, 04/02/2002**

***Wedding & Associates' or Thermo Environmental Instruments Inc. Model 600 PM<sub>10</sub> High-Volume Sampler*****Manual Reference Method: RFPS-1087-062**

"Wedding & Associates' or Thermo Environmental Instruments, Inc. Model 600 PM<sub>10</sub> Critical Flow High-Volume Sampler," consisting of the following W&A/TEII components: PM<sub>10</sub> Inlet; Critical Flow Device; Anodized Aluminum Shelter; Blower Motor Assembly for 115, 220 or 240 VAC and 50/60 Hz; Mechanical Timer; Elapsed Time Indicator; and Filter Cartridge/Cassette, and with or without the following options: Digital Timer, 6 or 7 Day Timer, and 1 or 7 Day Pressure Recorder.

**Federal Register: Vol. 52, page 37366, 10/06/1987**

***Wedding & Associates' or Thermo Environmental Instruments Inc. Model 650 PM<sub>10</sub> Beta Gauge*****Automated Equivalent Method: EQPM-0391-081**

"Wedding & Associates' or Thermo Environmental Instruments, Inc. Model 650 PM<sub>10</sub> Beta Gauge Automated Particle Sampler," consisting of the following W&A/TEII components: Particle Sampling Module, PM<sub>10</sub> Inlet (18.9 liter/min), Inlet Tube and Support Ring, Vacuum Pump (115, 220 or 240 VAC and 50/60 Hz); and operated for 24-hour average measurements with glass fiber filter tape.

**Federal Register: Vol. 56, page 9216, 03/05/1991**

**Particulate Matter – PM<sub>2.5</sub>****Andersen Model RAAS2.5-200 PM<sub>2.5</sub> Ambient Audit Air Sampler****Manual Reference Method: RFPS-0299-128**

“Andersen Instruments, Incorporated Model RAAS2.5-200 PM<sub>2.5</sub> Audit Sampler,” configured as a PM<sub>2.5</sub> reference method and operated with software (firmware) version 4B, 5.0.1 - 6.09, 6.0A, or 6.0B, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the Model RAAS2.5-200 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 64, page 12167, 03/11/1999**

**BGI Inc. Models PQ200 or PQ200A PM<sub>2.5</sub> Ambient Fine Particle Sampler****Manual Reference Method: RFPS-0498-116**

“BGI Incorporated Models PQ200 and PQ200A PM<sub>2.5</sub> Ambient Fine Particle Sampler,” operated with firmware version 3.88 or 3.89R, for 24-hour continuous sample periods, in accordance with the Model PQ200/PQ200A Instruction Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L, and with or without the optional Solar Power Supply or the optional dual-filter cassette (P/N F-21/6) and associated lower impactor housing (P/N B2027), where the upper filter is used for PM<sub>2.5</sub>. The Model PQ200A is described as a portable audit sampler and includes a set of three carrying cases.

**Federal Register: Vol. 63, page 18911, 04/16/1998**

**BGI Inc. Models PQ200-VSCC™ or PQ200A-VSCC™ PM<sub>2.5</sub> Sampler****Manual Reference Method: RFPS-0498-116 or Manual Equivalent Method: EQPM-0202-142**

“BGI Incorporated Models PQ200-VSCC™ or PQ200A-VSCC™ PM<sub>2.5</sub> Ambient Fine Particle Sampler,” configured with a BGI VSCC™ Very Sharp Cut Cyclone particle size separator and operated with firmware version 3.88, 3.91, 3.89R, or 3.91R, for 24-hour continuous sample periods, in accordance with the Model PQ200/PQ200A Instruction Manual and VSCC™ supplemental manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L, and with or without the optional Solar Power Supply or the optional dual-filter cassette (P/N F-21/6) and associated lower impactor housing (P/N B2027), where the upper filter is used for PM<sub>2.5</sub>. The Model PQ200A VSCC™ is described as a portable audit sampler and includes a set of three carrying cases.

**Federal Register: Vol. 67, page 15567, 04/02/2002**

**Graseby Andersen Model RAAS2.5-100 PM<sub>2.5</sub> Ambient Air Sampler****Manual Reference Method: RFPS-0598-119**

“Graseby Andersen Model RAAS2.5-100 PM<sub>2.5</sub> Ambient Air Sampler,” operated with software version 4B, 5.0.1 - 6.09, 6.0A, or 6.0B, configured for “Single 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the Model RAAS2.5-100 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 63, page 31991, 06/11/1998**

**Graseby Andersen Model RAAS2.5-300 PM<sub>2.5</sub> Sequential Ambient Air Sampler****Manual Reference Method: RFPS-0598-120**

“Graseby Andersen Model RAAS2.5-300 PM<sub>2.5</sub> Sequential Ambient Air Sampler,” operated with software version 4B, 5.0.1 - 6.09, 6.0A, or 6.0B, configured for “Multi 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the Model RAAS2.5-300 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 63, page 31991, 06/11/1998**

**Grimm Model EDM 180 PM<sub>2.5</sub> Monitor****Automated Equivalent Method: EQPM-0311-195**

“Grimm Technologies, Inc. Model EDM 180 PM<sub>2.5</sub> or Tisch Environmental TE-EDM 180 PM<sub>2.5</sub> Monitor,” light scattering continuous ambient particulate monitor operated for 24 hours at a volumetric flow rate of 1.2 L/min, configured with a Nafion®-type air sample dryer, complete for operation with firmware version 7.80 or later, in accordance with the Grimm Technologies, Inc. Model EDM 180 Operation and Instruction Manual. The optional graphic presentation can be made with the software model 1.177 version 3.30 or later.

**Federal Register:** Vol. 76, page 15974, 03/22/2011

**Latest Modification:** 01/2012

**Met One BAM-1020 Monitor – PM<sub>2.5</sub> FEM Configuration, Horiba APDA-371– PM<sub>2.5</sub> Configuration, or Ecotech Spirant BAM1100****Automated Equivalent Method: EQPM-0308-170**

“Met One Instruments, Inc. BAM-1020 Beta Attenuation Mass Monitor – PM<sub>2.5</sub> FEM Configuration, Horiba Instruments APDA-371 Beta Attenuation Mass Monitor – PM<sub>2.5</sub> FEM Configuration, or Ecotech Spirant BAM1100”, configured with the standard BX-802 EPA PM<sub>10</sub> inlet (or alternative louvered PM<sub>10</sub> inlet meeting 40 CFR 50 Appendix L specifications) and with a BGI VSCC™ Very Sharp Cut Cyclone particle size separator, operated for 24 1-hour average measurements with firmware revision 3.2.4 or later, with or without an inlet tube extension (BX-823), with or without external enclosures BX-902 or BX-903 and equipped with BX-596 ambient temperature and barometric pressure combination sensor, internal BX-961 automatic flow controller operated in Actual (volumetric) flow control mode, the standard BX-802 EPA PM<sub>10</sub> inlet head and a BGI VSCC™ Very Sharp Cut Cyclone (VSCC-A), BX-827 (110V) or BX-830 (230V) Smart Inlet Heater, with the heater RH set to 35% and the temperature control set to "off", the 8470-1 revision D or later tape control transport assembly with close geometry beta source configuration, used with standard glass fiber filter tape, COUNT TIME parameter set for 8 minutes, the SAMPLE TIME parameter set for 42 minutes, BX-302 zero filter calibration kit required and with or without BX-970 touch-screen display with USB interface. Instrument must be operated in accordance with the BAM 1020 Particulate Monitor operation manual, revision F or later, the Horiba APDA-371 Monitor operation manual or the Ecotech Spirant BAM Manual (rev. A or later), as appropriate, and the VSCC™ supplemental manual.

**Federal Register:** Vol. 73, page 13224, 03/12/2008

**Latest modifications:** 7/2010; 8/2010; 8/2012

**Opsis SM200- Dust Monitor****Automated Equivalent Method: EQPM-0812-203**

“Opsis SM200- Dust Monitor” configured for PM<sub>2.5</sub> with the US EPA PM<sub>10</sub> inlet specified in 40 CFR 50 Appendix L, followed by a BGI Inc. Very Sharp Cut Cyclone (VSCC™) particle size separator, operated for a 24-hour continuous sample period at a total actual flow rate of 16.67 L/min. using 47mm PTFE membrane filters, a TS200 temperature stabilizer and software version 1.04.16 or later, in accordance with the Opsis SM200 Dust Monitor Operation and Instruction Guide.

**Federal Register:** Vol. 77, page 55832, 09/11/2012

**Rupprecht & Patashnick Partisol®-FRM Model 2000 PM<sub>2.5</sub> Air Sampler****Manual Reference Method: RFPS-0498-117**

“Rupprecht & Patashnick Company, Incorporated Partisol®-FRM Model 2000 PM<sub>2.5</sub> Air Sampler,” operated with software versions 1.102 - 1.202, with either R&P-specified machined or molded filter cassettes, with or without the optional insulating jacket for cold weather operation, for 24-hour continuous sample periods, in accordance with the Model 2000 Instruction Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register:** Vol. 63, page 18911, 04/16/1998

**Rupprecht & Patashnick Partisol® Model 2000 PM<sub>2.5</sub> Audit Sampler****Manual Reference Method: RFPS-0499-129**

“Rupprecht & Patashnick Company, Inc. Partisol® Model 2000 PM<sub>2.5</sub> Audit Sampler,” configured as a PM<sub>2.5</sub> reference method and operated with software (firmware) version 1.2 - 1.202, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, in accordance with the Partisol® Model 2000 Operating Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register:** Vol. 64, page 19153, 04/19/1999

***Rupprecht & Patashnick Partisol<sup>®</sup> Model 2000 PM<sub>2.5</sub> FEM Audit Sampler*****Manual Reference Method: RFPS-0499-129 or Manual Equivalent Method: EQPM-0202-144**

“Rupprecht & Patashnick Co., Inc. Partisol<sup>®</sup> Model 2000 PM<sub>2.5</sub> FEM Audit Sampler,” configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator, and operated with software (firmware) version 1.2 - 1.202, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, in accordance with the Partisol<sup>®</sup> Model 2000 Operating Manual and VSCC<sup>™</sup> supplemental manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 67, page 15567, 04/02/2002**

***Rupprecht & Patashnick Partisol<sup>®</sup>-Plus Model 2025 Sequential Air Sampler*****Manual Reference Method: RFPS-0498-118**

“Rupprecht & Patashnick Company, Incorporated Partisol<sup>®</sup>-Plus Model 2025 PM<sub>2.5</sub> Sequential Air Sampler,” operated with any software version 1.003 through 1.4.16, with either R&P-specified machined or molded filter cassettes, for 24-hour continuous sample periods, in accordance with the Model 2025 Instruction Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 63, page 18911, 04/16/1998**

***Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System or SWAM 5a Dual Channel Monitor*****Automatic Equivalent Method: EQPM-0912-204**

“Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System” or “SWAM 5a Dual Channel Monitor” configured for 1-hour measurements of PM<sub>2.5</sub> by beta attenuation, on either a single (Line A or B) or both sampling lines (Line A and B) simultaneously, using 47 mm glass fiber filters, at a sample flow set to 16.67 liters/min and software version 05-02.07.63 or later and with an inlet system comprised of a PM<sub>10</sub> pre-impactor inlet (based on European PM<sub>10</sub> inlet design) combined with a BGI VSCC<sup>™</sup> PM<sub>2.5</sub> cyclone separator. Operated in accordance with the Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System Operation Manual.

**Federal Register: Vol. 77, page 60985, 10/5/2012**

***Thermo Electron Model RAAS2.5-100 FEM PM<sub>2.5</sub> Ambient Air Sampler*****Manual Reference Method: RFPS-0598-119 or Manual Equivalent Method: EQPM-0804-153**

“Thermo Electron Corporation Model RAAS2.5-100 FEM PM<sub>2.5</sub> Ambient Air Sampler,” configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator and operated with software version 06.0B.00 configured for “Single 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, in accordance with the Model RAAS2.5-100 FEM Operator’s Manual and VSCC<sup>™</sup> supplemental manual, and in accordance with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 69, page 47924, 08/06/2004**

***Thermo Electron Model RAAS2.5-200 FEM PM<sub>2.5</sub> Audit Air Sampler*****Manual Reference Method: RFPS-0299-128 or Manual Equivalent Method: EQPM-0804-154**

“Thermo Electron Corporation Model RAAS2.5-200 FEM PM<sub>2.5</sub> Audit Air Sampler,” configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator and operated with software version 06.0B.00, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, in accordance with the Model RAAS2.5-200 FEM Operator’s Manual and VSCC<sup>™</sup> supplemental manual, and in accordance with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 69, page 47924, 08/06/2004**

***Thermo Electron Model RAAS2.5-300 FEM PM<sub>2.5</sub> Sequential Ambient Air Sampler*****Manual Reference Method: RFPS-0598-120 or Manual Equivalent Method: EQPM-0804-155**

“Thermo Electron Corporation Model RAAS2.5-300 FEM PM<sub>2.5</sub> Sequential Ambient Air Sampler,” configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator and operated with software version 06.0B.00 configured for “Multi 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, in accordance with the Model RAAS2.5-300 FEM Operator’s Manual and VSCC<sup>™</sup> supplemental manual, and in accordance with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 69, page 47924, 08/06/2004**

***Thermo Environmental Instruments, Incorporated Model 605 “CAPS” Sampler*****Manual Reference Method: RFPS-1098-123**

“Thermo Environmental Instruments, Incorporated Model 605 “CAPS” Computer Assisted Particle Sampler,” configured as a PM<sub>2.5</sub> reference method and operated with software version 1.02A, for 24-hour continuous sample periods, in accordance with the Model 605 Instruction Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register: Vol. 63, page 58036, 10/29/1998**

***Thermo Scientific Model 5014i or Thermo Scientific FH62C14-DHS Continuous Ambient Particle Monitor*****Automated Equivalent Method: EQPM-0609-183**

“Thermo Scientific Model 5014i or FH62C14-DHS Continuous Ambient Particle Monitor,” operated at a flow rate of 16.67 liters per minute for 24-hour average measurements configured for PM<sub>2.5</sub> with a louvered PM<sub>10</sub> size selective inlet as specified in 40 CFR 50 Appendix L, Figs. L-2 through L-19, a PM<sub>2.5</sub> BGI Inc. Very Sharp Cut Cyclone (VSCC™) particle size separator, inlet connector, sample tube, DHS heater with 35% RH threshold, mass foil kit, GF10 filter tape, 8-hour filter change, and operational calibration and servicing as outlined in the 5014i Continuous Ambient Particulate Monitor or FH62C14-DHS Continuous Ambient Particulate Monitor operating manual.

**Federal Register: Vol. 74, page 28696, 06/17/2009.**

**Latest modification: 03/2010**

***Thermo Scientific Model 5030i SHARP Monitor or Model 5030 SHARP Monitor*****Automated Equivalent Method: EQPM-0609-184**

“Thermo Scientific Model 5030 SHARP Monitor,” operated at a flow rate of 16.67 liters per minute for 24-hour average measurements configured for PM<sub>2.5</sub> with a louvered PM<sub>10</sub> size selective inlet as specified in 40 CFR 50 Appendix L, Figs. L-2 through L-19, a PM<sub>2.5</sub> BGI Inc. Very Sharp Cut Cyclone (VSCC™) particle size separator, inlet connector, sample tube, DHS heater with 58% RH threshold, mass foil kit, GF10 filter tape, nephelometer zeroing kit, 8-hour filter change, and operational calibration and servicing as outlined in the Model 5030i or Model 5030 SHARP instructional manual.

**Federal Register: Vol. 74, page 28696, 06/17/2009**

**Latest modification: 06/2013**

***Thermo Scientific TEOM® 1400a Ambient Particular Monitor with Series 8500C FDMS®; Thermo Scientific TEOM® 1405-F Ambient Particular Monitor with FDMS®*****Automated Equivalent Method: EQPM-0609-181**

“Thermo Scientific TEOM® 1400a Ambient Particular Monitor with Series 8500C FDMS® (Filter Dynamics Measurement System) or Thermo Scientific TEOM® 1405-F Ambient Particular Monitor with FDMS®,” configured for PM<sub>2.5</sub> with the US EPA PM<sub>10</sub> inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19, followed by a BGI Inc. Very Sharp Cut Cyclone (VSCC™) particle size separator, operated with a total actual flow of 16.67 L/min., loaded with Series FDMS® 8500 module operating software and an FDMS® kit. TEOM® 1400a with Series 8500C FDMS® operated with firmware version 3.20 and later and TEOM® 1405-F with FDMS® operated with version 1.55 or later and according to the appropriate operating manual.

**Federal Register: Vol. 74, page 28696, 06/17/2009**

**Latest Modification: 09/2010**

***Thermo Scientific TEOM® 1405-DF Dichotomous Ambient Particular Monitor with FDMS®*****Automated Equivalent Method: EQPM-0609-182**

“Thermo Scientific TEOM® 1405-DF Dichotomous Ambient Particular Monitor with FDMS®,” configured for dual filter sampling of fine (PM<sub>2.5</sub>) and coarse particles using the US EPA PM<sub>10</sub> inlet specified in 40 CFR 50 Appendix L, Figs. L-2 thru L-19 and a virtual impactor, with a total flow rate of 16.67 L/min, fine sample flow of 3 L/min, and coarse sample flow rate of 1.67 L/min, and operating with firmware version 1.50 and later, operated with or without external enclosures, and operated in accordance with the Thermo Scientific TEOM® 1405-DF Dichotomous Ambient Particulate Monitor Instruction Manual (designated for PM<sub>2.5</sub> measurements only).

**Federal Register: Vol. 74, page 28696, 06/17/2009**

***Thermo Scientific Partisol® 2000-D Dichotomous Air Sampler or Thermo Fisher Scientific Partisol® 2000i-D Dichotomous Air Sampler*****Manual Equivalent Method: EQPS-0509-177**

“Thermo Scientific Partisol® 2000-D Dichotomous Air Sampler” or “Thermo Fisher Scientific Partisol® 2000i-D Dichotomous Air Sampler,” configured for dual-filter, single-event sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particles, operated with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate fine and coarse PM into two samples for

collection on two separate filter membranes, for a 24-hour sampling period, in accordance with the Partisol<sup>®</sup> 2000-D or Partisol<sup>®</sup> 2000i-D instruction manual, as appropriate. Partisol<sup>®</sup> 2000i-D operated with firmware version 2.0 or greater.

**Federal Register:** Vol. 74, page 26395, 06/02/2009

**Latest modification:** 06/ 2011

***Thermo Scientific Partisol<sup>®</sup>-Plus 2025-D Dichotomous Sequential Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2025i-D Dichotomous Sequential Air Sampler***

**Manual Equivalent Method: EQPS-0509-179**

“Thermo Scientific Partisol<sup>®</sup>-Plus 2025-D Dichotomous Sequential Air Sampler” or “Thermo Fisher Scientific Partisol<sup>®</sup> 2025i-D Dichotomous Sequential Air Sampler,” configured for dual-filter sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particle components, with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate the fine and coarse PM into two samples for collection on two separate filter membranes, and operated with the modified filter shuttle mechanism implemented May 31, 2008, and firmware version 1.500 or greater for the Partisol<sup>®</sup>-Plus 2025-D and version 2.0 or greater for the Partisol<sup>®</sup> 2025i-D, for 24-hour continuous sampling periods, in accordance with the Partisol<sup>®</sup>-Plus 2025-D or Partisol<sup>®</sup> 2025i-D instruction manual, as appropriate.

**Federal Register:** Vol. 74, page 26395, 06/02/2009

**Latest modification:** 06/ 2011

***Thermo Scientific Partisol<sup>®</sup> 2000-FRM PM<sub>2.5</sub> Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2000i PM<sub>2.5</sub> Air Sampler or Rupprecht & Patashnick Partisol<sup>®</sup>-FRM 2000 PM<sub>2.5</sub> Air Sampler***

**Manual Reference Method: RFPS-0498-117 or Manual Equivalent Method: EQPM-0202-143**

“Thermo Scientific Partisol<sup>®</sup> 2000-FRM PM<sub>2.5</sub> Air Sampler” or “Thermo Fisher Scientific Partisol<sup>®</sup> 2000i PM<sub>2.5</sub> Air Sampler” or “Rupprecht & Patashnick Partisol<sup>®</sup>-FRM 2000 PM<sub>2.5</sub> [FEM] Air Sampler configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator with either R&P-specified machined or molded filter cassettes, for 24-hour continuous sampling periods, in accordance with the Partisol<sup>®</sup> 2000-FRM or Partisol<sup>®</sup> 2000i Instruction Manual, as appropriate, and VSCC<sup>™</sup> supplemental manual. Partisol<sup>®</sup> 2000-FRM operated with software versions 1.102 through 1.202 and Partisol<sup>®</sup> 2000i with firmware version 2.0 or greater. Method is operated with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L, and with or without the optional insulating jacket for cold weather operation.

**Federal Register:** Vol. 67, page 15567, 04/02/2002

**Latest modification:** 06/ 2011

***Thermo Scientific Partisol<sup>®</sup>-Plus 2025 Sequential PM<sub>2.5</sub> Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2025i Sequential PM<sub>2.5</sub> Air Sampler or Rupprecht & Patashnick Partisol<sup>®</sup>-Plus 2025 PM<sub>2.5</sub> Sequential Sampler***

**Manual Reference Method: RFPS-0498-118 or Manual Equivalent Method: EQPM-0202-145**

“Thermo Scientific Partisol<sup>®</sup>-Plus 2025 PM<sub>2.5</sub> Sequential Air Sampler” or “Thermo Fisher Scientific Partisol<sup>®</sup> 2025i PM<sub>2.5</sub> Sequential Air Sampler” or “Rupprecht & Patashnick Partisol<sup>®</sup>-Plus 2025 PM<sub>2.5</sub> (FEM) Sequential Air Sampler,” configured with a BGI VSCC<sup>™</sup> Very Sharp Cut Cyclone particle size separator with either R&P-specified machined or molded filter cassettes, for 24-hour continuous sampling periods. Partisol<sup>®</sup>-Plus 2025 to be operated with any software version 1.003 through 1.5 and Partisol<sup>®</sup> 2025i with firmware version 2.0 or greater, and with the modified filter shuttle mechanism. Method to be operated in accordance with the Partisol<sup>®</sup>-Plus 2025 or Partisol<sup>®</sup> 2025i instruction manual, as appropriate, with the VSCC<sup>™</sup> supplemental manual, and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register:** Vol. 67, page 15567, 04/02/2002

**Latest modification:** 06/ 2011

***URG-MASS100 Single PM<sub>2.5</sub> FRM Sampler***

**Manual Reference Method: RFPS-0400-135**

“URG-MASS100 Single PM<sub>2.5</sub> FRM Sampler,” operated with software (firmware) version 4B or 5.0.1, configured for “Single 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the URG-MASS100 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

**Federal Register:** Vol. 65, page 26603, 05/08/2000

***URG-MASS300 Sequential PM<sub>2.5</sub> FRM Sampler*****Manual Reference Method: RFPS-0400-136**

“URG-MASS300 Sequential PM<sub>2.5</sub> FRM Sampler,” operated with software (firmware) version 4B or 5.0.1, configured for “Multi 2.5” operation, for 24-hour continuous sample periods at a flow rate of 16.67 liters/minute, and in accordance with the URG-MASS300 Operator’s Manual and with the requirements and sample collection filters specified in 40 CFR Part 50, Appendix L.

***Federal Register: Vol. 65, page 26603, 05/08/2000***

**Particulate Matter – PM<sub>10-2.5</sub>*****BGI Incorporated Model PQ200 Sampler Pair*****Manual Reference Method: RFPS-1208-173**

“BGI Incorporated Model PQ200 PM<sub>10-2.5</sub> Sampler Pair,” for the determination of coarse particulate matter as PM<sub>10-2.5</sub> consisting of a pair of BGI Model PQ200 samplers, with one configured for sampling PM<sub>2.5</sub> (RFPS-0498-116) and the other configured for sampling PM<sub>10c</sub> (RFPS-1298-125) with the PM<sub>2.5</sub> separator replaced with a BGI WINS Eliminator and operated in accordance with the Model PQ200 Instruction manual supplement Appendix O.

**Federal Register: Vol. 73, page 77024, 12/18/2008**

***Met One Instruments BAM-1020 PM<sub>10-2.5</sub> Measurement System*****Automated Equivalent Method: EQPM-0709-185**

“Met One Instruments BAM-1020 PM<sub>10-2.5</sub> Measurement System,” consisting of 2 BAM-1020 monitors, the first of which (PM<sub>2.5</sub> measurement) is configured as a PM<sub>2.5</sub> FEM (EQPM-0308-170). The second BAM-1020 monitor (PM<sub>10</sub> measurement) is configurable as a PM<sub>2.5</sub> FEM (EQPM-0308-170), but set to monitor PM<sub>10</sub>. The BAM-1020 monitors are collocated to within 1-4 meters of one another. The BAM-1020 performing the PM<sub>2.5</sub> measurement is equipped with Met One Instruments, Inc. P/N BX-Coarse interface board and accessories; the units are interconnected to provide concurrent sampling and to report PM<sub>10-2.5</sub> concentrations directly to the user. Both units are operated in accordance with BAM-1020 PM-Coarse Addendum Rev. 5-5 or later and the BAM-1020 Operations Manual Rev. D or later.

**Federal Register: Vol. 74, page 28241, 06/15/2009**

***Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System or SWAM 5a Dual Channel Monitor*****Automated Equivalent Method: EQPM -0912-206**

“Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System” or “SWAM 5a Dual Channel Monitor” configured for 1-hour measurements of PM<sub>10</sub> and PM<sub>2.5</sub> by beta attenuation, with the standard, louvered US EPA PM<sub>10</sub> size selective inlet specified in 40 CFR 50 Appendix L on one channel (Line A or B) and with an inlet system comprised of a PM<sub>10</sub> pre-impactor inlet (based on European PM<sub>10</sub> inlet design) combined with a BGI VSCC<sup>TM</sup> PM<sub>2.5</sub> cyclone separator on the second channel (Line A or B, but always with PM<sub>10</sub> on the opposite Line). The PM<sub>10-2.5</sub> mass measurement is performed using the resultant subtraction of PM<sub>10</sub> minus PM<sub>2.5</sub>. Operated in accordance with the Teledyne Model 602 Beta<sup>PLUS</sup> Particle Measurement System Operation Manual.

**Federal Register: Vol. 77, page 60985, 10/5/2012**

***Thermo Scientific Partisol<sup>®</sup> 2000-D Dichotomous Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2000i-D Dichotomous Air Sampler*****Manual Equivalent Method: EQPS-0509-178**

“Thermo Scientific Partisol<sup>®</sup> 2000-D Dichotomous Air Sampler” or “Thermo Fisher Scientific Partisol<sup>®</sup> 2000i-D Dichotomous Air Sampler,” configured for dual-filter, single-event sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particles, operated with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate fine and coarse PM into two samples for collection on two separate filter membranes, for a 24-hour sampling period, in accordance with the Partisol<sup>®</sup> 2000-D or Partisol<sup>®</sup> 2000i-D instruction manual, as appropriate. Partisol<sup>®</sup> 2000i-D operated with firmware version 2.0 or greater.

**Federal Register: Vol. 74, page 26395, 06/02/2009**

**Latest modification: 06/ 2011**

***Thermo Scientific Partisol<sup>®</sup>-Plus 2025-D Dichotomous Sequential Air Sampler or Thermo Fisher Scientific Partisol<sup>®</sup> 2025i-D Dichotomous Sequential Air Sampler*****Manual Equivalent Method: EQPS-0509-180**

“Thermo Scientific Partisol<sup>®</sup>-Plus 2025-D Dichotomous Sequential Air Sampler” or “Thermo Scientific Partisol<sup>®</sup> 2025i-D Dichotomous Sequential Air Sampler,” configured for dual-filter sampling of fine (PM<sub>2.5</sub>) and coarse (PM<sub>10-2.5</sub>) particle components, with a U.S. EPA PM<sub>10</sub> inlet and using a virtual impactor to separate the fine and coarse PM into two samples for collection on two separate filter membranes, and operated with the modified filter shuttle mechanism implemented May 31, 2008, and firmware version 1.500 or greater for the Partisol<sup>®</sup>-Plus 2025-D and version 2.0 or greater for the Partisol<sup>®</sup> 2025i-D, for 24-hour continuous sampling periods, in accordance with the Partisol<sup>®</sup>-Plus 2025-D or Partisol<sup>®</sup> 2025i-D instruction manual, as appropriate.

**Federal Register: Vol. 74, page 26395, 06/02/2009**

**Latest modification: 06/ 2011**



***Thermo Scientific Partisol® 2000-FRM PM<sub>10-2.5</sub> Sampler Pair or Thermo Fisher Scientific Partisol® 2000i PM<sub>10-2.5</sub> Air Sampler Pair*****Manual Reference Method: RFPS-0509-175**

“Thermo Scientific Partisol® 2000-FRM PM<sub>10-2.5</sub> Sampler Pair” or “Thermo Fisher Scientific Partisol® 2000i PM<sub>10-2.5</sub> Air Sampler Pair,” for the determination of coarse particulate matter as PM<sub>10-2.5</sub>, consisting of a pair of Thermo Scientific Partisol® 2000-FRM or 2000i samplers, with one configured as a PM<sub>2.5</sub> sampler (RFPS-0498-117) and the other configured as a PM<sub>10c</sub> sampler with the PM<sub>2.5</sub> separator replaced with a Thermo Scientific WINS bypass downtube (RFPS-1298-126), with U.S. EPA PM<sub>10</sub> inlets on both samplers and operated in accordance with the Partisol® 2000-FRM or 2000i instruction manual supplement, as appropriate, and the 2000i operated with firmware version 2.0 or greater.

**Federal Register: Vol. 74, page 26395, 06/02/2009**

**Latest modification: 06/ 2011**

***Thermo Scientific Partisol®-Plus 2025 Sequential PM<sub>10-2.5</sub> Air Sampler Pair or Thermo Fisher Scientific Partisol® 2025i Sequential PM<sub>10-2.5</sub> Air Sampler Pair*****Manual Reference Method: RFPS-0509-176**

“Thermo Scientific Partisol®-Plus 2025 Sequential PM<sub>10-2.5</sub> Air Sampler Pair” or “Thermo Fisher Scientific Partisol® 2025i Sequential PM<sub>10-2.5</sub> Air Sampler Pair,” for the determination of coarse particulate matter as PM<sub>10-2.5</sub>, consisting of a pair of Thermo Scientific Partisol®-Plus 2025 sequential samplers or a pair of Thermo Fisher Scientific Partisol® 2025i sequential samplers, with one configured as a PM<sub>2.5</sub> sampler (RFPS-0498-118) and the other configured as a PM<sub>10c</sub> sampler with the PM<sub>2.5</sub> separator replaced with a Thermo Scientific Partisol® 2025 downtube (RFPS-1298-127). Partisol®-Plus 2025 to be operated with any software version 1.003 through 1.5 and Partisol® 2025i with firmware version 2.0 or greater, with the modified filter shuttle mechanism. Method to be operated in accordance with the Partisol®-Plus 2025 or Partisol® 2025i instruction manual supplement, as appropriate.

**Federal Register: Vol. 74, page 26395, 06/02/2009**

**Latest modification: 06/ 2011**

**Sulfur Dioxide*****Reference Method for SO<sub>2</sub> (Pararosaniline Method)*****Manual Reference Method: 40 CFR Part 50, Appendix A**

Reference Method for the Determination of Sulfur Dioxide in the Atmosphere (Pararosaniline Method)

***Federal Register: Vol. 47, page 54899, 12/06/1982 and Vol. 48, 17355, 04/22/1983******Pararosaniline Method for SO<sub>2</sub> - Technicon I*****Manual Equivalent Method: EQS-0775-001**

“Pararosaniline Method for the Determination of Sulfur Dioxide in the Atmosphere-Technicon I Automated Analysis System”

***Federal Register: Vol. 40, page 34024, 08/13/1975******Pararosaniline Method for SO<sub>2</sub> - Technicon II*****Manual Equivalent Method: EQS-0775-002**

“Pararosaniline Method for the Determination of Sulfur Dioxide in the Atmosphere-Technicon II Automated Analysis System”

***Federal Register: Vol. 40, page 34024, 08/13/1975******Advanced Pollution Instrumentation, Inc. Model 100 SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-0990-077**“Advanced Pollution Instrumentation, Inc. Model 100 Fluorescent SO<sub>2</sub> Analyzer,” operated on the 0-0.1 ppm<sup>1</sup>, the 0-0.2 ppm<sup>1</sup>, the 0-0.5 ppm, or the 0-1.0 ppm range with a 5-micron TFE filter element installed in the rear-panel filter assembly, either a user- or vendor-supplied vacuum pump capable of providing 20 inches of mercury vacuum at 2.5 L/min, with or without any of the following options: Internal Zero/Span; Pump Pack; Rack Mount With Slides; RS-232 Interface; Status Output; TFE Zero/Span Valves; Zero Air Scrubber; Dual Range.<sup>2</sup>***Federal Register: Vol. 55, page 38149, 09/17/1990******ASARCO Model 500 SO<sub>2</sub> Monitor*****Automated Equivalent Method: EQSA-0877-024**

“ASARCO Model 500 Sulfur Dioxide Monitor,” operated on a 0-0.5 ppm range; or “ASARCO Model 600 Sulfur Dioxide Monitor,” operated on a 0-1.0 ppm range. (Both models are identical except for the range.) NOTE: This method is not now commercially available.

***Federal Register: Vol. 42, page 44264, 09/02/77 and Vol. 44, page 67522, 11/26/1979******Beckman Model 953 Fluorescent Ambient SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-0678-029**“Beckman Model 953 Fluorescent Ambient SO<sub>2</sub> Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with a time constant setting of 2, 2.5, or 3 minutes, a 5 to 10 micron membrane filter element installed in the rear-panel filter assembly, with or without any of the following options: Remote Operation Kit, Catalog No. 641984; Digital Panel Meter, Catalog No. 641710; Rack Mount Kit, Catalog No. 641709; Panel Mount Kit, Catalog No. 641708.***Federal Register: Vol. 43, page 35995, 08/14/1978******Bendix Model 8303 Sulfur Analyzer*****Automated Equivalent Method: EQSA-1078-030**“Bendix Model 8303 Sulfur Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with a Teflon filter installed on the sample inlet of the H<sub>2</sub>S scrubber assembly.***Federal Register: Vol. 43, page 50733, 10/31/1978***

***Columbia Scientific Industries Model 5700 SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-0494-095**

“Columbia Scientific Industries Model 5700 Sulfur Dioxide Analyzer,” operated with software version 1.0 on any full scale range between 0-250 ppb<sup>1</sup> and 0-1000 ppb, at any integration time setting from 20 to 99 seconds, at any temperature in the range of 15°C to 30°C, at any AC line voltage in the range of 105 to 130 volts, and with or without any of the following options:

|                                      |                               |                             |
|--------------------------------------|-------------------------------|-----------------------------|
| 964-0121 Alarm Relay Contacts        | 964-0125 Dual Current Outputs | 964-0131 Rack Mount         |
| 964-0122 Input Solenoids             | 964-0126 Printer              | 964-0012 Single Headed Pump |
| 964-0124 Dual Analog Voltage Outputs |                               |                             |

**Federal Register: Vol. 59, page 18818, 04/20/1994**

***Dasibi Model 4108 U.V. Fluorescence SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-1086-061**

“Dasibi Model 4108 U.V. Fluorescence SO<sub>2</sub> Analyzer,” operated with a range of 0-100 ppb<sup>1</sup>, 0-200 ppb<sup>1</sup>, 0-500 ppb, or 0-1000 ppb, with a Teflon-coated particulate filter and continuous hydrocarbon removal system, with or without any of the following options: Rack Mounting Brackets and Slides; RS-232-C Interface; Temperature Correction.

**Federal Register: Vol. 51, page 32244, 09/10/1986**

***DKK-TOA Corp. Model GFS-32 U.V. Fluorescent SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-0701-115**

“DKK-TOA Corporation Model GFS-32 Ambient Air SO<sub>2</sub> Ultraviolet Fluorescent Analyzer,” operated within the 0.000 to 0.500 ppm range in the temperature range of 20°C to 30°C.

**Federal Register: Vol. 62, page 44007, 08/18/1997**

***DKK-TOA Corp. Model GFS-112E U. V. Fluorescent SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-0100-133**

“DKK-TOA Corporation Models GFS-112E and GFS-112E-1 U.V. Fluorescence SO<sub>2</sub> Analyzer,” operated at any temperature ranging from 15°C to 35°C, on any of the following measurement ranges: 0-0.05 ppm<sup>1</sup>, 0-0.100 ppm<sup>1</sup>, 0-0.200 ppm<sup>1</sup>, 0-0.5 ppm, or 0-1.000 ppm, and with or without the optional Internal zero air supply and permeation tube oven.<sup>2</sup>

**Federal Register: Vol. 65, page 2610, 01/18/2000**

***DKK-TOA Corp. Model GFS-312E Ambient SO<sub>2</sub> Analyzer*****Automated Equivalent Method: EQSA-1107-168**

“DKK-TOA Corporation Model GFS-312E Ambient SO<sub>2</sub> Analyzer,” operated at any environmental temperature in the range of 20°C to 30°C on any of the following measurement ranges<sup>1</sup>: 0-0.1 ppm, 0-0.2 ppm and 0-0.5 ppm.<sup>2</sup>

**Federal Register: Vol. 72, page 63176, 11/08/2007**

***Ecotech Serinus 50 Sulfur Dioxide Analyzer or Opsis AB OPS50 Sulfur Dioxide Analyzer or Teledyne Analytical Instruments 6400E Sulfur Dioxide Analyzer*****Automated Equivalent Method: EQSA-0809-188**

“Ecotech Serinus 50 Sulfur Dioxide Analyzer” or “Opsis AB OPS50 Sulfur Dioxide Analyzer” or “Teledyne Analytical Instruments 6400E Sulfur Dioxide Analyzer,” operated in the range of 0–0.5 ppm, with a five-micron Teflon® filter element installed, and with the following selected: Background-Enabled, Control Loop-Enabled, Diagnostic Mode-Operate, Pres/Temp/Flow Compensation-Enabled, Span Compensation-Disabled, with concentration automatically corrected for temperature and pressure changes, and operated according to the Serinus 50 Sulfur Dioxide Analyzer User Manual or the Opsis OPS50 Sulfur Dioxide Analyzer User Manual or the Teledyne Analytical Instruments 6400E Sulfur Dioxide Analyzer Instruction Manual, as appropriate.

**Federal Register: Vol. 74, page 38184, 07/31/2009**

**Latest Modifications: 05/2010, 05/2011, 05/2012**

**Environnement S.A. Model AF21M SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0292-084**

“Environnement S.A Model AF21M Sulfur Dioxide Analyzer,” operated on a range of 0-0.5 ppm with a response time coefficient setting of 01, a Teflon filter installed in the rear-panel filter assembly, and with or without any of the following options: Rack Mount/Slides; RS-232-C Interface.

**Federal Register: Vol. 57, page 5444, 02/14/1992**

**Environnement S.A. Model AF22M SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0802-149**

“Environnement S.A Model AF22M UV Fluorescence Sulfur Dioxide Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 10°C to 35°C, with a 5-micron PTFE sample particulate filter, with a response time setting of 11 (Automatic response time), with the automatic “ZERO-REF” cycle ON and set for activation every 24 hours, and with or without either of the following options: Permeation oven, Rack mount slides.<sup>2</sup>

**Federal Register: Vol. 67, page 57811, 09/12/2002**

**Environnement S.A. SANOVA Multigas Longpath Monitoring System****Automated Equivalent Method: EQSA-0400-138**

“Environnement S.A. Model SANOVA Multigas Longpath Air Quality Monitoring System,” consisting of a receiver, one or more projectors, interface unit, a user-provided control unit computer running the SANOVA VisionAIR software, and associated incidental equipment; configured for measuring SO<sub>2</sub>, with the temperature control and internal calibration cell options installed, operated with a measurement range of 0 to 0.5 ppm, over an installed monitoring path length of between 27 and 500 meters, within an ambient air temperature range of -30 to +45°C, with a measurement (integrating) time of 180 seconds, and with or without external temperature and barometric pressure sensors or any of the following options: external (meteo) input connection, series 1M bus connection, OGR type projector, analog outputs.

**Federal Register: Vol. 65, page 26603, 05/08/2000**

**Horiba Models APSA-360, APSA-360-CE, or APSA-360A-CE SO<sub>2</sub> Monitors****Automated Equivalent Method: EQSA-0197-114**

“Horiba Instruments, Inc. Models APSA-360, APSA-360-CE or APSA-360A-CE Ambient Sulfur Dioxide Monitor,” operated with a full scale range of 0 - 0.50 ppm, at any temperature in the range of 5°C to 40°C, with a Line Setting of “MEASURE,” an Analog Output Setting of “MOMENTARY VALUE”, and with or without any of the following options:<sup>2</sup> 1) Rack Mounting Plate and Side Rails, 2) RS-232 Communications Port, and 3) Internal zero gas and span gas generator.

“Horiba Instruments, Inc. Model APSA-360A-CE Ambient Sulfur Dioxide Monitor,” operated with one of the following measurement ranges: 0-0.05 ppm, 0-0.1 ppm, 0-0.2 ppm, 0-0.5 ppm or 0-1.0 ppm; with selectable time constants from 10 to 300 seconds; at any temperature in the range of 5°C to 40°C; and with or without the optional internal zero gas and span gas generator.

**Federal Register: Vol. 62, page 6968, 02/14/97; Vol. 63, page 31992, 06/11/1998**

**Horiba Model APSA-370 Ambient SO<sub>2</sub> Monitor****Automated Equivalent Method: EQSA-0506-159**

“Horiba Instruments Incorporated Model APSA-370 Ambient SO<sub>2</sub> Monitor,” operated with a full scale fixed measurement range of 0 - 0.50 ppm, with the automatic range switching off, at any environmental temperature in the range of 20°C to 30°C.<sup>2</sup>

**Federal Register: Vol. 71, page 25587, 05/01/2006**

**Lear Siegler Model AM2020 SO<sub>2</sub> Monitor****Automated Equivalent Method: EQSA-0486-049**

“Lear Siegler Model AM2020 Ambient SO<sub>2</sub> Monitor,” operated on a range of either 0-0.5 or 0-1.0 ppm, at a wavelength of 299.5 nm, with a 5 minute integration period, over any 10°C temperature range between 20°C and 45°C, with or without the automatic zero and span correction feature.

**Federal Register: Vol. 45, page 79574, 12/01/1980 and Vol. 46, page 9997, 01/30/1981**

**Lear Siegler Model SM1000 SO<sub>2</sub> Monitor****Automated Equivalent Method: EQSA-1275-005**

"Lear Siegler Model SM1000 SO<sub>2</sub> Ambient Monitor," operated on the 0-0.5 ppm range, at a wavelength of 299.5 nm, with the "slow" (300 second) response time, with or without any of the following options: SM-1 Internal Zero/Span; SM-2 Span Timer Card; SM-3 0-0.1 Volt Output; SM-4 0-5 Volt Output; SM-5 Alternate Sample Pump; SM-6 Outdoor Enclosure.

**Federal Register: Vol. 41, page 3893, 01/27/1976; Vol. 41, page 32946, 08/06/1976; Vol. 42, page 13044, 03/08/1977; Vol. 45, page 1147, 01/04/1980**

**Meloy Model SA185-2A SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-1275-006**

"Meloy Model SA185-2A Sulfur Dioxide Analyzer," operated on the 0-0.5 ppm range, with or without any of the following options:

|                                                  |                                                |                                    |
|--------------------------------------------------|------------------------------------------------|------------------------------------|
| S-1 Linearized Output                            | S-2 Modified Recorder Output                   | S-18 Rack Mount Conversion         |
| S-24 Dual Range Linearized Output                | S-5 Teflon-Coated Block                        | S-18A Rack Mount Conversion        |
| S-33 Remote Range Control And Status             | S-6A Reignite Timer Circuit                    | S-21 Front Panel Digital (Signals) |
| S-7 Press To Read Volt Meter                     | S-34 Remote Control                            | S-11A Manual Zero And Span         |
| S-22 Remote Zero/Span Control And Status (Timer) | S-35 Front Panel Digital Meter With BCD Output |                                    |
| S-11B Automatic Zero And Span                    | S-13 Status Lights                             | S-22A Remote Zero/Span Control     |
| S-36 Dual Range Log-Linear Output                | S-14 Output Booster Amplifier                  | S-23 Automatic Zero Adjust         |
| S-38 Sampling Mode Status                        | S-14B Line Transmitter Board                   | S-23A Automatic/Manual Zero Adjust |

or operated on the 0-1.0 ppm range with either option S-36 or options S-1 and S-24, with or without any of the other options.

**Federal Register: Vol. 41, page 3893, 01/27/1976 and Vol. 43, page 38088, 08/25/1978**

**Meloy Model SA285E SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-1078-032**

"Meloy Model SA285E Sulfur Dioxide Analyzer," operated on the following ranges and time constant switch positions:

|                        |                   |                    |              |              |
|------------------------|-------------------|--------------------|--------------|--------------|
| Range, ppb:            | 0-50 <sup>1</sup> | 0-100 <sup>1</sup> | 0-500        | 0-1000       |
| Time Constant Setting: | 1 or 10           | 1 or 10            | off, 1 or 10 | off, 1 or 10 |

The analyzer may be operated at temperatures between 10°C and 40°C and at line voltages between 105 and 130 volts, with or without any of the following options:

|                                                  |                                                   |                                                |
|--------------------------------------------------|---------------------------------------------------|------------------------------------------------|
| S-5 Teflon Coated Block                          | S-22B Remote Zero/Span Control and Status (Pulse) | S-30 Auto Reignite                             |
| S-14B Line Transmitter Board                     | S-23 Auto Zero Adjust                             | S-32 Remote Range Control And Status           |
| S-18 Rack Mount Conversion                       | S-23A Auto/Manual Zero Adjust                     | S-35 Front Panel Digital Meter With BCD Output |
| S-18A Rack Mount Conversion                      | S-25 Press To Read                                | S-37 Temperature Status Lights                 |
| S-21 Front Panel Digital Meter                   | S-26 Manual Zero And Span                         | S-38 Sampling Mode Status                      |
| S-22 Remote Zero/Span Control and Status (Timer) | S-27 Auto Manual Zero/Span                        |                                                |
| S-22A Remote Zero/Span Control                   | S-28 Auto Range And Status                        |                                                |

**Federal Register: Vol. 43, page 50733, 10/31/1978**

**Meloy Model SA 700 Fluorescence Sulfur Dioxide Analyzer****Automated Equivalent Method: EQSA-0580-046**

"Meloy Model SA 700 Fluorescence Sulfur Dioxide Analyzer," operated on the 0-250 ppb<sup>1</sup>, the 0-500 ppb, or the 0-1000 ppb range with a time constant switch position of either 2 or 3. The analyzer may be operated at temperatures between 20°C and 30°C and at line voltages between 105 and 130 volts, with or without any of the following options: FS-1 Current Output; FS-2 Rack Mount Conversion; FS-2A Rack Mount Conversion; FS-2B Rack Mount Conversion; FS-3 Front Panel Mounted Digital Meter; FS-5 Auto/Manual Zero/Span With Status; FS-6 Remote/Manual Zero/Span With Status; FS-7 Auto Zero Adjust.

**Federal Register: Vol. 45, page 31488, 05/13/1980**

**Monitor Labs Model 8450 Sulfur Monitor****Automated Equivalent Method: EQSA-0876-013**

“Monitor Labs Model 8450 Sulfur Monitor,” operated on a range of either 0-0.5 or 0-1.0 ppm, with a 5 second time constant, a model 8740 hydrogen sulfide scrubber in the sample line, with or without any of the following options: BP Bipolar Signal Processor; IZS Internal Zero/Span Module; V Zero/Span Valves; CLO Current Loop Output; TF TFE Sample Particulate Filter; VT Zero/Span Valves And Timer; DO Status Remote Interface.

**Federal Register: Vol. 41, page 36245, 08/27/1976 and Vol. 44, page 33476, 06/11/1979**

**Monitor Labs/Lear Siegler Model 8850 SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0779-039**

“Monitor Labs or Lear Siegler Model 8850 Fluorescent SO<sub>2</sub> Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with an internal time constant setting of 55 seconds, a TFE sample filter installed on the sample inlet line, with or without any of the following options: 03A Rack; 03B Slides; 05A Valves Zero/Span; 06A IZS Internal Zero/Span Source; 06B,C,D NIST-Traceable Permeation Tubes; 08A Pump; 09A Rack Mount For Option 08A; 010 Status Output W/Connector; 013 Recorder Output Options; 014 DAS Output Options; 017 Low Flow Option; 018 Kicker.

**Federal Register: Vol. 44, page 44616, 07/30/1979**

**Monitor Labs/Lear Siegler Model 8850S SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0390-075**

“Monitor Labs or Lear Siegler Model 8850S SO<sub>2</sub> Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm.

**Federal Register: Vol. 55, page 5264, 02/14/1990**

**Opsis Model AR 500 and System 300 Open Path Ambient Air Monitoring Systems for SO<sub>2</sub>****Automated Equivalent Method: EQSA-0495-101**

“Opsis Model AR 500 System” or “System 300 Open Path (long path) Ambient Air Monitoring Systems,” configured for measuring SO<sub>2</sub>, with one detector and movable grating, operated with a measurement range of 0 to 0.5 ppm or 0 to 1.0 ppm, an installed monitoring path length between 20 and 500 meters (or 20 and 1000 meters with the ER 150 option, AR 500 System only), xenon lamp type B (150 watt), fiber optic cable length between 3 to 20 meters; operating within an ambient air temperature range of -50 to +50°C, an analyzer temperature range of 20 to 30°C, a measurement (integrating) time setting between 30 and 120 seconds (0 min:30 sec. to 2 min:00 sec.), and with a complete cycle time of not more than 200 seconds (3 min, 20 sec.). Under this method designation, the Model AR 500 System or System 300 consists of: AR 500 opto-analyzer; emitter EM 110 and receiver RE 110 (together identified as ER 110); optic fibre cable OF60-S; power supply PS 150; Opsis operational software, version 7.0 or 7.1; and initial on-site installation, setup, and limited operator training.<sup>2</sup>

**Optional components that can be used with the Model AR 500 only**, in addition to or as alternative to corresponding components listed above: • AR 503 opto-analyzer configured as Model AR 500 (only the center detector active, sequential monitoring) • Emitter/receiver ER 150 (for monitoring path lengths up to 1 kilometer) • Transceiver ER 130 and Retroreflector RE 090 with 7 prisms (max. monitoring path length 150 meters) or 12 prisms (max. monitoring path length 250 meters) • Receiver RE 130 • Xenon lamp type A (higher short-wavelength UV output) • Optic fibre cable OF60-R (low-loss for short wavelengths) • Multiplexers MX 004 and MX 024 • Dataloggers DL 010 and DL 016 • Analogue and digital input/output cards AO 008, AI 016, and DI 032 • Analogue and digital isolation cards IA 008, ID 008, OA 008, and OD 008 • Window heaters HF 110 and HF 150 • Mirror heaters HM 110 and HM 150 • Auto calibration unit CU 007 • Software packages IO 80 (for the analogue and digital input/output adapters), DL10 and DL16 (for data loggers), COMVISION, and STAT 500;

**Recommended calibration and accuracy audit components (or equivalent) for either Model AR 500 or System 300:** • Wavelength calibration lamp CA 004 • Calibration bench CB 100 • Receiver unit RE 060 (two required) • Calibration unit CA 150, with same type lamp as used in the monitoring path emitter • Power supply PS 150 for calibration unit CA 150 • Calibration cells CC 001-X, where X represents various cell lengths from 1 to 900 mm • Special calibration cells CC 110 or CC 150 (for mounting directly on receiver) • Light meter LM 010.

**Federal Register: Vol. 60, page 21518, 05/02/1995**

**Philips PW9755 SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0676-010**

“Philips PW9755 SO<sub>2</sub> Analyzer,” consisting of the following components: PW9755/02 SO<sub>2</sub> Monitor with PW9741/00 SO<sub>2</sub> Source, PW9721/00 Filter Set SO<sub>2</sub>, PW9711/00 Electrolyte SO<sub>2</sub>, PW9750/00 Supply Cabinet, PW9750/10 Supply Unit/Coulometric, either PW9731/00 Sampler or PW9731/20 Dust Filter (or vendor-approved alternate particulate filter); operated with a 0-0.5 ppm range and with a reference voltage setting of 760 millivolts; with or without any of the following options: PW9750/30 Frame For MTT; PW9752/00 Air Sampler Manifold; PW9753/00 Mounting Rack For Accessories; PW9750/41 Control Clock 60 Hz; PW9754/00 Air Distributor.

**Federal Register: Vol. 41, page 26252, 6/25/1976; Vol. 41, page 46019, 10/19/1976; Vol. 42, page 28571, 06/03/1977**

**Philips PW9700 SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0876-011**

“Philips PW9700 SO<sub>2</sub> Analyzer,” consisting of the following components: PW9710/00 Chemical Unit with PW9711/00, Electrolyte SO<sub>2</sub>, PW9721/00 Filter Set SO<sub>2</sub>, PW9740/00 SO<sub>2</sub> Source; PW9720/00 Electrical Unit; PW9730/00 Sampler Unit (or vendor-approved alternate particulate filter); operated with a 0-0.5 ppm range and with a reference voltage of 760 millivolts.

**Federal Register: Vol. 41, page 34105, 08/12/1976**

**SERES Model SF 2000 G Sulfur Dioxide Analyzer****Automated Equivalent Method: EQSA-0810-194**

“SERES model SF 2000 G Sulfur Dioxide Analyzer,” UV fluorescence method using a wavelength source approaching 215 nm and a selective membrane for aromatic hydrocarbon removal, operated with a full scale measurement range of 0 - 0.5 ppm at any ambient temperature in the range of 20°C to 30°C, with tabletop or rack mounts, microprocessor controlled menu-driven user interface, onboard diagnostics and system test functions, analog output signals of 4-20 mA or user selectable voltage ranges up to 10 V, printer port, modem port and 32 pin data/control/alarm port, user selectable manual and automatic zero/span and calibrate modes; with or without a permeation tube system (optional equipment) for internal calibration; operated in accordance with the SF 2000 G User and Maintenance Manual.

**Federal Register: Vol. 75, page 51039, 08/18/2010**

**SIR S.A. Model S-5001 U.V. Fluorescence SO<sub>2</sub> Analyzer****Automated Equivalent Method: EQSA-0507-166**

“SIR S.A. Model S-5001 U.V. Fluorescence SO<sub>2</sub> Analyzer,” operated with a full-scale measurement range of 0 - 0.5 ppm, with an integration time setting of 1 minute, and with or without an optional PCMCIA card or the optional Internal permeation oven.<sup>2</sup>

**Federal Register: Vol. 72, page 26627, 05/10/2007**

**Teledyne - Advanced Pollution Instrumentation, Inc. Models 100A, 100AS, 100E, 100EU, T100, T100U; Teledyne Analytical Instruments Model 6400A; or Teledyne Monitor Labs sensor-e™ Model TML-50 SO<sub>2</sub> Analyzers; or recordum airpointer® system module 801-001000**

**Automated Equivalent Method: EQSA-0495-100**

“Teledyne - Advanced Pollution Instrumentation, Inc. Models 100A, 100AS, 100E, 100EU, T100 or T100U; Teledyne Analytical Instruments Model 6400A; or Teledyne Monitor Labs, Inc. sensor-e™ Model TML-50 UV Fluorescent Sulfur Dioxide Analyzer; operated on any full scale range between 0-50 ppb<sup>1</sup> and 0-1000 ppb, at any temperature in the range of 5 to 40 degrees C, with a TFE filter element installed in the filter assembly, with either the vendor-supplied internal pump or a user- or vendor-supplied external vacuum pump capable of maintaining an absolute pressure of 35 cm (14 inches) of mercury (or less) at 1.0 standard liter per minute flow rate, with the following software settings: Dynamic zero: OFF or ON; Dynamic span: OFF; AutoCal: ON or OFF; Dual range: ON or OFF; Autorange: ON or OFF; Temp/pressure compensation: ON; dilution factor: OFF or 1.0; and with or without any of the following options (if available for the various models):<sup>2</sup> Rack mount with or without chassis slides; Fluorocarbon zero/span valves; Internal zero/span (IZS); Three-point internal zero/span (IZS, option 51C); 4-20 mA, isolated analog outputs; analog input option; External pump; Status outputs; Control inputs; Rack mount for external pump with tray; RS-232 output; Ethernet output; Zero air scrubber; Combustion Filter; SO<sub>2</sub> Permeation tube, certified or uncertified, 0.4 ppm @ 0.7 L/min; SO<sub>2</sub> Permeation tube, certified or uncertified, 0.8 ppm @ 0.7 L/min. **Airpointer® module 801-001000 only:** operated on any full scale range between 0-0.05 ppm and 0-1.0 ppm, with a PTFE filter element installed in the internal filter assembly, with the software setting: FRM/FEM conform mode; at any temperature in the range of 10°C to 45°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 16 inches mercury or less at 2.5 slpm; installed in the compact. thermally controlled (-40°C to + 45°C)

and weather proof airpointer base unit with integrated data acquisition and management system mounted on a frame, pole, or wall; with or without wireless telemetry; with or without internal span option as module supplement consisting of permeation oven and permeation tube; with or without modules for other criteria pollutants; with or without analyzer for particulate matter; with or without additional 3<sup>rd</sup> party sensors for e.g. meteorology, noise, or traffic counting. Operated with the appropriate instrument manual.

**Federal Register: Vol. 60, page 17061, 04/04/1995**

**Latest Modification: 08/2010; 05/2013**

**Teledyne Monitor Labs/Casella/Ecotech Model ML9850/CM2050/EC9850/EC9850T; Teledyne Monitor Labs/Casella/Ecotech/Model ML9850B/CM2050B/EC9850B; or Wedding & Associates Model 1040 SO<sub>2</sub> Analyzers**

**Automated Equivalent Method: EQSA-0193-092**

“Teledyne Monitor Labs, Casella Monitor, or Ecotech Models ML9850/CM2050/EC9850, or ML9850B/CM2050B/EC9850B, Ecotech Model EC9850T, or Wedding & Associates, Inc. Model 1040 Sulfur Dioxide Analyzers,” operated on any full scale range between 0-0.050 ppm<sup>1</sup> and 0-1.0 ppm, at any temperature in the range of 15°C to 35°C, with the service switch on the secondary panel set to the *In* position; with the following menu choices selected: Range: *0.05 ppm to 1.0 ppm*; Over-ranging: *Enabled or Disabled*; Background: *Not Disabled*; Calibration: *Manual or Timed*; Diagnostic Mode: *Operate*; Filter Type: *Kalman*; Pres/Temp/Flow Comp: *On*; Span Comp: *Disabled*; and as follows: **Model ML9850/CM2050/EC9850/EC9850T** - with a five-micron Teflon® filter element installed internally, with the 50-pin I/O board installed on the rear panel configured at any of the following output range settings: Voltage, 0.1 V, 1 V, 5 V, 10 V; Current, 0-20 mA, 2-20 mA, 4-20 mA; and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Rack Mount Assembly; Internal Floppy Disk Drive. **Models ML9850B/CM2050B/EC9850B and 1040** - with either a vendor-supplied or equivalent user supplied five-micron Teflon® filter, zero air scrubber, and exhaust pump, and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Rack Mount Assembly; 50-pin I/O board; Exhaust Scrubber; Internal Zero/Span Assembly (IZS); hinged, fold-down front panel. Operated with the appropriate instrument manual.

**Federal Register: Vol. 58, page 6964, 02/03/1993**

**Latest Modification: 03/2011**

**Thermo Electron Model 43 SO<sub>2</sub> Analyzer**

**Automated Equivalent Method: EQSA-0276-009**

“Thermo Electron Model 43 Pulsed Fluorescent SO<sub>2</sub> Analyzer,” equipped with an aromatic hydrocarbon cutter and operated on a range of either 0-0.5 or 0-1.0 ppm, with or without any of the following options: 001 Rack Mounting For Standard 19 Inch Relay Rack; 002 Automatic Actuation Of Zero And Span Solenoid Valves; 003 Type S Flash Lamp Power Supply; 004 Low Flow.

**Federal Register: Vol. 41, page 8531, 02/27/1976; Vol. 41, page 15363, 04/12/1976; Vol. 42, page 20490, 04/20/1977 Vol. 44, page 21861, 04/12/1979; Vol. 45, page 2700, 01/14/1980; Vol. 45, page 32419, 05/16/1980**

**Thermo Environmental Instruments, Inc./Thermo Electron Models 43A, 43B, 43C, 43C-TLE, 43i, 43i-TLE SO<sub>2</sub> Analyzers**

**Automated Equivalent Method: EQSA-0486-060**

“Thermo Electron or Thermo Environmental Instruments, Inc. **Model 43A or 43B** Pulsed Fluorescence SO<sub>2</sub> Analyzer,” operated on the 0-0.1 ppm<sup>1</sup>, the 0-0.2 ppm<sup>1</sup>, the 0-0.5 ppm, or the 0-1.0 ppm range, with either a high or a low time constant setting (Model 43A) and with or without any of the following options:<sup>2</sup>

|                                     |                                     |                                                   |
|-------------------------------------|-------------------------------------|---------------------------------------------------|
| 001 Teflon Particulate Filter       | 004 High Flow Rate (1 LPM)          | 007 Remote Activation of Zero/Span Valves         |
| 002 19" Rack Mounting Configuration | 005 Current Output                  | 008 RS-232 Interface (Model 43B)                  |
| 003 Internal Zero/Span Valves       | 006 Internal Permeation Span Source | 009 Pressure/Temperature Compensation (Model 43B) |

“Thermo Environmental Instruments, Inc. **Models 43C, 43C-TLE, 43i, 43i-TLE**, Pulsed Fluorescence SO<sub>2</sub> Analyzer,” operated on any measurement range between 0-50 ppb<sup>1</sup> (0-10<sup>1</sup> ppb for Model TLE) and 0-1000 ppb, with any time average setting from 10 to 300 seconds, with temperature and/or pressure compensation on or off, operated at temperatures between 20°C and 30°C, and with or without any of the following options:<sup>2</sup> Teflon particulate filter, 4-20 mA current output or I/O expansion board, Rack mounts, Internal permeation span source/Permeation oven, Internal zero/span and sample solenoid valves, High flow rate (0.5-1.0 LPM); **Models 43C, 43C-TLE**: Remote activation of zero/span and sample valves, RS-232/485 interface.

**Federal Register: Vol. 51, page 12390, 04/10/1986**



**Ozone****2B Technologies Model 202 and 205 Ozone Monitors****Automated Equivalent Method: EQOA-0410-190**

“2B Technologies Models 202 (single beam) and 205 (dual beam) Ozone Monitors,” operated in a range of 0 - 0.5 ppm in an environment of 10 - 40°C, with temperature and pressure compensation, internal DewLine for humidity control, using a 10 second average, with a 110-220V AC power adapter or a 12V DC source 4.0 to 6.0 watt power consumption, operated according to the Model 202 or 205 Ozone Monitor Operation Manual with or without the following: cigarette lighter adapter or a 12V DC battery for portable operation, external TFE inlet filter and holder, serial data port with computer cable, BNC connector for 0-2.5V scalable analog output, internal data logger, 3-analog inputs for external signals (such as temperature, relative humidity or pressure), rack mount hardware, on-board backup sample pump.

**Federal Register: Vol.75, pages 22126-22127, 04/27/2010**

**Latest Modification: 12/2010**

**Beckman Model 950A Ozone Analyzer****Automated Reference Method: RFOA-0577-020**

“Beckman Model 950A Ozone Analyzer,” operated on a range of 0-0.5 ppm and with the "SLOW" (60 second) response time, with or without any of the following options: Internal Ozone Generator; Computer Adaptor Kit; Pure Ethylene Accessory.

**Federal Register: Vol. 42, page 28571, 06/03/1977**

**Bendix or Combustion Engineering Model 8002 Ozone Analyzer****Automated Reference Method: RFOA-0176-007**

“Bendix or Combustion Engineering Model 8002 Ozone Analyzer,” operated on the 0-0.5 ppm range, with a 40 second time constant, with or without any of the following options: Rack Mounting With Chassis Slides; Rack Mounting Without Chassis Slides; Zero And Span Timer; Ethylene/CO<sub>2</sub> Blend Reactant Gas.

**Federal Register: Vol. 41, page 5145, 02/04/1976 and Vol. 45, page 18474, 03/21/1980**

**Columbia Scientific Industries Model 2000 Ozone Meter****Automated Reference Method: RFOA-0279-036**

“Columbia Scientific Industries Model 2000 Ozone Meter,” when operated on the 0-0.5 ppm range with either AC or battery power: The BCA 952 battery charger/AC adapter M952-0002 (115V) or M952-0003 (230V) is required for AC operation; an internal battery M952-0006 or 12 volt external battery is required for portable non-AC powered operation.

**Federal Register: Vol. 44, page 10429, 02/20/1979**

**Dasibi Models 1003-AH, 1003-PC, or 1003-RS Ozone Analyzers****Automated Equivalent Method: EQOA-0577-019**

“Dasibi Model 1003-AH, 1003-PC, or 1003-RS Ozone Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with or without any of the following options: Adjustable Alarm; Aluminum Coated Absorption Tubes, Integrated Output; Vycor-Jacketed U.V. Source Lamp; BCD Digital Output; Rack Mounting Ears And Slides; 0-10 mV, 0-100 mV, 0-1 V, Or 0-10 V; Glass (Pyrex) Absorption Tubes; Teflon-based Solenoid Valve; Analog Output.

**Federal Register: Vol. 42, page 28571, 06/03/1977**

**Dasibi Models 1008-AH, 1008-PC, or 1008-RS Ozone Analyzers****Automated Equivalent Method: EQOA-0383-056**

“Dasibi Model 1008-AH, 1008-PC, or 1008-RS Ozone Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with or without any of the following options: Aluminum Coated Absorption Tubes; BCD Digital Output; RS232 Interface; Glass (Pyrex) Absorption Tubes; Vycor-Jacketed U.V. Source Lamp; Ozone Generator; Teflon-based Solenoid Valve; Photometer Flow Restrictor (2 LPM); 4-20 mA, Isolated, Or Dual Analog Outputs; Rack Mounting Brackets Or Slides; 20 Second Update Software.

**Federal Register: Vol. 48, page 10126, 03/10/1983**

**DKK-TOA Corp. Model GUX-113E Ozone Analyzer****Automated Equivalent Method: EQOA-0200-134**

“DKK-TOA Corporation Models GUX-113E and GUX-113E-1 Ozone Analyzer,” operated at any temperature in the range of 15°C to 35°C, on any of the following measurement ranges: 0-0.100 ppm<sup>1</sup>, 0-0.200 ppm<sup>1</sup>, 0-0.5 ppm, or 0-1.000 ppm, and with or without the optional Internal Ozone Generator.<sup>2</sup>

**Federal Register: Vol. 65, page 11308, 03/02/2000**

**DKK-TOA Corp. Model GUX-313E Ambient O<sub>3</sub> Analyzer****Automated Equivalent Method: EQOA-1107-169**

“DKK-TOA Corporation Model GUX-313E Ambient O<sub>3</sub> Analyzer,” operated at any environmental temperature in the range of 20°C to 30°C on any of the following measurement ranges<sup>1</sup>: 0-0.1 ppm, 0-0.2 ppm and 0-0.5 ppm.<sup>2</sup>

**Federal Register: Vol. 72, page 63176, 11/08/2007**

**Ecotech Serinus 10 Ozone Analyzer or Opsis AB OPS10 Ozone Analyzer****Automated Equivalent Method: EQOA-0809-187**

“Ecotech Serinus 10 Ozone Analyzer” or “Opsis AB OPS10 Ozone Analyzer,” operated in the range of 0–0.5 ppm, with a five-micron Teflon® filter element installed, and with the following selected: Control Loop-Enabled, Diagnostic Mode-Operate, Pres/Temp/Flow Compensation-Enabled, Span Compensation-Disabled, with concentration automatically corrected for temperature and pressure changes, and operated according to the Serinus 10 Ozone Analyzer User Manual or the Opsis AB OPS10 Ozone Analyzer User Manual, as appropriate.

**Federal Register: Vol. 74, page 38184, 07/31/2009**

**Latest Modifications: 05/2010, 05/2012**

**EnviroNics Series 300 Ozone Analyzer****Automated Equivalent Method: EQOA-0990-078**

“EnviroNics Series 300 Computerized Ozone Analyzer,” operated on the 0-0.5 ppm range, with the following parameters entered into the analyzer's computer system: Absorption Coefficient = 308 √ 4; Flush Time = 3; Integration Factor = 1; Offset Adjustment = 0.025 ppm; Ozone Average Time = 4; Signal Average = 0; Temp/Press Correction = On; and with or without the RS-232 Serial Data Interface.

**Federal Register: Vol. 55, page 38386, 09/18/1990**

**Environnement S.A. Model O341M UV Ozone Analyzer****Automated Equivalent Method: EQOA-0895-105**

“Environnement S.A. Model O<sub>3</sub>41M UV Photometric Ozone Analyzer,” operated on a full scale range of 0 - 500 ppb, at any temperature in the range of 15°C to 35°C, with the response time set to 50 seconds, and with or without any of the following options:<sup>2</sup> Internal Ozone Generator; Span External Control; RS232-422 Serial Interface; Internal Printer.

**Federal Register: Vol. 60, page 39382, 08/02/1995**

**Environnement S.A. Model O342M UV Ozone Analyzer****Automated Equivalent Method: EQOA-0206-148**

“Environnement S.A Model O<sub>3</sub>42M UV Photometric Ozone Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 10°C to 35°C, with a 5-micron PTFE sample particulate filter, with response time setting of 11 (Automatic response time), and with or without any of the following options:<sup>2</sup> c) Internal ozone generator, d) Span external control (zero/span solenoid valve).

**Federal Register: Vol. 67, page 42557, 06/24/2002**

**Environnement S.A. SANOA Multigas Longpath Monitoring System****Automated Equivalent Method: EQOA-0400-137**

“Environnement S.A. Model SANOA Multigas Longpath Air Quality Monitoring System, consisting of a receiver, one or more projectors, interface unit, a user-provided control unit computer running the SANOA VisionAIR software, and associated incidental equipment; configured for measuring O<sub>3</sub>, with the temperature control and internal calibration cell options installed, operated with a measurement range of 0 to 0.5 ppm, over an installed monitoring path length of between 27 and 500 meters, within an ambient air temperature range of -30 to +45°C, with a measurement (integrating) time of 180 seconds, and with or without external temperature and barometric pressure sensors or any of the following options: external (meteo) input connection, series 1M bus connection, OGR type projector, analog outputs. A high-concentration ozone generator, part # 80-231-03, or the SONIMIX 7121B calibration system is recommended for calibration or accuracy auditing.  
**Federal Register: Vol. 65, page 26603, 05/08/2000**

**Horiba Instruments Models APOA-360 or APOA-360-CE Ozone Monitor****Automated Equivalent Method: EQOA-0196-112**

“Horiba Instruments, Inc. Model APOA-360 or APOA-360-CE Ambient Ozone Monitor,” operated with a full scale range of 0 - 0.50 ppm, at any temperature in the range of 10°C to 40°C, with a Line Setting of "MEASURE," and an Analog Output of "MOMENTARY VALUE," and with or without any of the following options:<sup>2</sup> 1) Rack Mounting Plate and Side Rails 2) RS-232 Communications Port, and 3) Optional Internal Zero/Span Check  
**Federal Register: Vol. 61, page 11404, 03/20/1996**

**Horiba Instruments Model APOA-370 Ozone Monitor****Automated Equivalent Method: EQOA-0506-160**

“Horiba Instruments Incorporated APOA-370 Ambient O<sub>3</sub> Monitor,” standard specification, operated with a full-scale fixed measurement range of 0 - 0.5 ppm, with the automatic range switching off, at any temperature in the range of 20 to 30°C.<sup>2</sup>  
**Federal Register: Vol. 71, page 25587, 05/01/2006**

**McMillan (MEC) Models 1100-1, 1100-2, and 1100-3 Ozone Meters****“MEC Model 1100-1 Ozone Meter,” Automated Reference Method: RFOA-1076-014****“MEC Model 1100-2 Ozone Meter,” Automated Reference Method: RFOA-1076-015****“MEC Model 1100-3 Ozone Meter,” Automated Reference Method: RFOA-1076-016**

Operated on a 0-0.5 ppm range, with or without any of the following options: 0011 Rack Mounting Ears; 0026 Alarm Set Feature; 0012 Instrument Bail; 0033 Local-Remote Sample; Zero, Span Kit Blend Feature; 0016 Chassis Slide Kit; 0040 Ethylene/CO<sub>2</sub>.

**Federal Register: Vol. 41, page 46647, 10/22/1976 and Vol. 42, page 30235, 06/13/1977**

**Meloy Model OA325-2R Ozone Analyzer****Automated Reference Method: RFOA-1075-003**

“Meloy Model OA325-2R Ozone Analyzer,” operated with a scale range of 0-0.5 ppm, with or without any of the following options: 0-4 Output Booster Amplifier; 0-18 Rack Mount Conversion; 0-18A Rack Mount Conversion.

**Federal Register: Vol. 40, page 54856, 11/26/1975**

**Meloy Model OA350-2R Ozone Analyzer****Automated Reference Method: RFOA-1075-004**

“Meloy Model OA350-2R Ozone Analyzer,” operated with a scale range of 0-0.5 ppm, with or without any of the following options: 0-2 Automatic Zero And Span; 0-3 Remote Control Zero And Span; 0-4 Output Booster Amplifier; 0-18 Rack Mount Conversion; 0-18A Rack Mount Conversion.

**Federal Register: Vol. 40, page 54856, 11/26/1975**

**Monitor Labs Model 8410E Ozone Analyzer****Automated Reference Method: RFOA-1176-017**

“Monitor Labs Model 8410E Ozone Analyzer,” operated on a range of 0-0.5 ppm with a time constant setting of 5 seconds, with or without any of the following options: DO Status Outputs; ER Ethylene Regulator Assembly; V TFE Zero/Span Valves; TF TFE Sample Particulate Filter; VT TFE Zero/Span Valves and Timer.

**Federal Register: Vol. 41, page 53684, 12/08/1976**

**Monitor Labs/Lear Siegler Model 8810 Ozone Analyzer****Automated Equivalent Method: EQOA-0881-053**

“Monitor Labs or Lear Siegler Model 8810 Photometric Ozone Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with selectable electronic time constant settings from 20 through 150 seconds, with or without any of the following options: 05 Pressure Compensation; 06 Averaging Option; 07 Zero/Span Valves; 08 Internal Zero/Span (Valve And Ozone Source); 09 Status; 10 Particulate Filter; 15 through 20 DAS/REC Output.

**Federal Register: Vol. 46, page 52224, 10/26/1981**

**Opsis Model AR 500 and System 300 Open Path Ambient Air Monitoring Systems for Ozone****Automated Equivalent Method: EQOA-0495-103**

“Opsis Model AR 500 System” or “System 300 Open Path (long path) Ambient Air Monitoring Systems,” configured for measuring O<sub>3</sub>, with one detector and moveable grating, operated with a measurement range of 0 to 0.5 ppm, an installed monitoring path length between 20 and 500 meters (or 20 and 1000 meters with the ER 150 option, AR 500 System only), xenon lamp type B (150 watt), fiber optic cable length between 3 to 20 meters; operating within an ambient air temperature range of -50 to +50°C, an analyzer temperature range of 20 to 30°C, a measurement (integrating) time setting between 30 and 120 seconds (0 min:30 sec. to 2 min:00 sec.), and with a complete cycle time of not more than 200 seconds (3 min, 20 sec.). Under this method designation, the Model AR 500 System or System 300 consists of: AR 500 opto-analyser; emitter EM 110 and receiver RE 110 (together identified as ER 110); optic fibre cable OF60-S; power supply PS 150, Opsis operational software, version 7.0 or 7.1; and initial on-site installation, setup, and limited operator training.<sup>2</sup>

**Optional components that can be used with the Model AR 500 only**, in addition to or as alternative to corresponding components listed above: • AR 503 opto-analyser configured as Model AR 500 (only the center detector active, sequential monitoring) • Emitter/receiver ER 150 (for monitoring path lengths up to 1 kilometer) • Transceiver ER 130 and Retroreflector RE 090 with 7 prisms (max. monitoring path length 150 meters) or 12 prisms (max. monitoring path length 250 meters) • Receiver RE 130 • Optic fibre cable OF60-R (low-loss for short wavelengths) • Multiplexers MX 004 and MX 024 • Dataloggers DL 010 and DL 016 • Analogue and digital input/output cards AO 008, AI 016, and DI 032 • Analogue and digital isolation cards IA 008, ID 008, OA 008, and OD 008 • Window heaters HF 110 and HF 150 • Mirror heaters HM 110 and HM 150 • Auto calibration unit CU 007 • Software packages IO 80 (for the analogue and digital input/output adapters), DL10 and DL16 (for data loggers), ComVision, and STAT 500;

**Recommended calibration and accuracy audit components (or equivalent) for either Model AR 500 or System 300:** • Wavelength calibration lamp CA 004 • Calibration bench CB 100 • Receiver unit RE 060 (two required) • Calibration unit CA 150, with same type lamp as used in the monitoring path emitter • Power supply PS 150 for calibration unit CA 150 • Calibration cells CC 001-X, where X represents various cell lengths from 1 to 900 mm • Special calibration cells CC 110 or CC 150 (for mounting directly on receiver) • Ozone generator OC 500 • Light meter LM 010.

**Federal Register: Vol. 60, page 21518, 05/02/1995**

**PCI Ozone Corporation Model LC-12 Ozone Analyzer****Automated Equivalent Method: EQOA-0382-055**

“PCI Ozone Corporation Model LC-12 Ozone Analyzer,” operated on a range of 0-0.5 ppm.

**Federal Register: Vol. 47, page 13572, 03/31/1982**

**Philips PW9771 03 Analyzer****Automated Equivalent Method: EQOA-0777-023**

“Philips PW9771 03 Analyzer,” consisting of the following components: PW9771/00 03 Monitor with PW9724/00 Disc.-Set; PW9750/00 Supply Cabinet; PW9750/20 Supply Unit operated on a range of 0-0.5 ppm, with or without any of the following accessories: PW9732/00 Sampler Line Heater; PW9750/30 Frame For MTT; PW9750/41 Control Clock 60 Hz; PW9733/00 Sampler; PW9752/00 Air Sampler Manifold.

**Federal Register: Vol. 42, page 38931, 08/01/1977; Vol. 42, page 57156, 11/01/1977**

**Seres Model OZ 2000 G Ozone Analyzer****Automated Equivalent Method: EQOA-0506-161**

“Seres OZ 2000 G Ozone Ambient Air Analyzer,” operated with a full scale range of 0 - 0.5 ppm, at any temperature in the range of 20°C to 30°C, and with or without either of the following options: internal ozone generator, teletransmission interface.<sup>2</sup>

**Federal Register: Vol. 71, page 25587, 05/01/2006**

**SIR S.A. Model S-5014 O<sub>3</sub> Analyzer****Automated Equivalent Method: EQOA-0207-164**

“SIR S.A. Model S-5014 Photometric O<sub>3</sub> Analyzer,” operated on the 0 - 500 ppb measurement range, within an ambient temperature range of 20°C to 30°C, with a sample inlet particulate filter, and with or without an optional PCMCIA card.

**Federal Register: Vol. 72, page 8985, 02/28/2007**

**Tanabyte Models 722, 723, 724, 725, or 726 Ambient Ozone Analyzer****Automated Equivalent Method: EQOA-0407-165**

“Tanabyte Models 722, 723, 724, 725, or 726 Ambient Ozone Analyzer,” enclosed in either a Dual-Bay Chassis or a Single-Bay Chassis and operated on either the 0 - 0.5 ppm or 0 - 1.0 ppm measurement range, within an ambient temperature range of 20 to 30 degrees C, and with a sample inlet particulate filter installed in the sample filter holder.

**Federal Register: Vol. 72, page 20846, 04/26/2007**

**Teledyne - Advanced Pollution Instrumentation, Inc. Model 400E or T400; Advanced Pollution Instrumentation, Inc. Model 400/400A; Teledyne Monitor Labs sensor-e™ Model TML-10 Ozone Analyzers; or recordum airpointer® system module 801-004000;**

**Automated Equivalent Method: EQOA-0992-087**

“Teledyne - Advanced Pollution Instrumentation, Inc. Model 400E or T400; Advanced Pollution Instrumentation, Inc. Model 400 or 400A; or Teledyne Monitor Labs sensor-e™ Model TML-10 Ozone Analyzer” operated on any full scale range between 0-100 ppb<sup>1</sup> and 0-1000 ppb, with any range mode (Single, Dual, or AutoRange), at any ambient temperature in the range of 5°C to 40°C, and with a TFE filter. **Models 400E, T400 and TML-50:** operated with a sample flow rate of 800 ± 80 cm<sup>3</sup>/min (sea level), with the dilution factor set to 1, with Dynamic Zero ON or OFF, with Dynamic Span OFF, with Temp/Press compensation ON, and with or without any of the following options: Internal or external sample pump, Sample/Cal valve option, Internal Zero/Span (IZS), Rack mount with or without slides, analog input option, 4-20 mA isolated current loop output.<sup>2</sup> **Models 400/400A:** operated with the dynamic zero and span adjustment feature (some Model 400 units only) set to OFF, and with or without any of the following options: Zero/Span Valve option, Internal Zero/Span (IZS) option, IZS ozone generator reference feedback option, standard serial port or Multi-drop RS-232, digital status outputs, analog outputs: 100 mV, 1V, 5V, 10V, 4-20 mA current loop, optional metal wool ozone scrubber, optional external sample pump, optional 47 mm diameter filter, optical bench heater, rack mount with slides. **airpointer® system module 801-004000 only:** operated on any full scale range between 0-0.100 ppm and 0-1.0 ppm, with a PTFE filter element installed in the internal filter assembly, with the software setting: FRM/FEM conform mode; at any temperature in the range of 10°C to 45°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 16 inches mercury or less at 2.5 slpm; installed in the compact, thermally controlled (-40°C to + 45°C) and weather proof airpointer base unit with integrated data acquisition and management system mounted on a frame, pole, or wall; with or without wireless telemetry; with or without the internal span option as module supplement consisting of ozone generator; with or without modules for other criteria pollutants; with or without analyzer for particulate matter; with or without additional 3<sup>rd</sup> party sensors for e.g. meteorology, noise, or traffic counting. Operated with the appropriate instrument manual.

**Federal Register: Vol. 57, page 44565, 09/28/1992; Vol. 63, page 31992, 06/11/1998**

**Federal Register: Vol. 67, page 57811, 09/12/2002**

**Latest Modification: 08/2010; 05/2013**

**Teledyne - Advanced Pollution Instrumentation, Inc. Model 265E or T265 Chemiluminescence Ozone Analyzers****Automated Equivalent Method: EQOA-0611-199**

“Teledyne - Advanced Pollution Instrumentation, Inc. Model 265E or T265 Chemiluminescence Ozone Analyzer,” operated on any full scale range between 0-100 ppb and 0-1000 ppb, with any range mode (Single, Dual, or AutoRange), at any ambient temperature in the range of 5°C to 40°C, and with a TFE filter in the sample air inlet, operated with a sample flow rate of  $500 \pm 50 \text{ cm}^3/\text{min}$  (sea level), with the dilution factor set to 1, with Temp/Press compensation ON, and in accordance with the appropriate associated instrument manual, and with or without any of the following options: Internal or external sample pump, Sample/Cal valve option, Rack mount with or without slides, analog input option, 4-20 mA isolated current loop output.

**Federal Register: Vol. 76, page 62402, 10/07/2011**

**Latest Modification: 10/2012**

**Teledyne Monitor Labs/Casella/Ecotech Models ML9810/CM2010/EC9810, -11, or -12, Teledyne Monitor Labs/Casella/Ecotech Model ML9810B/CM2010B/EC9810B, or Wedding & Associates Model 1010 Ozone Analyzers****Automated Equivalent Method: EQOA-0193-091**

“Teledyne Monitor Labs, Casella Monitor, or Ecotech Models ML9810/CM2010/EC9810, ML9811/CM2011/EC9811, ML9812/CM2012/EC9812, or ML9810B/CM2010B/EC9810B or Wedding & Associates, Inc. Model 1010 Ozone Analyzers,” operated on any full scale range between 0-0.05 ppm<sup>1</sup> and 0-1.0 ppm, at any temperature in the range of 15°C to 35°C, with the service switch on the secondary panel set to the *In* position; with the following menu choices selected: Range: 0.05 ppm to 1.0 ppm; Over-ranging: *Enabled* or *Disabled*; Calibration: *Manual* or *Timed*; Diagnostic Mode: *Operate*; Filter Type: *Kalman*; Pres/Temp/Flow Comp: *On*; Span Comp: *Disabled*; and as follows: **Models ML9810/CM2010/EC9810, -11, and -12** - with a five-micron Teflon® filter element installed internally, with the 50-pin I/O board installed on the rear panel configured at any of the following output range settings: Voltage, 0.1V, 1V, 5V, 10V; Current, 0-20 mA, 2-20 mA, 4-20 mA; and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Rack Mount Assembly; Internal Floppy Disk Drive. **Models ML9810B/CM2010B/EC9810B and 1010** - with either a vendor-supplied or equivalent user-supplied five micron Teflon® filter and exhaust pump, and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Rack Mount Assembly; 50-pin I/O board; Internal Zero/Span Assembly (IZS); hinged, fold-down front panel. Operated with appropriate instrument manual.

**Federal Register: Vol. 58, page 6964, 02/03/1993**

**Latest Modification: 03/2011**

**Thermo Electron/Thermo Environmental Instruments Models 49, 49C, 49i****Automated Equivalent Method: EQOA-0880-047**

“Thermo Electron or Thermo Environmental Instruments, Inc. **Model 49** U.V. Photometric Ambient O<sub>3</sub> Analyzer,” operated on a measurement range of either 0-0.5 or 0-1.0 ppm with or without any of the following options: 49-001 Teflon Particulate Filter; 49-002 19 Inch Rack Mount; 49-100 Internal Ozone Generator for Zero, Precision, and Level 1 Span Check; 49-103 Internal Ozone Generator for Zero, Precision, and Level 1 Span Checks With Remote Activation; 49-488 GPIB (General Purpose Interface Bus) IEEE-488.

“Thermo Electron or Thermo Environmental Instruments, Inc. **Model 49C or 49i** U.V. Photometric Ambient O<sub>3</sub> Analyzer,” operated on any measurement range between 0-0.05<sup>1</sup> to 1.0 ppm, with any time average setting between 10 and 300 seconds, with the temperature and/or pressure compensation on or off, within a temperature range of 20°C to 30°C, with or without any of the following options: <sup>2</sup> Teflon particulate filter, Internal Zero Air Scrubber, Internal Ozonator with remote activation, Rack mounts; **Model 49C**: Internal Ozonator, Carrying Handle, 4-20 mA current output, RS-232 Interface, RS-485 Interface; **Model 49i**: I/O expansion board.

**Federal Register: Vol. 45, page 57168, 08/27/1980**

**Latest Modification: 12/2011**

**Carbon Monoxide*****Beckman Model 866 CO Monitoring System*****Automated Reference Method: RFCA-0876-012**

“Beckman Model 866 Ambient CO Monitoring System,” consisting of the following components: Pump/Sample-Handling Module; Gas Control Panel; Model 865-17 Analyzer Unit; Automatic Zero/Span Standardizer; operated with a 0-50 ppm range, a 13 second electronic response time, with or without any of the following options: Current Output Feature; Bench Mounting Kit; Linearizer Circuit.

**Federal Register: Vol. 41, page 36245, 08/27/1976**

***Bendix/Combustion Engineering Model 8501-5CA CO Analyzer*****Automated Reference Method: RFCA-0276-008**

“Bendix or Combustion Engineering Model 8501-5CA Infrared CO Analyzer,” operated on the 0-50 ppm range and with a time constant setting between 5 and 16 seconds, with or without any of the following options: Rack Mounting with Chassis Slides; Rack Mounting without Chassis Slides; External Sample Pump.

**Federal Register: Vol. 41, page 7450, 02/18/1976**

***Dasibi Model 3003 CO Analyzer*****Automated Reference Method: RFCA-0381-051**

“Dasibi Model 3003 Gas Filter Correlation Dasibi Environmental CO Analyzer,” operated on the 0-50 ppm range, with a sample particulate filter installed on the sample inlet line, with or without any of the following options:

|                            |                            |                              |
|----------------------------|----------------------------|------------------------------|
| 3-001 Rack Mount           | 3-003 BCD Digital Output   | 3-007 Zero/Span Module Panel |
| 3-002 Remote Zero and Span | 3-004 4-20 Milliamp Output |                              |

**Federal Register: Vol. 46, page 20773, 04/07/1981**

***Dasibi Model 3008 CO Analyzer*****Automated Reference Method: RFCA-0488-067**

“Dasibi Model 3008 Gas Filter Correlation CO Analyzer,” operated on the 0-50 ppm range, with a time constant setting of 60 seconds, a particulate filter installed in the analyzer sample inlet line, with or without use of the auto zero or auto zero/span feature, and with or without any of the following options: N-0056-A RS-232-C Interface; S-0132-A Rack Mounting Slides; Z-0176-S Rack Mounting Brackets.

**Federal Register: Vol. 53, page 12073, 04/12/1988**

***DKK-TOA Corporation Model GFC-311E Ambient CO Analyzer*****Automated Reference Method: RFCA-0907-167**

“DKK-TOA Corporation Model GFC-311E Ambient CO Analyzer,” operated with full scale fixed measurement ranges of 0-5, 0-20, and 0-50 ppm at any environmental temperature in the range of 20°C to 30°C.

**Federal Register: Vol. 72, page 56339, 10/03/2007**

***Ecotech Serinus 30 Carbon Monoxide Analyzer or Opsis AB OPS 30 Carbon Monoxide Analyzer or Teledyne Analytical Instruments GFC7001E Carbon Monoxide Analyzer*****Automated Reference Method: RFCA-0509-174**

“Ecotech Serinus 30 Carbon Monoxide Analyzer” or “Opsis AB OPS30 Carbon Monoxide Analyzer” or “Teledyne Analytical Instruments GFC7001E Carbon Monoxide Analyzer,” operated in the range of 0-50 ppm, with a five-micron Teflon® filter element installed, and with the following selected: Background–Enabled, Control Loop–Enabled, Diagnostic Mode–Operate, Pres/Temp/Flow Compensation–Enabled, Span Compensation–Disabled, with concentration automatically corrected for temperature and pressure changes, and operated according to the Serinus 30 Carbon Monoxide Analyzer User Manual or the OPS30 Carbon Monoxide Analyzer User Manual or the Teledyne Analytical Instruments GFC7001E Carbon Monoxide Analyzer Instruction Manual, as appropriate.

**Federal Register: Vol. 74, page 26395, 06/02/2009**

**Latest Modifications: 05/2010, 05/2011, 05/2012**

***Environnement S.A. Model CO11M CO Analyzer*****Automated Reference Method: RFCA-0995-108**

“Environnement S.A. Model CO11M Ambient Carbon Monoxide Analyzer,” operated on a full scale range of 0 - 50 ppm, at any temperature in the range of 15°C to 35°C, with a 5-micron PTFE sample particulate filter, with the following software settings: Automatic response time ON; Minimum response time set to 40 seconds (RT 13); Automatic ZERO-REF cycle programmed every 24 hours; and with or without any of the following options: <sup>2</sup> RS232-422 Serial Interface; Internal Printer.

**Federal Register: Vol. 60, page 54684, 10/25/1995**

***Environnement S.A. Model CO12M CO Analyzer*****Automated Reference Method: RFCA-0206-147**

“Environnement S.A. Model CO12M Gas Filter Correlation Carbon Monoxide Analyzer,” operated with a full scale range of 0 - 50 ppm, at any temperature in the range of 10°C to 35°C, with a 5-micron PTFE sample particulate filter, with response time ON, and with the automatic “ZERO-REF” cycle either ON or OFF.<sup>2</sup>

**Federal Register: Vol. 67, page 42557, 06/24/2002**

***Horiba Models AQM-10, AQM-11, or AQM12 CO Monitoring Systems*****Automated Reference Method: RFCA-1278-033**

“Horiba Models AQM-10, AQM-11, or AQM12 Ambient CO Monitoring Systems,” operated on the 0-50 ppm range, with a response time setting of 15.5 seconds, with or without any of the following options: AIC-101 Automatic Indication Corrector; VIT-3 Non-Isolated Current Output; ISO-2 And DCS-3 Isolated Current Output.

**Federal Register: Vol. 43, page 58429, 12/14/1978**

***Horiba Model APMA-300E CO Monitoring System*****Automated Reference Method: RFCA-1180-048**

“Horiba Model APMA-300E Ambient Carbon Monoxide Monitoring System,” operated on the 0-20 ppm<sup>1</sup>, the 0-50 ppm, or the 0-100 ppm range with a time constant switch setting of No. 5. The monitoring system may be operated at temperatures between 10°C and 40°C. (This method was originally designated as “Horiba Model APMA 300E/300SE Ambient Carbon Monoxide Monitoring System.”)

**Federal Register: Vol. 45, page 72774, 11/03/1980**

***Horiba Models APMA-360 or APMA-360-CE CO Monitor*****Automated Reference Method: RFCA-0895-106**

“Horiba Instruments Incorporated, Models APMA-360 or APMA-360-CE Ambient Carbon Monoxide Monitor,” operated on the 0-50 ppm range, with the Line Setting set to “MEASURE,” with the Analog Output set to “MOMENTARY VALUE,” and with or without the following options:<sup>2</sup> 1) Rack Mounting Plate and Side Rails 2) RS-232 Com Port.

**Federal Register: Vol. 60, page 39382, 08/02/1995**

***Horiba Model APMA-370 CO Monitor*****Automated Reference Method: RFCA-0506-158**

“Horiba Instruments Incorporated Model APMA-370 Ambient CO Monitor,” operated with a full scale fixed measurement range of 0 - 50 ppm, with the automatic range switching off, at any environmental temperature in the range of 20°C to 30°C.<sup>2</sup>

**Federal Register: Vol. 71, page 25587, 05/01/2006**

***MASS-CO, Model 1 CO Analyzer*****Automated Reference Method: RFCA-1280-050**

“MASS-CO, Model 1 Carbon Monoxide Analyzer,” operated on a range of 0-50 ppm, with automatic zero and span adjustments at time intervals not to exceed 4 hours, with or without the 100 millivolt and 5 volt output options. The method consists of the following components: (1) Infra-2 (Uras 2) Infrared Analyzer Model 5611-200-35, (2) Automatic Calibrator Model 5869-111, (3) Electric Gas Cooler Model 7865-222 or equivalent with prehumidifier, (4) Diaphragm Pump Model 5861-214 or equivalent, (5) Membrane Filter Model 5862-111 or equivalent, (6) Flow Meter Model SK 1171-U or equivalent, (7) Recorder Model Mini Comp DN 1/192 or equivalent. NOTE: This method is not now commercially available.

**Federal Register: Vol. 45, page 81650, 12/11/1980**



**Monitor Labs Model 8310 CO Analyzer****Automated Reference Method: RFCA-0979-041**

"Monitor Labs Model 8310 CO Analyzer," operated on the 0-50 ppm range, with a sample inlet filter, with or without any of the following options:

|                      |                  |                                   |
|----------------------|------------------|-----------------------------------|
| 02A Zero/Span Valves | 04B Pump (50 Hz) | 07A Zero/Span Valve Power Supply  |
| 03A Floor Stand      | 05A CO Regulator | 08A Calibration Valves            |
| 04A Pump (60 Hz)     | 06A CO Cylinder  | 09A,B,C,D Input Power Transformer |

**Federal Register: Vol. 44, page 54545, 09/20/1979 and Vol. 45, page 2700, 01/14/1980**

**Monitor Labs/Lear Siegler Model 8830 CO Analyzer****Automated Reference Method: RFCA-0388-066**

"Monitor Labs or Lear Siegler Model 8830 CO Analyzer," operated on the 0-50 ppm range, with a five micron Teflon filter element installed in the rear-panel filter assembly, with or without any of the following options: 2 - Zero/Span Valve Assembly; 3 - Rack Assembly; 4 - Slide Assembly; 7 - 230 VAC, 50/60 Hz.

**Federal Register: Vol. 53, page 7233, 03/07/1988**

**MSA/LIRA Model 202S CO Analyzer System****Automated Reference Method: RFCA-0177-018**

"LIRA Model 202S Air Quality Carbon Monoxide Analyzer System," consisting of a LIRA Model 202S optical bench (P/N 459839), a regenerative dryer (P/N 464084), and rack-mounted sampling system; operated on a 0-50 ppm range, with the slow response amplifier, with or without any of the following options: Remote Meter; Remote Zero And Span Controls; 0-1, 5, 20, Or 50 mA Output; 1-5, 4-20, Or 10-50 mA Output; 0-10 Or 100 mV Output; 0-1, 5, Or 10 Volt Output.

**Federal Register: Vol. 42, page 5748, 01/31/1977**

**SIR S.A. Model S-5006 CO Analyzer****Automated Reference Method: RFCA-0708-172**

"SIR S.A. Model S-5006 CO Analyzer," operated with full scale fixed measurement ranges 0-50 ppm at any environment temperature in the range of 20°C to 30°C.

**Federal Register: Vol. 73, page 40866, 07/16/2008**

**Teledyne Advanced Pollution Instrumentation, Inc. Models 300, 300E, 300EU, T300, T300U or Teledyne Monitor Labs sensor-e™ Model TML-30 CO Analyzer; or recordum airpointer® system module 801-003000;**

**Automated Reference Method: RFCA-1093-093**

"Teledyne Advanced Pollution Instrumentation, Inc. Models 300, 300E, 300EU, T300, T300U or Teledyne Monitor Labs, Inc. sensor-e™ Model TML-30, Gas Filter Correlation Carbon Monoxide Analyzer," operated on any full scale range between 0-10 ppm and 0-50 ppm (0 - 0.1 ppm for **Models 300EU and T300U**), at any temperature in the range of 15°C to 35°C for **Model 300** or 10°C to 40°C for **Models 300E, 300 EU, T300, T300U and TML-30**, with a 5-micron TFE filter element installed in the sample filter assembly, with the dynamic zero and span adjustment set to *Off* for **Model 300**, and with or without any of the following options:<sup>2</sup> Option 50, Zero/Span Valves with pressurized span gas and shutoff valve; Option 51, Zero/Span Valves with pressurized span gas and shutoff valve and Internal Zero Air Generator; Option 52, Zero/Span Valves; Option 53, Zero/Span Valves with Internal Zero Air Generator; Rack Mount with slides; RS-232 serial port with status outputs; analog input option; and (for **Models 300E, 300EU, T300, T300U and TML-30**) 4-20 mA isolated outputs. **Airpointer® model 801-003000 only:** operated on any full scale range between 0-0.100 ppm and 0-1.0 ppm, with a PTFE filter element installed in the internal filter assembly, with the software setting: FRM/FEM conform mode; at any temperature in the range of 10°C to 45°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 16 inches mercury or less at 2.5 slpm; installed in the compact. thermally controlled (-40°C to + 45°C) and weather proof airpointer base unit with integrated data acquisition and management system mounted on a frame, pole, or wall; with or without wireless telemetry;; with or without internal dilution system with internal span gas bottle; with or without modules for other criteria pollutants; with or without analyzer for particulate matter; with or without additional 3<sup>rd</sup> party sensors for e.g. meteorology, noise, or traffic counting. Operated with the appropriate instrument manual.

**Federal Register: Vol. 58, page 58166, 10/29/1993**

**Latest Modification: 08/2010; 05/2013**

**Teledyne Monitor Labs/Casella/Ecotech Model ML9830/CM2030/EC9830/EC9830T, Teledyne Monitor Labs/Casella/Ecotech Model ML9830B/CM2030B/EC9830B, or Wedding & Associates Model 1020 CO Analyzers**

**Automated Reference Method: RFCA-0992-088**

"Teledyne Monitor Labs, Casella Monitor, or Ecotech Models ML9830/CM2030/EC9830 or ML9830B/CM2030B/

EC9830B, Ecotech Model EC9830T, or Wedding & Associates, Inc. Model 1020 Carbon Monoxide Analyzer,” operated on any full scale range between 0-5.0 ppm<sup>1</sup> and 0-100 ppm, at any temperature in the range of 15°C to 35°C, with the service switch on the secondary panel set to the *In* position, with the following menu choices selected: Range: *5.0 ppm* to *100.0 ppm*; Over-ranging: *Enabled or Disabled*; Background: *Not Disabled*; Calibration: *Manual* or *Timed*; Diagnostic Mode: *Operate*; Filter Type: *Kalman*; Pres/Temp/Flow Comp: *On*; Span Comp: *Disabled*; and as follows: **Model ML9830/CM2030/EC9830/EC9830T**: with a five-micron Teflon® filter element installed internally, with the 50-pin I/O board installed on the rear panel configured at any of the following output range settings: Voltage, 0.1V, 1V, 5V, 10V; Current, 0-20 mA, 2-20 mA and 4-20 mA; and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Valve Assembly for Internal Zero/Span (IZS); Rack Mount Assembly; Internal Floppy Disk Drive. **Models ML9830B/CM2030B/EC9830B and 1020**: with either a vendor-supplied or equivalent user-supplied five micron Teflon® filter and exhaust pump, and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); 50-pin I/O board; Rack Mount Assembly; High Pressure Span Valve; hinged, fold-down front panel. Operated with appropriate instrument manual.

**Federal Register: Vol. 57, page 44565, 09/28/1992**

**Latest Modification: 03/2011**

**Thermo Electron/Thermo Environmental Instruments Models 48, 48C, 48CTLE, 48i, 48iTLE**

**Automated Reference Method: RFCA-0981-054**

“Thermo Electron or Thermo Environmental Instruments, Inc. **Model 48** Gas Filter Correlation Ambient CO Analyzer,” operated on the 0-50 ppm range, with a time constant setting of 30 seconds, with or without any of the following options:

48-001 Teflon Particulate Filter

48-010 Internal Zero Air Package

48-002 19 Inch Rack Mount

48-488 GPIB (General Purpose Interface Bus) EEEE-488

48-003 Internal Zero/Span Valves with Remote Activation

“Thermo Electron or Thermo Environmental Instruments, Inc. **Models 48C or 48i** Gas Filter Correlation Ambient CO Analyzer,” operated on any measurement range between 0-1 ppm<sup>1</sup> and 0-100 ppm, with any averaging time setting from 10 to 300 seconds, with temperature and/or pressure compensation on or off, operated at temperatures between 20°C and 30°C, with or without any of the following options:<sup>2</sup> Teflon particulate filter, Internal zero air scrubber, I/O Expansion board; **Model 48C**: Carrying handle, 4-20 mA current output, Rack mounts, RS-232 interface, Internal zero/span and sample/calibration solenoid valves, RS-485 interface, Internal zero/span and sample/calibration solenoid valves with remote I/O activation; **Models 48C or 48i Trace Level-Enhanced (TLE) Gas Filter Correlation Ambient CO Analyzers** operated between 0-1 and 100 ppm with averaging time from 10 to 300 seconds, operated at temperatures between 20° and 30°C at line voltages of 90-110, 105-125, and 210-250 VAC @ 50/60 Hz, with or without any of the following options: rack mounts, Teflon® particulate filter, I/O Expansion board.

**Federal Register: Vol. 46, page 47002, 09/23/1981**

**Latest modifications: 04/2009; 10/2010**

## Nitrogen Dioxide

### ***Sodium Arsenite Method for NO<sub>2</sub>***

**Manual Equivalent Method:** EQN-1277-026

“Sodium Arsenite Method for the Determination of Nitrogen Dioxide in the Atmosphere.”

**Federal Register:** Vol. 42, page 62971, 12/14/1977

### ***Sodium Arsenite Method for NO<sub>2</sub> - Technicon II***

**Manual Equivalent Method:** EQN-1277-027

“Sodium Arsenite Method for the Determination of Nitrogen Dioxide in the Atmosphere-Technicon II Automated Analysis System.”

**Federal Register:** Vol. 42, page 62971, 12/14/1977

### ***TGS-ANSA Method for NO<sub>2</sub>***

**Manual Equivalent Method:** EQN-1277-028

“TGS-ANSA Method for the Determination of Nitrogen Dioxide in the Atmosphere.”

**Federal Register:** Vol. 42, page 62971, 12/14/1977

### ***Advanced Pollution Instrumentation, Inc. Model 200 NO<sub>2</sub> Analyzer***

**Automated Reference Method:** RFNA-0691-082

“Advanced Pollution Instrumentation, Inc. Model 200 Nitrogen Oxides Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with a 5-micron TFE filter element installed in the rear-panel filter assembly, with either a user- or vendor-supplied vacuum pump capable of providing 5 inches mercury absolute pressure at 5 slpm, with either a user- or vendor-supplied dry air source capable of providing air at a dew point of 0°C or lower, with the following settings of the adjustable setup variables:

|                         |                                     |                                 |
|-------------------------|-------------------------------------|---------------------------------|
| Adaptive Filter = On    | PMT Temperature Set Point = 15°C    | Normal Filter Size = 12 samples |
| Dwell Time = 7 seconds  | Rate of Change(ROC) Threshold = 10% | Dynamic Span = Off              |
| Sample Time = 8 seconds | Reaction Cell Temperature = 50°C    | Dynamic Zero = Off              |

and with or without any of the following options:

|                            |                                          |                             |
|----------------------------|------------------------------------------|-----------------------------|
| 180 Stainless Steel Valves | 283 Internal Zero/Span With Valves (IZS) | 356 Level One Spares Kit    |
| 184 Pump Pack              | 325 RS-232/Status Output                 | 357 Level Two Spares Kit    |
| 280 Rack Mount With Slides | 355 Expendables                          | PE5 Permeation Tube for IZS |

**Federal Register:** Vol. 56, page 27014, 06/12/1991

### ***Beckman Model 952-A NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer***

**Automated Reference Method:** RFNA-0179-034

“Beckman Model 952-A NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer,” operated on the 0-0.5 ppm range with the 5-micron Teflon sample filter (Beckman P/N 861072 supplied with the analyzer) installed on the sample inlet line, with or without the Remote Operation Option (Beckman No. 635539).

**Federal Register:** Vol. 44, page 7806, 02/07/1979

### ***Bendix Model 8101-B Oxides of Nitrogen Analyzer***

**Automated Reference Method:** RFNA-0479-038

“Bendix Model 8101-B Oxides of Nitrogen Analyzer,” operated on a 0-0.5 ppm range with a Teflon sample filter installed on the sample inlet line and with the following post-manufacture modifications: 1) Ozone generator and reaction chamber input-output tubing modification per Bendix Service Bulletin 8101B-2; 2) The approved converter material; 3) The revised and EPA-approved operation and service manual. These items are mandatory and must be obtained from ABB Process Analytics. The analyzer may be operated with or without any of the following optional modifications: a. Perma Pure dryer/ambient air modification; b. Valve cycle time modification; c. Zero potentiometer centering modification per Bendix Service Bulletin 8101B-1; d. Reaction chamber vacuum gauge modification.

**Federal Register:** Vol. 44, page 26792, 05/07/1979

**Bendix/Combustion Engineering Model 8101-C Oxides of Nitrogen Analyzer****Automated Reference Method: RFNA-0777-022**

“Bendix or Combustion Engineering Model 8101-C Oxides of Nitrogen Analyzer,” operated on a 0-0.5 ppm range with a Teflon sample filter (Bendix P/N 007163) installed on the sample inlet line.

**Federal Register: Vol. 42, page 37435, 07/21/1977**

**Columbia Scientific Industries Models 1600 and 5600 Analyzers****Automated Reference Method: RFNA-0977-025**

“CSI Model 1600 Oxides of Nitrogen Analyzer,” operated on a 0-0.5 ppm range with a Teflon sample filter (CSI P/N M951-8023) installed on the sample inlet line, with or without any of the following options:

|                                                  |                                                     |
|--------------------------------------------------|-----------------------------------------------------|
| 951-0103 Rack Ears                               | 951-0114 Recorder Output, 5 V                       |
| 951-0104 Rack Mounting Kit (Ears & Slides)       | 951-0115 External Pump (115 V, 60 Hz)               |
| 951-0106 Current Output, 4-20 mA (Non-Insulated) | 951-8072 Molybdenum Converter Assembly (Horizontal) |
| 951-0108 Diagnostic Output Option                | 951-8074 Copper Converter Assembly (Horizontal)     |
| 951-0111 Recorder Output, 10 V                   | 951-8079 Copper Converter Assembly (Vertical)       |
| 951-0112 Remote Zero/Span Sample Control         | 951-8085 Molybdenum Converter Assembly (Vertical)   |

NOTE: The vertical molybdenum converter assembly is standard on all new analyzers as of 1-1-87; however, use of any of the other converter assemblies is optional. Also, the above options reflect new CSI part numbers.

“CSI Model 5600 Oxides of Nitrogen Analyzer,” operated on a 0-0.5 ppm range, with any signal integration time in the range of 20 to 99 seconds, with a Teflon sample filter (CSI P/N M951-8023) installed on the sample inlet line, and with or without any of the following options: 954-0121 - Status Contacts; 964-0126 - Printer; 954-0131 - Rack Mounting Kit (ears and slides); 954-0122 - Input Solenoids; 954-8024 - Cartridge Dryer; 964-0012 - Single Headed Pump - Gast; 954-0125 - Current Output, 4-20 mA; 951-0115 - Single Headed Pump - KNF.

**Federal Register: Vol. 42, page 46574, 09/16/1977**

**Dasibi Model 2108 Oxides of Nitrogen Analyzer****Automated Reference Method: RFNA-1192-089**

“Dasibi Model 2108 Oxides of Nitrogen Analyzer,” operated on the 0-500 ppb range, with software revision 3.6 installed in the analyzer, with the auto thumbwheel switch and the diag thumbwheel switch settings at 0, with the following internal CPU dipswitch settings:

| <u>switch</u> | <u>position</u> | <u>function</u>                           |
|---------------|-----------------|-------------------------------------------|
| 1             | open (down)     | Recorder outputs are NO & NO <sub>2</sub> |
| 5             | open (down)     | 3 minute time constant                    |
| 6             | closed (up)     | 3 minute time constant;                   |

with a 5-micron Teflon filter element installed in the filter holder, and with or without any of the following options:

|                                        |                               |
|----------------------------------------|-------------------------------|
| Built-in Permeation Oven Rack Mounting | Three-Channel Recorder Output |
| RS-232 Interface 4-20 mA Output        |                               |

**Federal Register: Vol. 57, page 55530, 11/25/1992**

**DKK-TOA Corporation Model GLN-114E Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-0798-121**

“DKK-TOA Corporation Models GLN-114E and GLN-114E-1 Nitrogen Oxides Analyzer,” operated within a temperature range of 20 to 30 degrees C, on any of the following measurement ranges: 0-0.050<sup>1</sup>, 0-0.100<sup>1</sup>, 0-0.200<sup>1</sup>, 0-0.500, and 0-1.000 ppm, and with or without the optional Internal zero air supply and permeation tube oven.<sup>2</sup>

**Federal Register: Vol. 63, page 41253, 08/03/1998**

**DKK-TOA Corporation Model GLN-314E Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-0508-171**

“DKK-TOA Corporation Model GLN-314E Nitrogen Oxides Analyzer,” operated at any temperature in the range of 20°C to 30°C, on any of the following measurement ranges:<sup>1</sup> 0-0.100 ppm, 0-0.200 ppm, 0-0.500 ppm.<sup>2</sup>

**Federal Register: Vol. 73, page 28819, 05/19/2008**

***Ecotech Serinus 40 Oxides of Nitrogen Analyzer or Opsis AB OPS40 Oxides of Nitrogen Analyzer or Teledyne Analytical Instruments 9110E Oxides of Nitrogen Analyzer*****Automated Reference Method: RFNA-0809-186**

“Ecotech Serinus 40 Oxides of Nitrogen Analyzer” or “Opsis AB OPS40 Oxides of Nitrogen Analyzer” or “Teledyne Analytical Instruments 9110E Oxides of Nitrogen Analyzer,” operated in the range of 0–0.5 ppm, with a five-micron Teflon® filter element installed, with either a molybdenum or a vitreous carbon NO<sub>2</sub> to NO catalyst, and with the following selected: Control Loop-Enabled, Diagnostic Mode-Operate, Pres/Temp/Flow Compensation-Enabled, Span Compensation-Disabled, with concentration automatically corrected for temperature and pressure changes, and operated according to the Serinus 40 Oxides of Nitrogen Analyzer User Manual or the OPS40 Oxides of Nitrogen Analyzer User Manual or the Teledyne Analytical Instruments 9110E Oxides of Nitrogen Analyzer Instruction Manual as appropriate.

**Federal Register: Vol. 74, page 38184, 07/31/2009**

**Latest Modifications: 05/2010, 05/2011, 05/2012, 03/2013**

***Environnement S. A. Model AC31M NO<sub>2</sub> Analyzer*****Automated Reference Method: RFNA-0795-104**

“Environnement S. A. Model AC31M Chemiluminescent Nitrogen Oxide Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 15°C to 35°C, with a 5-micron PTFE sample particulate filter, with the following software settings: Automatic response time ON; Minimum response time set to 60 seconds (RT ) 2); and with or without any of the following options:<sup>2</sup> Internal Permeation Oven; Connection for Silica Gel Dryer; RS232-422 interface; EV3 valve; Internal Printer.

**Federal Register: Vol. 60, page 38326, 07/26/1995**

***Environnement S. A. Model AC32M NO<sub>2</sub> Analyzer*****Automated Reference Method: RFNA-0202-146**

“Environnement S. A. Model AC32M Chemiluminescent Nitrogen Oxides Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 10°C to 35°C, with a 5-micron PTFE sample particulate filter, with response time setting 11 (automatic response time), and with or without the following option: Internal permeation oven.

**Federal Register: Vol. 67, page 15567, 04/02/2002**

***Environnement S.A. SANO A Multigas Longpath Monitoring System*****Automated Reference Method: EQNA-0400-139**

“Environnement S.A. Model SANO A Multigas Longpath Air Quality Monitoring System,” consisting of a receiver, one or more projectors, interface unit, a user-provided control unit computer running the SANO A VisionAIR software, and associated incidental equipment; configured for measuring NO<sub>2</sub>, with the temperature control and internal calibration cell options installed, operated with a measurement range of 0 to 0.5 ppm, over an installed monitoring path length of between 27 and 500 meters, within an ambient air temperature range of -30 to +45°C, with a measurement (integrating) time of 180 seconds, and with or without external temperature and barometric pressure sensors or any of the following options: external (meteo) input connection, series 1M bus connection, OGR type projector, analog outputs.

**Federal Register: Vol. 65, page 26603, 05/08/2000**

***Horiba Instruments Models APNA-360 or APNA-360-CE NO-NO<sub>2</sub>-NO<sub>x</sub> Monitor*****Automated Reference Method: RFNA-0196-111**

“Horiba Instruments, Inc. Models APNA-360 or APNA-360-CE Ambient NO-NO<sub>2</sub>-NO<sub>x</sub> Monitor,” operated with a full scale range of 0 - 0.50 or 0 - 1.0 ppm, at any temperature in the range of 10°C to 40°C, with a Line Setting of "MEASURE", and an Analog Output of "MOMENTARY VALUE", and with or without the following options:<sup>2</sup> 1) Rack Mounting Plate and Side Rails 2) RS-232 Communications Port.

**Federal Register: Vol. 61, page 11404, 03/20/1996**

***Horiba Instruments Model APNA-370 NO<sub>2</sub> Monitor*****Automated Reference Method: RFNA-0506-157**

“Horiba Instruments Incorporated Model APNA-370 Ambient NO<sub>x</sub> Monitor,” standard specification, operated with a full scale fixed measurement range of 0 - 0.50 ppm with the automatic range switching off, at any ambient temperature in the range of 20°C to 30°C, and with a 0.3 micrometer sample particulate filter installed.<sup>2</sup>

**Federal Register: Vol. 71, page 25587, 05/01/2006**

**Meloy Model NA530R Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-1078-031**

“Meloy Model NA530R Nitrogen Oxides Analyzer,” operated on the following ranges and time constant switch positions:

|                        |                    |                     |           |           |
|------------------------|--------------------|---------------------|-----------|-----------|
| Range, ppm:            | 0-0.1 <sup>1</sup> | 0-0.25 <sup>1</sup> | 0-0.5     | 0-1.0     |
| Time Constant Setting: | 4                  | 3 or 4              | 2,3, or 4 | 2,3, or 4 |

Operation of the analyzer requires an external vacuum pump, either Meloy Option N-10 or an equivalent pump capable of maintaining a vacuum of 200 torr (22 inches mercury vacuum) or better at the pump connection at the specified sample and ozone-air flow rates of 1200 and 200 cm<sup>3</sup>/min, respectively. The analyzer may be operated at temperatures between 10°C and 40°C and at line voltages between 105 and 130 volts, with or without any of the following options: N-1A Automatic Zero And Span; N-2 Vacuum Gauge; N-4 Digital Panel Meter; N-6 Remote Control For Zero And Span; N-6B Remote Zero/Span Control And Status (Pulse); N-6C Remote Zero/Span Control And Status (Timer); N-9 Manual Zero/Span; N-10 Vacuum Pump Assembly (See Alternate Requirement Above); N-11 Auto Ranging; N-14B Line Transmitter; N-18 Rack Mount Conversion; N-18A Rack Mount Conversion.

**Federal Register: Vol. 43, page 50733, 10/31/1978 and Vol. 44, page 8327, 02/09/1979**

**Monitor Labs Model 8440E Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-0677-021**

“Monitor Labs Model 8440E Nitrogen Oxides Analyzer,” operated on a 0-0.5 ppm range (position 2 of range switch) with a time constant setting of 20 seconds, with or without any of the following options:

|                               |                    |                     |                                   |
|-------------------------------|--------------------|---------------------|-----------------------------------|
| TF- Sample Particulate Filter | DO- Status Outputs | 018A- Ozone Dry Air | 018B- Ozone Dry Air - No Drierite |
| With TFE Filter Element       | R- Rack Mount      | V- Zero/Span Valves | FM- Flow meters                   |

**Federal Register: Vol. 42, page 37434, 07/21/1977; Vol. 42, page 46575, 09/16/1977; Vol. 46, page 29986, 06/04/1981**

**Monitor Labs/Lear Siegler Model 8840 Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-0280-042**

“Monitor Labs or Lear Siegler Model 8840 Nitrogen Oxides Analyzer,” operated on a range of either 0-0.5 or 0-1.0 ppm, with an internal time constant setting of 60 seconds, a TFE sample filter installed on the sample inlet line, with or without any of the following options:

|                                               |                                              |                                  |
|-----------------------------------------------|----------------------------------------------|----------------------------------|
| 02 Flowmeter                                  | 08A Pump Pac Assembly with 09A (115 VAC)     | 011A Recorder Output 1 Volt      |
| 03A Rack Ears                                 | 08B Pump Pac Assembly with 09B (100 VAC)     | 011B Recorder Output 100 mV      |
| 03B Slides                                    | 08C Pump Pac Assembly with 09C (220/240 VAC) | 011C Recorder Output 10 mV       |
| 05A Zero/Span Valves                          | 08D Rack Mount Panel Assembly                | 012A DAS Output 1 Volt           |
| 05B Valve/Relay                               | 09A Pump 115 VAC 50/60 Hz                    | 012B DAS Output 100 mV           |
| 06 Status                                     | 09B Pump 100 VAC 50/60 Hz                    | 012C DAS Output 10 mV            |
| 07A Input Power Transformer 100 VAC, 50/60 Hz | 09C Pump 220/240 VAC 50 Hz                   | 013A Ozone Dry Air               |
| 07B Input Power Transformer 220/240 VAC 50 Hz |                                              | 013B Ozone Dry Air - No Drierite |

**Federal Register: Vol. 45, page 9100, 02/11/1980 and Vol. 46, page 29986, 06/04/1981**

**Monitor Labs/Lear Siegler Model 8841 Nitrogen Oxides Analyzer****Automated Reference Method: RFNA-0991-083**

“Monitor Labs or Lear Siegler Model 8841 Nitrogen Oxides Analyzer,” operated on the 0-0.05 ppm,<sup>1</sup> 0-0.1 ppm,<sup>1</sup> 0-0.2 ppm,<sup>1</sup> 0 - 0.5 ppm, or 0-1.0 ppm range, with manufacturer-supplied vacuum pump or alternative user-supplied vacuum pump capable of providing 200 torr or better absolute vacuum while operating with the analyzer.

**Federal Register: Vol. 56, page 47473, 09/19/1991**

**Opsis Model AR 500 and System 300 Open Path Ambient Air Monitoring Systems for NO<sub>2</sub>****Automated Equivalent Method: EQNA-0495-102**

“Opsis Model AR 500 System” or “System 300 Open Path (long path) Ambient Air Monitoring Systems,” configured for measuring NO<sub>2</sub>, with one detector and movable grating, operated with a measurement range of 0 to 0.5 ppm, an installed monitoring path length between 50 and 500 meters (or 50 and 1000 meters with the ER 150 option, AR 500 System only), xenon lamp type B (150 watt), fiber optic cable length between 3 and 20 meters; operating within an ambient air temperature range of -50 to +50°C, an analyzer temperature range of 20 to 30°C, a measurement (integrating) time setting between 30 and 120 seconds (0 min:30 sec. to 2 min:00 sec.), and with a complete cycle time of not more than 200 seconds (3 min, 20 sec.). Under this method designation, the Model AR 500 System or System 300 consists of: AR 500 opto-analyser; emitter EM 110 and receiver RE 110 (together identified as ER 110); optic fibre cable OF60-S; power supply PS 150; Opsis operational software, version 7.0 or 7.1; and initial on-site installation, setup, and limited operator training.<sup>2</sup>

**Optional components that can be used with the Model AR 500 only**, in addition to or as alternative to corresponding components listed above: • AR 503 opto-analyzer configured as Model AR 500 (only the center detector active, sequential monitoring) • Emitter/receiver ER 150 (for monitoring path lengths up to 1 kilometer) • Transceiver ER 130 and Retroreflector RE 090 with 7 prisms (max. monitoring path length 150 meters) or 12 prisms (max. monitoring path length 250 meters) • Receiver RE 130 • Xenon lamp type A (higher short-wavelength UV output) • Optic fibre cable OF60-R (low-loss for short wavelengths) • Multiplexers MX 004 and MX 024 • Dataloggers DL 010 and DL 016 • Analogue and digital input/output cards AO 008, AI 016, and DI 032 • Analogue and digital isolation cards IA 008, ID 008, OA 008, and OD 008 • Window heaters HF 110 and HF 150 • Mirror heaters HM 110 and HM 150 • Auto calibration unit CU 007 • Software packages IO 80 (for the analogue and digital input/output adapters), DL10 and DL16 (for data loggers), ComVision, and STAT 500.

**Recommended calibration and accuracy audit components (or equivalent) for either Model AR 500 or System 300:** • Wavelength calibration lamp CA 004 • Calibration bench CB 100 • Receiver unit RE 060 (two required) • Calibration unit CA 150, with same type lamp as used in the monitoring path emitter • Power supply PS 150 for calibration unit CA 150 • Calibration cells CC 001-X, where X represents various cell lengths from 1 to 900 mm • Filter GG 400 • Special calibration cells CC 110 or CC 150 (for mounting directly on receiver) • Light meter LM 010

**Federal Register: Vol. 60, page 21518, 05/02/1995**

**Philips Model PW9762/02 NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer**

**Automated Reference Method: RFNA-0879-040**

“Philips Model PW9762/02 NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer,” consisting of the following components: PW9762/02 Basic Analyzer; PW9729/00 Converter Cartridge; PW9731/00 Sampler or PW9731/20 Dust Filter; operated on a range of 0-0.5 ppm, with or without any of the following accessories: PW9752/00 Air Sampler Manifold; PW9732/00 Sample Line Heater; PW9011/00 Remote Control Set.

**Federal Register: Vol. 44, page 51683, 09/04/1979**

**Seres Model NO<sub>x</sub> 2000 G Nitrogen Dioxide Analyzer**

**Automated Reference Method: RFNA-0706-163**

“Seres Model NO<sub>x</sub> 2000 G Nitrogen Dioxide Ambient Air Analyzer,” operated with a full scale measurement range of 1 - 0.50 ppm, at any ambient temperature in the range of 20°C to 30°C.

**Federal Register: Vol. 71, page 42089, 07/25/2006**

**SIR S.A. Model S-5012 Nitrogen Oxides Analyzer**

**Automated Reference Method: RFNA-0804-152**

“SIR S.A. Model S-5012 Chemiluminescent Nitrogen Oxides Analyzer,” operated with a full scale range of 0 - 500 ppb, at any temperature in the range of 20°C to 30°C, with the integration time set to 1 minute, with the “initial zero” disabled, and with a specified Teflon particulate filter installed in the sample inlet filter holder.<sup>2</sup>

**Federal Register: Vol. 69, page 47924, 08/06/2004**

**Teledyne - Advanced Pollution Instrumentation, Inc. Models 200A, 200AU, 200E, 200EU, T200, T200U; Teledyne Analytical Instruments Model 9110A; or Teledyne Monitor Labs sensor-e™ Model TML-41 NO<sub>2</sub> Analyzers; or recordum airpointer® System, Module 801-002000;**

**Automated Reference Method: RFNA-1194-099**

“Teledyne - Advanced Pollution Instrumentation, Inc. Models 200A, 200AU, 9110A, 200E, 200EU, T200, T200U; Teledyne Analytical Instruments Model 9110A; or Teledyne Monitor Labs, Inc. sensor-e™ Model TML-41 Chemiluminescence Nitrogen Oxides Analyzer,” operated on any full scale range between 0-0.05 ppm and 0-1.0 ppm, with a PTFE filter element installed in the internal filter assembly, with the following software settings: dynamic zero: OFF or ON; dynamic span: OFF; cal-on-NO<sub>2</sub>: OFF; dilution factor: OFF or set to 1.0; autocal: ON or OFF; independent range: ON or OFF; autorange: ON or OFF; temperature/pressure compensation: ON; and with or without any of the following options (if available): rack mounts with or without slides, rack mount for external pump, zero/span valves, 4-20 mA analog outputs, status outputs, RS-232 output. **Models 200A, 200E<sup>2</sup>, and T200 and TML-41 only:** operated at any temperature in the range of 5°C to 40°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 10 inches mercury or less at 2 slpm, with or without optional internal zero/span (IZS) and permeation tubes for IZS, gold-plated reaction chamber, or Nafion-type sample gas conditioner, ethernet output, control input, analog input option, RS-485 output. **Model 200AU, 200EU, and T200U only:** operated at any temperature in the range of 20°C to 30°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 4 inches mercury or less at 1 slpm. **Airpointer® system module 801-002000 only:** operated on any full scale range between 0-0.05 ppm and 0-1.0 ppm, with a PTFE filter element installed in the internal filter assembly, with the software setting: FRM/FEM conform mode; at any temperature in the range of 10°C

to 45°C, with either a user- or vendor-supplied vacuum pump capable of providing an absolute pressure of 16 inches mercury or less at 2.5 slpm; installed in the compact, thermally controlled (-40°C to + 45°C) and weather proof airpointer base unit with integrated data acquisition and management system mounted on a frame, pole, or wall; with or without wireless telemetry; with or without internal span option as module supplement consisting of permeation oven and permeation tube; with or without modules for other criteria pollutants; with or without analyzer for particulate matter; with or without additional 3<sup>rd</sup> party sensors for e.g. meteorology, noise, or traffic counting. Operated with the appropriate instrument manual.

**Federal Register: Vol. 59, page 61892, 12/02/1994**

**Latest modifications: 03/2009; 08/2010; 10/2012; 5/2013**

**Teledyne - Advanced Pollution Instrumentation, Inc. Model 200EUP or T200UP Chemiluminescence Nitrogen Oxides Analyzer**

**Automated Equivalent Method: EQNA-0512-200**

“Teledyne - Advanced Pollution Instrumentation, Inc. Model 200EUP or T200UP Chemiluminescence Nitrogen Oxides Analyzer,” operated on any full scale range between 0-50 ppb and 0-1000 ppb, with any range mode (Single, Independent, or AutoRange), at any ambient temperature in the range of 20°C to 30°C, with software Temperature and Pressure compensation ON, in accordance with the associated instrument manual; and with or without any of the following options: Zero/Span Valves, standard serial port (RS232/RS485) or Multi-drop RS-232, Ethernet port, USB COM port, analog inputs, digital status outputs, analog outputs: 100 mV, 1V, 5V, 10V, 4-20 mA current loop outputs.

**Federal Register: Vol 77, page 32632, 06/01/2012**

**Latest Modification: 10/2012**

**Teledyne Monitor Labs/Casella/Ecotech Models ML9841/CM2041, ML9841A/CM2041A/EC9841A/EC9841T, Teledyne Monitor Labs/Casella/Ecotech Model ML9841B/CM2041B/EC9841B, or Wedding & Associates Model 1030 NO<sub>2</sub> Analyzers**

**Automated Reference Method: RFNA-1292-090**

“Teledyne Monitor Labs, Casella Monitor, or Ecotech Models ML9841/CM2041, ML9841A/CM2041A/EC9841A, or ML9841B/CM2041B/EC9841B, Ecotech Model 9841T, or Wedding & Associates, Inc. Model 1030 Nitrogen Oxides Analyzers,” with either a molybdenum or a vitreous carbon NO<sub>2</sub> to NO catalyst, operated on any full scale range between 0-0.05 ppm<sup>1</sup> and 0-1.0 ppm, at any temperature in the range of 15°C to 35°C, with the service switch on the secondary panel set to the *In* position; with the following menu choices selected: Range: *0.05 ppm to 1.0 ppm*; Over-ranging: *Enabled or Disabled*; Calibration: *Manual or Timed*; Diagnostic Mode: *Operate*; Filter Type: *Kalman*; Pres/Temp/Flow Comp: *On*; Span Comp: *Disabled*; and as follows: **Models ML9841/CM2041, ML9841A/CM2041A/EC9841A, and EC9841T** - with a five-micron Teflon® filter element installed internally, with the 50-pin I/O board installed on the rear panel configured at any of the following output range setting: Voltage, 0.1V, 1V, 5V, 10V; Current, 0-20 mA, 2-20 mA, 4-20 mA; and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); Internal Zero/Span (IZS) Assembly for; Rack Mount Assembly; Internal Floppy Disk Drive. **Models ML9841B/CM2041B/EC9841B and 1030** - with a vendor-supplied or equivalent user-supplied five-micron Teflon® filter and exhaust pump, and with or without any of the following options: Valve Assembly for External Zero/Span (EVS); 50-pin I/O board; Internal Zero/Span (IZS) Assembly; Rack Mount Assembly; Charcoal exhaust scrubber; hinged, fold-down front panel. Operated with appropriate instrument manual.

**Federal Register: Vol. 57, page 60198, 12/18/1992**

**Latest Modification: 03/2011, 03/2013**

**Thermo Electron/Thermo Environmental Instruments Model 14 B/E**

**Automated Reference Method: RFNA-0179-035**

“Thermo Electron or Thermo Environmental Instruments, Inc. Model 14 B/E Chemiluminescent NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer,” operated on the 0-0.5 ppm range, with or without any of the following options:

|                                  |                                          |                         |
|----------------------------------|------------------------------------------|-------------------------|
| 14-001 Teflon Particulate Filter | 14-003 Long-Time Signal Integrator       | 14-005 Sample Flowmeter |
| 14-002 Voltage Divider Card      | 14-004 Indicating Temperature Controller | 14-006 Air Filter       |

**Federal Register: Vol. 44, page 7805, 02/07/1979 and Vol.44, page 54545, 09/20/1979**

**Thermo Electron/Thermo Environmental Instruments Model 14 D/E**

**Automated Reference Method: RFNA-0279-037**

“Thermo Electron or Thermo Environmental Instruments, Inc. Model 14 D/E Chemiluminescent NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer,” operated on the 0-0.5 ppm range, with or without any of the following options: 14-001 Teflon Particulate Filter; 14-002 Voltage Divider Card.

**Federal Register: Vol. 44, page 10429, 02/20/1979**



***Thermo Environmental Instruments Models 42, 42C, 42i NO/NO<sub>2</sub>/NO<sub>x</sub> Analyzer*****Automated Reference Method: RFNA-1289-074**

“Thermo Environmental Instruments Inc. Model 42, Model 42C, or Model 42i Chemiluminescence NO-NO<sub>2</sub>-NO<sub>x</sub> Analyzer,” operated on any measurement range between 0-50 ppb and 0-1000 ppb, with any time average setting from 10 to 300 seconds, with temperature and/or pressure compensation on or off, operated at temperatures between 15°C and 35°C, and with or without an exhaust ozone scrubber or any of the following options: 2 Rack mounts; Internal Zero/span and sample valves with remote activation; Ozone particulate filter; Teflon particulate filter; Ozone permeation dryer; Permeation Oven; RS-232/485 interface, 4-20 mA current output, or I/O expansion board; Model 42 only: Pressure transducer, Sample/ozone flow meters. Model 42i-TL operated between 10 and 1000 ppb with averaging times from 10 to 300 seconds, operated at temperatures between 15°C and 35°C at line voltages of:

- a) 90-110 VAC @ 50/60/Hz
- b) 105-125 VAC @ 50/60/Hz
- c) 210-250 VAC @ 50/60/Hz

and with or without the following options: rack mounts, Teflon® Particle Filter, I/O Expansion Board.

***Federal Register: Vol. 54, page 50820, 12/11/1989***

***Latest modification: 01/2010***

**Lead*****Reference Method for Lead*****Manual Reference Method: 40 CFR Part 50, Appendix G**

Reference Method for the Determination of Lead in Suspended Particulate Matter Collected from Ambient Air.

***Federal Register: Vol. 43, page 46258, 10/05/1978***

***Energy-Dispersive X-Ray Fluorescence Spectrometry (TNRCC)*****Manual Equivalent Method: EQL-0783-058**

“Determination of Lead Concentration in Ambient Particulate Matter by Energy-Dispersive X-Ray Fluorescence Spectrometry (Texas Natural Resource Conservation Commission),” Texas Natural Resource Conservation Commission, P.O. Box 13087, Austin, TX 78711-3087.

***Federal Register: Vol. 48, page 29742, 06/28/1983***

***Energy-Dispersive X-Ray Fluorescence Spectrometry (NEA, Inc.)*****Manual Equivalent Method: EQL-0589-072**

“Determination of Lead Concentration in Ambient Particulate Matter by Energy-Dispersive X-Ray Fluorescence Spectrometry (NEA, Inc.),” Nuclear Environmental Analysis, Inc., Suite 260, 10950 SW 5th Street, Beaverton, OR 97005.

***Federal Register: Vol. 54, page 20193, 05/10/1989***

***Flame Atomic Absorption Spectrometry*****Manual Equivalent Method: EQL-0380-043**

“Determination of Lead Concentration in Ambient Particulate Matter by Flame Atomic Absorption Spectrometry Following Ultrasonic Extraction with Heated HNO<sub>3</sub>-HCl”

***Federal Register: Vol. 45, page 14648, 03/06/1980***

***Flameless Atomic Absorption Spectrometry (EPA/RTP, N.C.)*****Manual Equivalent Method: EQL-0380-044**

“Determination of Lead Concentration in Ambient Particulate Matter by Flameless Atomic Absorption Spectrometry (EPA/RTP, NC)”

***Federal Register: Vol. 45, page 14648, 03/06/1980***

***Flameless (Graphite Furnace) Atomic Absorption (Houston, Texas)*****Manual Equivalent Method: EQL-0895-107**

“Determination of Lead Concentration in Ambient Particulate Matter by Flameless (Graphite Furnace) Atomic Absorption (City of Houston, Texas),” Health and Human Services Department, Environmental Chemistry Service, 1115 S. Braeswood, Houston, TX 77030.

***Federal Register: Vol. 60, page 39383, 08/02/1995***

***Flameless Atomic Absorption Spectrometry (Omaha)*****Manual Equivalent Method: EQL-0785-059**

“Determination of Lead Concentration in Ambient Particulate Matter by Flameless Atomic Absorption Spectrometry (Omaha-Douglas County Health Department),” Omaha-Douglas County Health Department, 1819 Farnam Street, Omaha, NE 68183.

***Federal Register: Vol. 50, page 37909, 09/18/1985***

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Doe Run)*****Manual Equivalent Method: EQL-0196-113**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Doe Run Co.),” Doe Run Company, Smelting Division, 881 Main Street Herculaneum, MO 63048

***Federal Register: Vol. 61, page 11404, 03/20/1996***

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (EPA/RTP)*****Manual Equivalent Method: EQL-0380-045**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (EPA/RTP, NC)”

**Federal Register: Vol. 45, page 14648, 03/06/1980**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (IL)*****Manual Equivalent Method: EQL-1193-094**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of Illinois),” State of Illinois, Environmental Protection Agency, Champaign Inorganic Laboratory, 2120 South First Street, Champaign, IL 61820

**Federal Register: Vol. 58, page 61902, 11/23/1993**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Kansas)*****Manual Equivalent Method: EQL-0592-085**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of Kansas),” State of Kansas, Department of Health and Environment, Forbes Field, Building 740, Topeka, KS 66620-0001.

**Federal Register: Vol. 57, page 20823, 05/15/1992**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Montana)*****Manual Equivalent Method: EQL-0483-057**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of Montana),” State of Montana, Department of Health and Environmental Sciences, Cogswell Building, Helena, MT 59620.

**Federal Register: Vol. 48, page 14748, 04/05/1983**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (NETI)*****Manual Equivalent Method: EQL-1188-069**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Northern Engineering and Testing, Inc.),” Northern Engineering and Testing, Inc., P.O. Box 30615, Billings, MT 59107.

**Federal Register: Vol. 53, page 44947, 11/07/1988**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (NH)*****Manual Equivalent Method: EQL-1290-080**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of New Hampshire),” State of New Hampshire, Department of Environmental Services, Laboratory Service Unit, 6 Hazen Drive (P.O. Box 95), Concord, NH 03302-0095.

**Federal Register: Vol. 55, page 49119, 11/26/1990**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (PA)*****Manual Equivalent Method: EQL-0592-086**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Commonwealth of Pennsylvania),” Commonwealth of Pennsylvania, Department of Environmental Resources, P.O. Box 2357, Harrisburg, PA 17105-2357.

**Federal Register: Vol. 57, page 20823, 05/15/1992**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Pima, AZ)*****Manual Equivalent Method: EQL-0995-109**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Pima County, Arizona),” Pima County, Wastewater Management Department, 201 North Stone Avenue, Tucson, Arizona 85701-1207.

**Federal Register: Vol. 60, page 54684, 10/25/1995**

***Inductively Coupled Argon Plasma-Mass Spectrometry (Pima Co., AZ)*****Manual Equivalent Method: EQL-0995-110**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Plasma-Mass Spectrometry (Pima County, Arizona),” Pima County, Wastewater Management Department, 201 North Stone Avenue, Tucson, Arizona 85701-1207.

**Federal Register: Vol. 60, page 54684, 10/25/1995**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (RI)*****Manual Equivalent Method: EQL-0888-068**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of Rhode Island),” State of Rhode Island Department of Health, Air Pollution Laboratory, 50 Orms Street, Providence, RI 02904

**Federal Register: Vol. 53, page 30866, 08/16/1988**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Silver Valley)*****Manual Equivalent Method: EQL-1288-070**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (Silver Valley Laboratories),” Silver Valley Laboratories, Inc., P.O. Box 929, Kellogg, ID 83837.

**Federal Register: Vol. 53, page 48974, 12/05/1988**

***Inductively Coupled Argon Plasma-Atomic Emission Spectrometry (TNRCC)*****Manual Equivalent Method: EQL-0400-140**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Atomic Emission Spectrometry (TNRCC),” Texas Natural Resource Conservation Commission Laboratory, 5144 E. Sam Houston Parkway N., Houston, TX 77030.

**Federal Register: Vol. 65, page 26603, 05/8/2000**

***Inductively Coupled Argon Plasma-Optical Emission Spectrometry (WV)*****Manual Equivalent Method: EQL-0694-096**

“Determination of Lead Concentration in Ambient Particulate Matter by Inductively Coupled Argon Plasma-Optical Emission Spectrometry (State of West Virginia),” State of West Virginia, Department of Commerce, Labor and Environmental Resources, Division of Environmental Protection, 1558 Washington Street East, Charleston, WV 25311-2599

**Federal Register: Vol. 59, page 29429, 06/07/1994**

***Wavelength Dispersive X-Ray Fluorescence Spectrometry (CA)*****Manual Equivalent Method: EQL-0581-052**

“Determination of Lead Concentration in Ambient Particulate Matter by Wavelength Dispersive X-Ray Fluorescence Spectrometry,” California Department of Health Services, Air & Industrial Hygiene Laboratory, 2151 Berkeley Way, Berkeley, CA 94704.

**Federal Register: Vol. 46, page 29986, 06/04/1981**

---

***Inductively Coupled Plasma - Mass Spectrometry (Inter-Mountain Labs, Inc.)*****Manual Equivalent Method: EQL-0310-189**

“Procedure for Determination of Lead in Ambient TSP by Hot Plate Acid Extraction and ICP-MS Analysis,” where total suspended particulate matter (TSP) is collected according to 40 CFR Appendix B to part 50, *EPA Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High-Volume Method)*, extracted on a hot plate with 3M HNO<sub>3</sub> according to 40 CFR Appendix G to part 50, *EPA Reference Method for the Determination of Lead in Suspended Particulate Matter Collected from Ambient Air*, and analyzed by Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) based on EPA SW-846 Method 6020A.

**Federal Register: Vol. 75, page 9894, 0304/2010**

***Inductively Coupled Plasma - Mass Spectrometry (US EPA/OAQPS)*****Manual Equivalent Method: EQL-0510-191**

“Determination of Lead Concentration in TSP by Inductively Coupled Plasma Mass Spectrometry (ICP-MS) with Heated Ultrasonic Nitric and Hydrochloric Acid Filter Extraction,” where total suspended particulate matter (TSP) is collected according to 40 CFR Appendix B to part 50, *EPA Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High-Volume Method)*, extracted with a solution of nitric and hydrochloric acids, heated to 80° C and sonicated for one hour, brought to a final volume of 40mL, and analyzed by Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) based on EPA SW-846 Method 6020A.

**Federal Register:** Vol.75, page 30022, 05/28/2010

***Inductively Coupled Plasma - Mass Spectrometry (US EPA/Region 9)*****Manual Equivalent Method: EQL-0710-192**

“Heated Nitric Acid Hot Block Digestion and ICP/MS Analysis for Lead (Pb) on TSP High-Volume Filters.” In this method, total suspended particulate matter (TSP) is collected on glass fiber filters according to 40 CFR Appendix B to part 50, *EPA Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High-Volume Method)*, extracted with a solution of nitric acid, heated on a hot block to 95°C for one hour, and brought to a final volume of 50 mL. The lead content of the sample extract is analyzed by Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) based on EPA Method 200.8 and SW-846 Method 6020A.

**Federal Register:** Vol.75, page 45627, 08/03/2010

***Inductively Coupled Plasma-Atomic Emission Spectroscopy (US EPA/OAQPS)*****Manual Equivalent Method: EQL-0311-196**

“Heated Ultrasonic Nitric and Hydrochloric Acid Digestion and ICP/AES Analysis for Lead (Pb) on TSP High-Volume Filters.” A sample of total suspended particulate matter (TSP) is collected on a glass fiber filter, using the sampler and procedure of the *EPA Reference Method for the Determination of Suspended Particulate Matter in the Atmosphere (High-Volume Method)* (40 CFR 50, Appendix B). The TSP sample is extracted with a solution of nitric and hydrochloric acid, heated in an ultrasonic bath to 80°C for one hour, and brought to a final volume of 40 mL. The lead content of the sample extract is analyzed by Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP-AES), based on EPA SW-846 Method 6010C.

**Federal Register:** Vol. 76, page 15974, 03/22/2011

***Inductively Coupled Plasma- Mass Spectrometry (Eastern Research Group, Inc.)*****Manual Equivalent Method: EQL-0512-201**

“Determination of Lead in TSP by Inductively Coupled Plasma Mass Spectrometry (ICP-MS) with Hot Block Dilute Acid and Hydrogen Peroxide Filter Extraction.” In this method, total suspended particulate matter (TSP) is collected on glass fiber filters according to 40 CFR Appendix G to part 50, *EPA Reference Method for the Determination of Lead in Suspended Particulate Matter Collected From Ambient Air*. The filter samples are extracted in a hot block at 95°C with a solution of dilute hydrochloric acid and nitric acid and two aliquots of hydrogen peroxide, for a total of two and a half hours extraction time. The samples are brought to a final volume of 50 mL and the lead content of the sample extract is analyzed by Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) based on EPA Compendium Method IO-3.5 and SW-846 Method 6020A.

**Federal Register:** Vol. 77, page 32632, 06/01/2012

***Inductively Coupled Plasma- Mass Spectrometry (Eastern Research Group, Inc.)*****Manual Equivalent Method: EQL-0512-202**

“Determination of Lead in PM<sub>10</sub> by Inductively Coupled Plasma Mass Spectrometry (ICP-MS) with Hot Block Dilute Acid and Hydrogen Peroxide Filter Extraction.” In this method, PM<sub>10</sub> particulate matter is collected on Teflon® membrane filters according to 40 CFR Appendix Q to part 50, *EPA Reference Method for the Determination of Lead in Particulate Matter as PM<sub>10</sub> Collected From Ambient Air*. The filter samples are extracted in a hot block at 95°C with a solution of hydrochloric acid, nitric acid, and hydrofluoric acid and an aliquot of hydrogen peroxide for a total of two and a half hours extraction time. Samples are brought to a final volume of 50 mL and analyzed by Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) based on EPA Compendium Method IO-3.5 and SW-846 Method 6020A.

**Federal Register:** Vol. 77, page 32632, 06/01/2012

**Notes**

<sup>1</sup> Users should be aware that designation of this analyzer for operation on ranges less than the range specified in the performance specifications for this analyzer (40 CFR 53, Subpart B) is based on meeting the same absolute performance specifications required for the specified range. Thus, designation of these lower ranges does not imply commensurably better performance than that obtained on the specified range.

<sup>2</sup> This analyzer is approved for use, with proper factory configuration (if applicable), on either 50 or 60 Hertz line frequency and nominal power line voltages of 115 VAC and 230 VAC, or similar voltages as specified in the operation or instruction manual associated with the method.

**U.S. EPA Reference and Equivalent Methods for Ambient Air****Particulate Matter – TSP**

| TSP Manual Method              |                 |             |
|--------------------------------|-----------------|-------------|
| Method                         | Designation No. | Method Code |
| Reference method (high-volume) | —               | 802         |

|                                             |
|---------------------------------------------|
| <b>Particulate Matter – PM<sub>10</sub></b> |
|---------------------------------------------|

| PM <sub>10</sub> Samplers                                                                    |                 |             |
|----------------------------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                                       | Designation No. | Method Code |
| Andersen Instruments RAAS10-100                                                              | RFPS-0699-130   | 130         |
| Andersen Instruments RAAS10-200                                                              | RFPS-0699-131   | 131         |
| Andersen Instruments RAAS10-300                                                              | RFPS-0699-132   | 132         |
| BGI Model PQ100                                                                              | RFPS-1298-124   | 124         |
| BGI Model PQ200                                                                              | RFPS-1298-125   | 125         |
| Ecotech Model 3000 PM <sub>10</sub> High Volume Sampler                                      | RFPS-0706-162   | 162         |
| New Star Environmental Model NS-6070                                                         | RFPS-0202-141   | 141         |
| Oregon DEQ Medium volume sampler                                                             | RFPS-0389-071   | 071         |
| Rupprecht & Patashnick Partisol <sup>®</sup> 2000                                            | RFPS-0694-098   | 098         |
| R & P Partisol <sup>®</sup> -FRM 2000                                                        | RFPS-1298-126   | 126         |
| R & P Partisol <sup>®</sup> -Plus 2025 Seq.                                                  | RFPS-1298-127   | 127         |
| Sierra-Andersen/GMW 1200                                                                     | RFPS-1287-063   | 063         |
| Sierra-Andersen/GMW 321-B                                                                    | RFPS-1287-064   | 064         |
| Sierra-Andersen/GMW 321-C                                                                    | RFPS-1287-065   | 065         |
| Sierra-Andersen/GMW SA241, SA241M Dichot.                                                    | RFPS-0789-073   | 073         |
| Thermo Scientific Partisol <sup>®</sup> 2000                                                 | RFPS-0694-098   | 098         |
| Thermo Scientific Partisol <sup>®</sup> 2000-D, Partisol <sup>®</sup> 2000i-D                | EQPS-311-197    | 197         |
| Thermo Scientific Partisol <sup>®</sup> 2000-FRM, Partisol <sup>®</sup> 2000i                | RFPS-1298-126   | 126         |
| Thermo Scientific Partisol <sup>®</sup> -Plus 2025 Seq., Partisol <sup>®</sup> 2025i Seq     | RFPS-1298-127   | 127         |
| Thermo Scientific Partisol <sup>®</sup> -Plus 2025-D Seq., Partisol <sup>®</sup> 2025i-D Seq | EQPS-0311-198   | 198         |
| Tisch Environmental Model TE-6070                                                            | RFPS-0202-141   | 141         |
| W&A/Thermo Electron Model 600 HVL                                                            | RFPS-1087-062   | 062         |

| PM <sub>10</sub> Analyzers                                          |                 |             |
|---------------------------------------------------------------------|-----------------|-------------|
| Method                                                              | Designation No. | Method Code |
| Andersen Instruments FH62I-N Beta                                   | EQPM-0990-076   | 076         |
| DKK-TOA FPM-222/222C/223/223C                                       | EQPM-0905-156   | 156         |
| DKK-TOA DUB-222(S)/223(S)                                           | EQPM-0905-156   | 156         |
| Environnement S.A. MP101M Beta                                      | EQPM-0404-151   | 151         |
| Horiba APDA-371                                                     | EQPM-0798-122   | 122         |
| Met One BAM1020, GBAM1020, BAM1020-1, GBAM1020-1 Beta               | EQPM-0798-122   | 122         |
| Opsis SM200                                                         | EQPM-0810-193   | 193         |
| R & P TEOM <sup>®</sup> 1400, 1400a                                 | EQPM-1090-079   | 079         |
| SWAM 5a Dual Channel Monitor                                        | EQPM-0912-205   | 205         |
| Teledyne Model 602 Beta <sup>PLUS</sup> Particle Measurement System | EQPM-0912-205   | 205         |
| Thermo Andersen Series FH 62 C14 Beta Monitor, 5014i Beta           | EQPM-1102-150   | 150         |
| Thermo Scientific TEOM <sup>®</sup> 1400AB, 1405                    | EQPM-1090-079   | 079         |
| W&A/Thermo Electron 650 Beta Gauge                                  | EQPM-0391-081   | 081         |



|                                              |
|----------------------------------------------|
| <b>Particulate Matter – PM<sub>2.5</sub></b> |
|----------------------------------------------|

| PM <sub>2.5</sub> Samplers                                                                 |                 |             |
|--------------------------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                                     | Designation No. | Method Code |
| Andersen Model RAAS2.5-200 Audit                                                           | RFPS-0299-128   | 128         |
| BGI PQ200/200A                                                                             | RFPS-0498-116   | 116         |
| BGI PQ200-VSCC™ or PQ200A-VSCC™                                                            | EQPM-0202-142   | 142         |
| BGI PQ200-VSCC™ or PQ200A-VSCC™                                                            | RFPS-0498-116   | 116         |
| Graseby Andersen RAAS2.5-100                                                               | RFPS-0598-119   | 119         |
| Graseby Andersen RAAS2.5-300                                                               | RFPS-0598-120   | 120         |
| R & P Partisol®-FRM 2000 PM-2.5                                                            | RFPS-0498-117   | 117         |
| R & P Partisol®-FRM 2000 PM-2.5 [FEM]                                                      | EQPM-0202-143   | 143         |
| R & P Partisol®-FRM 2000 PM-2.5 [FEM],                                                     | RFPS-0498-117   | 117         |
| R & P Partisol® 2000 PM-2.5 Audit                                                          | RFPS-0499-129   | 129         |
| R & P Partisol® 2000 PM-2.5 FEM Audit                                                      | EQPM-0202-144   | 144         |
| R & P Partisol® 2000 PM-2.5 FEM Audit                                                      | RFPS-0499-129   | 129         |
| R & P Partisol®-Plus 2025 PM-2.5 Seq.                                                      | RFPS-0498-118   | 118         |
| R & P Partisol®-Plus 2025 PM-2.5 [FEM] Seq.                                                | EQPM-0202-145   | 145         |
| R & P Partisol®-Plus 2025 PM-2.5 [FEM] Seq.                                                | RFPS-0498-118   | 118         |
| Thermo Electron RAAS2.5-100 FEM                                                            | EQPM-0804-153   | 153         |
| Thermo Electron RAAS2.5-100 FEM                                                            | RFPS-0598-119   | 119         |
| Thermo Electron RAAS2.5-200 FEM                                                            | EQPM-0804-154   | 154         |
| Thermo Electron RAAS2.5-200 FEM                                                            | RFPS-0299-128   | 128         |
| Thermo Electron RAAS2.5-300 FEM                                                            | EQPM-0804-155   | 155         |
| Thermo Electron RAAS2.5-300 FEM                                                            | RFPS-0598-120   | 120         |
| Thermo Environmental Model 605 CAPS                                                        | RFPS-1098-123   | 123         |
| Thermo Scientific Partisol® 2000-D Dichot., Partisol® 2000i-D Dichot.                      | EQPS-0509-177   | 177         |
| Thermo Scientific Dichot. Partisol®-Plus Model 2025-D Seq., Partisol® 2025i-D Dichot. Seq. | EQPS-0509-179   | 179         |
| Thermo Scientific Partisol® 2000-FRM, Partisol® 2000i                                      | RFPS-0498-117   | 117         |
| Thermo Scientific Partisol® 2000-FRM, Partisol® 2000i                                      | EQPM-0202-143   | 143         |
| Thermo Scientific Partisol®-Plus 2025 Seq., Partisol® 2025i Seq                            | RFPS-0498-118   | 118         |
| Thermo Scientific Partisol®-Plus 2025 Seq., Partisol® 2025i Seq                            | EQPM-0202-145   | 145         |
| URG-MASS100                                                                                | RFPS-0400-135   | 135         |
| URG-MASS300                                                                                | RFPS-0400-136   | 136         |

| PM <sub>2.5</sub> Analyzers                                         |                 |             |
|---------------------------------------------------------------------|-----------------|-------------|
| Method                                                              | Designation No. | Method Code |
| Grimm Model EDM 180 PM <sub>2.5</sub> Monitor                       | EQPM-0311-195   | 195         |
| Horiba APDA-371                                                     | EQPM-0308-170   | 170         |
| Met One BAM-1020 PM-2.5 [FEM]                                       | EQPM-0308-170   | 170         |
| Thermo Scientific FH62C14-DHS Continuous, 5014i                     | EQPM-0609-183   | 183         |
| Opsis SM200                                                         | EQPM-0812-203   | 203         |
| SWAM 5a Dual Channel Monitor                                        | EQPM-0912-204   | 204         |
| Teledyne Model 602 Beta <sup>PLUS</sup> Particle Measurement System | EQPM-0912-204   | 204         |
| Thermo Scientific Model 5030i Sharp, 5030 SHARP                     | EQPM-0609-184   | 184         |

| PM <sub>2.5</sub> Analyzers                                                   |                 |             |
|-------------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                        | Designation No. | Method Code |
| Thermo Scientific TEOM <sup>®</sup> 1400a with Series 8500C FDMS <sup>®</sup> | EQPM-0609-181   | 181         |
| Thermo Scientific TEOM <sup>®</sup> 1405-DF Dichot. with FDMS <sup>®</sup>    | EQPM-0609-182   | 182         |
| Tisch Environmental TE-EDM 180 PM <sub>2.5</sub>                              | EQPM-0311-195   | 195         |

|                                                 |
|-------------------------------------------------|
| <b>Particulate Matter – PM<sub>10-2.5</sub></b> |
|-------------------------------------------------|

| PM <sub>10-2.5</sub> Samplers                                                                                                |                 |             |
|------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                                                                       | Designation No. | Method Code |
| BGI Inc. Model PQ200 PM <sub>10-2.5</sub> sampler pair                                                                       | RFPS-1208-173   | 173         |
| Thermo Scientific Partisol <sup>®</sup> 2000-FRM sampler pair, Partisol <sup>®</sup> 2000 <i>i</i> sampler pair              | RFPS-0509-175   | 175         |
| Thermo Scientific Partisol <sup>®</sup> -Plus 2025 Seq. sampler pair, Partisol <sup>®</sup> 2025 <i>i</i> Seq sampler pair   | RFPS-0509-176   | 176         |
| Thermo Scientific Partisol <sup>®</sup> 2000-D Dichot., Partisol <sup>®</sup> 2000 <i>i</i> -D Dichot.                       | EQPS-0509-178   | 178         |
| Thermo Scientific Dichot. Partisol <sup>®</sup> -Plus Model 2025-D Seq., Partisol <sup>®</sup> 2025 <i>i</i> -D Dichot. Seq. | EQPS-0509-180   | 180         |

| PM <sub>10-2.5</sub> Analyzers                                      |                 |             |
|---------------------------------------------------------------------|-----------------|-------------|
| Method                                                              | Designation No. | Method Code |
| Met One Instruments BAM-1020 System                                 | EQPM-0709-185   | 185         |
| SWAM 5a Dual Channel Monitor                                        | EQPM-0912-206   | 206         |
| Teledyne Model 602 Beta <sup>PLUS</sup> Particle Measurement System | EQPM-0912-206   | 206         |

## Sulfur Dioxide

| SO <sub>2</sub> Manual Methods    |                 |             |
|-----------------------------------|-----------------|-------------|
| Method                            | Designation No. | Method Code |
| Reference method (pararosaniline) | —               | 097         |
| Technicon I (pararosaniline)      | EQS-0775-001    | 097         |
| Technicon II (pararosaniline)     | EQS-0775-002    | 097         |

| SO <sub>2</sub> Analyzers                |                 |             |
|------------------------------------------|-----------------|-------------|
| Method                                   | Designation No. | Method Code |
| Advanced Pollution Instr. 100            | EQSA-0990-077   | 077         |
| Advanced Pollution Instr. 100A/100AS     | EQSA-0495-100   | 100         |
| Asarco 500                               | EQSA-0877-024   | 024         |
| Beckman 953                              | EQSA-0678-029   | 029         |
| Bendix 8303                              | EQSA-1078-030   | 030         |
| Casella ML9850, ML9850B, CM2050, CM2050B | EQSA-0193-092   | 092         |
| Columbia Scientific Industries 5700      | EQSA-0494-095   | 095         |
| Dasibi 4108                              | EQSA-1086-061   | 061         |
| DKK-TOA Corp. GFS-32                     | EQSA-0701-115   | 115         |
| DKK-TOA Corp. GFS-112E, GFS-112E-1       | EQSA-0100-133   | 133         |
| DKK-TOA Corp. GFS-312E                   | EQSA-1107-168   | 168         |
| Ecotech ML9850/EC9850, ML9850B/EC9850B   | EQSA-0193-092   | 092         |
| Ecotech EC9850T                          | EQSA-0193-092   | 092         |
| Ecotech Serinus 50                       | EQSA-0809-188   | 188         |
| Environnement S.A AF21M                  | EQSA-0292-084   | 084         |
| Environnement S.A AF22M                  | EQSA-0802-149   | 149         |
| Environnement S.A. SANO A                | EQSA-0400-138   | 138         |
| Horiba Model APSA-360/APSA-360ACE        | EQSA-0197-114   | 114         |
| Horiba Model APSA-370                    | EQSA-0506-159   | 159         |
| Lear Siegler AM2020                      | EQSA-1280-049   | 049         |
| Lear Siegler SM1000                      | EQSA-1275-005   | 005         |
| Meloy SA185-2A                           | EQSA-1275-006   | 006         |
| Meloy SA285E                             | EQSA-1078-032   | 032         |
| Meloy SA700                              | EQSA-0580-046   | 046         |
| Monitor Labs 8450                        | EQSA-0876-013   | 513         |
| Monitor Labs or Lear Siegler 8850        | EQSA-0779-039   | 039         |
| Monitor Labs or Lear Siegler 8850S       | EQSA-0390-075   | 075         |
| Opsis AB OPS50                           | EQSA-0809-188   | 188         |
| Opsis AR 500, System 300 (open path)     | EQSA-0495-101   | 101         |
| Philips PW9700                           | EQSA-0876-011   | 511         |
| Philips PW9755                           | EQSA-0676-010   | 010         |
| recordum airpointer system               | EQSA-0495-100   | 100         |
| SERES SF 2000 G                          | EQSA-0810-194   | 194         |
| SIR S.A. S-5001                          | EQSA-0507-166   | 166         |

| SO <sub>2</sub> Analyzers                                                |                 |             |
|--------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                   | Designation No. | Method Code |
| recordum airpointer system                                               | EQSA-0495-100   | 100         |
| Teledyne-Advanced Pollution Inst. 100A, 100 AS, 100E, 100EU, T100, T100U | EQSA-0495-100   | 100         |
| Teledyne Analytical Instruments 6400A                                    | EQSA-0495-100   | 100         |
| Teledyne Analytical Instruments 6400E                                    | EQSA-0809-188   | 188         |
| Teledyne Monitor Labs ML9850, ML9850B                                    | EQSA-0193-092   | 092         |
| Teledyne Monitor Labs TML-50                                             | EQSA-0495-100   | 100         |
| Thermo Electron 43                                                       | EQSA-0276-009   | 009         |
| Thermo Electron 43A, 43C-TLE, 43i                                        | EQSA-0486-060   | 060         |
| Thermo Environmental Instruments 43B, 43C                                | EQSA-0486-060   | 060         |
| Wedding 1040                                                             | EQSA-0193-092   | 092         |

## Ozone

| O <sub>3</sub> Analyzers                                                 |                 |             |
|--------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                   | Designation No. | Method Code |
| 2B Technologies Model 202                                                | EQOA-0410-190   | 190         |
| Advanced Pollution Instr. 400/400A/400E                                  | EQOA-0992-087   | 087         |
| Beckman 950A                                                             | RFOA-0577-020   | 020         |
| Bendix 8002                                                              | RFOA-0176-007   | 007         |
| Casella ML9810, ML9810B, ML9811, ML9812, CM2010, CM2010B, CM2011, CM2012 | EQOA-0193-091   | 091         |
| Columbia Scientific Industries 2000                                      | RFOA-0279-036   | 036         |
| Dasibi 1003-AH, -PC, -RS                                                 | EQOA-0577-019   | 019         |
| Dasibi 1008-AH, -PC, -RS                                                 | EQOA-0383-056   | 056         |
| DKK-TOA Corp. GUX-113E, GUX-113E-1                                       | EQOA-0200-134   | 134         |
| DKK-TOA Corp. GUX-313E                                                   | EQOA-1107-169   | 169         |
| Ecotech ML9810/EC9810, -9810B, -9811, -9812                              | EQOA-0193-091   | 091         |
| Ecotech Serinus 10                                                       | EQOA-0809-187   | 187         |
| EnviroNics 300                                                           | EQOA-0990-078   | 078         |
| Environnement S.A O <sub>3</sub> 41M                                     | EQOA-0895-105   | 105         |
| Environnement S.A O <sub>3</sub> 42M                                     | EQOA-0206-148   | 148         |
| Environnement S.A SANOA                                                  | EQOA-0400-137   | 137         |
| Horiba APOA-360                                                          | EQOA-0196-112   | 112         |
| Horiba APOA-370                                                          | EQOA-0506-160   | 160         |
| McMillan 1100-1                                                          | RFOA-1076-014   | 514         |
| McMillan 1100-2                                                          | RFOA-1076-015   | 515         |
| McMillan 1100-3                                                          | RFOA-1076-016   | 016         |
| Meloy OA325-2R                                                           | RFOA-1075-003   | 003         |
| Meloy OA350-2R                                                           | RFOA-1075-004   | 004         |
| Monitor Labs 8410E                                                       | RFOA-1176-017   | 017         |
| Monitor Labs or Lear Siegler 8810                                        | EQOA-0881-053   | 053         |
| Opsis AB OPS10                                                           | EQOA-0809-187   | 187         |
| Opsis AR 500, System 300 (open path)                                     | EQOA-0495-103   | 103         |
| PCI Ozone Corp. LC-12                                                    | EQOA-0382-055   | 055         |
| Philips PW9771                                                           | EQOA-0777-023   | 023         |
| recordum airpointer system                                               | EQOA-0992-087   | 087         |
| Seres Model OZ 2000 G                                                    | EQOA-0506-161   | 161         |
| SIR S.A. S-5014                                                          | EQOA-0207-164   | 164         |
| Tanabyte 722, 723, 724, 725, 726                                         | EQOA-0407-165   | 165         |
| Teledyne - Advanced Pollution Instr. 265E, T265                          | EQOA-0611-199   | 199         |
| Teledyne - Advanced Pollution Instr. 400, 400A, 400E, T400               | EQOA-0992-087   | 087         |
| Teledyne Monitor Labs ML9810/9810B, ML9811, ML9812                       | EQOA-0193-091   | 091         |
| Teledyne Monitor Labs TML-10                                             | EQOA-0992-087   | 087         |
| Thermo Electron or Thermo Environmental Instruments 49, 49C, 49i         | EQOA-0880-047   | 047         |
| Wedding 1010                                                             | EQOA-0193-091   | 091         |

## Carbon Monoxide

| CO Analyzers                                                             |                 |             |
|--------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                   | Designation No. | Method Code |
| Beckman 866                                                              | RFCA-0876-012   | 012         |
| Bendix 8501-5CA                                                          | RFCA-0276-008   | 008         |
| Casella ML9830, ML9830B, CM2030, CM2030B                                 | RFCA-0992-088   | 088         |
| Dasibi 3003                                                              | RFCA-0381-051   | 051         |
| Dasibi 3008                                                              | RFCA-0488-067   | 067         |
| DKK-TOA Corp. GFC-311E                                                   | RFCA-0907-167   | 167         |
| Ecotech Serinus 30                                                       | RFCA-0509-174   | 174         |
| Ecotech ML9830/EC9830, ML9830B/EC9830B                                   | RFCA-0992-088   | 088         |
| Ecotech EC9830T                                                          | RFCA-0992-088   | 088         |
| Environnement S.A CO11M                                                  | RFCA-0995-108   | 108         |
| Environnement S.A CO12M                                                  | RFCA-0206-147   | 147         |
| Horiba AQM-10, -11, -12                                                  | RFCA-1278-033   | 033         |
| Horiba 300E/300SE                                                        | RFCA-1180-048   | 048         |
| Horiba APMA-360                                                          | RFCA-0895-106   | 106         |
| Horiba APMA-370                                                          | RFCA-0506-158   | 158         |
| MASS – CO 1 (Massachusetts)                                              | RFCA-1280-050   | 050         |
| Monitor Labs 8310                                                        | RFCA-0979-041   | 041         |
| Monitor Labs or Lear Siegler 8830                                        | RFCA-0388-066   | 066         |
| MSA 202S                                                                 | RFCA-0177-018   | 018         |
| Opsis AB OPS 30                                                          | RFCA-0509-174   | 174         |
| recordum airpointer system                                               | RFCA-1093-093   | 093         |
| SIR S.A. Model S-5006                                                    | RFCA-0708-172   | 172         |
| Teledyne Adv. Pollution Instr. 300, 300E, 300EU, T300, T300U             | RFCA-1093-093   | 093         |
| Teledyne Analytical Instruments GFC7001E                                 | RFCA-0509-174   | 174         |
| Teledyne Monitor Labs ML9830/9830B                                       | RFCA-0992-088   | 088         |
| Teledyne Monitor Labs TML-30                                             | RFCA-1093-093   | 093         |
| Thermo Electron or Thermo Environmental Instruments 48, 48C, 48i, 48/TLE | RFCA-0981-054   | 054         |
| Wedding 1020                                                             | RFCA-0992-088   | 088         |

|                         |
|-------------------------|
| <b>Nitrogen Dioxide</b> |
|-------------------------|

| NO <sub>2</sub> Manual Methods |                 |             |
|--------------------------------|-----------------|-------------|
| Method                         | Designation No. | Method Code |
| Sodium arsenite (orifice)      | EQN-1277-026    | 084         |
| Sodium arsenite/Technicon II   | EQN-1277-027    | 084         |
| TGS-ANSA (orifice)             | EQN-1277-028    | 098         |

| NO <sub>2</sub> Analyzers                                               |                 |             |
|-------------------------------------------------------------------------|-----------------|-------------|
| Method                                                                  | Designation No. | Method Code |
| Advanced Pollution Instr. 200                                           | RFNA-0691-082   | 082         |
| Advanced Pollution Instr. 200A/200AU                                    | RFNA-1194-099   | 099         |
| Beckman 952A                                                            | RFNA-0179-034   | 034         |
| Bendix 8101-B                                                           | RFNA-0479-038   | 038         |
| Bendix 8101-C                                                           | RFNA-0777-022   | 022         |
| Casella ML9841, ML9841A, ML9841B, CM2041, CM2041A, CM2041B              | RFNA-1292-090   | 090         |
| Columbia Scientific Indust.1600, 5600                                   | RFNA-0977-025   | 025         |
| Dasibi 2108                                                             | RFNA-1192-089   | 089         |
| DKK-TOA Corp GLN-114E, GLN-114E-1                                       | RFNA-0798-121   | 121         |
| DKK-TOA Corp GLN-314E                                                   | RFNA-0508-171   | 171         |
| Ecotech ML9841A/EC9841A,ML9841B/EC9841B                                 | RFNA-1292-090   | 090         |
| Ecotech EC9841T                                                         | RFNA-1292-090   | 090         |
| Ecotech Serinus 40                                                      | RFNA-0809-186   | 186         |
| Environnement S.A. AC31M                                                | RFNA-0795-104   | 104         |
| Environnement S.A. AC32M                                                | RFNA-0202-146   | 146         |
| Environnement S.A. SANO A                                               | EQNA-0400-139   | 139         |
| Horiba APNA-360                                                         | RFNA-0196-111   | 111         |
| Horiba APNA-370                                                         | RFNA-0506-157   | 157         |
| Meloy NA530R                                                            | RFNA-1078-031   | 031         |
| Monitor Labs 8440E                                                      | RFNA-0677-021   | 021         |
| Monitor Labs or Lear Siegler 8840                                       | RFNA-0280-042   | 042         |
| Monitor Labs or Lear Siegler 8841                                       | RFNA-0991-083   | 083         |
| Monitor Labs ML9841                                                     | RFNA-1292-090   | 090         |
| Opsis AB OPS40                                                          | RFNA-0809-186   | 186         |
| Opsis AR 500, System 300 (open path)                                    | EQNA-0495-102   | 102         |
| Philips PW9762/02                                                       | RFNA-0879-040   | 040         |
| recordum airpointer system                                              | RFNA-1194-099   | 099         |
| Seres Model NO <sub>x</sub> 2000 G                                      | RFNA-0706-163   | 163         |
| SIR S.A. S-5012                                                         | RFNA-0804-152   | 152         |
| Teledyne-Advanced Pollution Inst. 200A, 200AU, 200E, 200EU, T200, T200U | RFNA-1194-099   | 099         |
| Teledyne-Advanced Pollution Inst. 200EUP, T200UP                        | EQNA-0512-200   | 200         |
| Teledyne Analytical Instruments 9110A                                   | RFNA-1194-099   | 099         |
| Teledyne Analytical Instruments 9110E                                   | RFNA-0809-186   | 186         |
| Teledyne Monitor Labs ML9841, ML9841A, ML9841B                          | RFNA-1292-090   | 090         |



| NO <sub>2</sub> Analyzers                                 |                 |             |
|-----------------------------------------------------------|-----------------|-------------|
| Method                                                    | Designation No. | Method Code |
| Teledyne Monitor Labs TML-41                              | RFNA-1194-099   | 099         |
| Thermo Electron or Thermo Environmental Instruments 14B/E | RFNA-0179-035   | 035         |
| Thermo Electron or Thermo Environmental Instruments 14D/E | RFNA-0279-037   | 037         |
| Thermo Environmental Instr. 42, 42C, 42 <i>i</i>          | RFNA-1289-074   | 074         |
| Wedding 1030                                              | RFNA-1292-090   | 090         |

## Lead

| Pb Manual Methods                   |                 |             |
|-------------------------------------|-----------------|-------------|
| Method                              | Designation No. | Method Code |
| Reference method (hi-vol/AA spect.) | —               | 803         |
| Hi-vol/AAS (alt. extr.)             | EQL-0380-043    | 043         |
| Hi-vol/Energy-disp XRF (TX ACB)     | EQL-0783-058    | 058         |
| Hi-vol/Energy-disp XRF (NEA)        | EQL-0589-072    | 072         |
| Hi-vol/Flameless AA (EMSL/EPA)      | EQL-0380-044    | 044         |
| Hi-vol/Flameless AA (Houston)       | EQL-0895-107    | 107         |
| Hi-vol/Flameless AA (Omaha)         | EQL-0785-059    | 059         |
| Hi-vol/ICAP-OES (Doe Run Co.)       | EQL-0196-113    | 113         |
| Hi-vol/ICAP-OES (EMSL/EPA)          | EQL-0380-045    | 045         |
| Hi-vol/ICAP-OES (Illinois)          | EQL-1193-094    | 094         |
| Hi-vol/ICAP-OES (Kansas)            | EQL-0592-085    | 085         |
| Hi-vol/ICAP-OES (Montana)           | EQL-0483-057    | 057         |
| Hi-vol/ICAP-OES (NE&T)              | EQL-1188-069    | 069         |
| Hi-vol/ICAP-OES (New Hampshire)     | EQL-1290-080    | 080         |
| Hi-vol/ICAP-OES (Pennsylvania)      | EQL-0592-086    | 086         |
| Hi-vol/ICAP-OES (Pima Co.,AZ)       | EQL-0995-109    | 109         |
| Hi-vol/ICAP-MS (Pima Co.,AZ)        | EQL-0995-110    | 110         |
| Hi-vol/ICAP-OES (Rhode Island)      | EQL-0888-068    | 068         |
| Hi-vol/ICAP-OES (Silver Val. Labs)  | EQL-1288-070    | 070         |
| Hi-vol/ICAP-AES (TNRCC)             | EQL-0400-140    | 140         |
| Hi-vol/ICAP-OES (West Virginia)     | EQL-0694-096    | 096         |
| Hi-vol/WL-disp. XRF (CA A&IHL)      | EQL-0581-052    | 052         |
|                                     |                 |             |
| Hi-vol/ICP-MS (IML, Inc)            | EQL-0310-189    | 189         |
| Hi-Vol/ICP-MS (US EPA/OAQPS)        | EQL-0510-191    | 191         |
| Hi-Vol/ICP-MS (US EPA/Region 9)     | EQL-0710-192    | 192         |
| Hi-Vol/ICP-AES (US EPA/OAQPS)       | EQL-0311-196    | 196         |
| Hi-Vol/ICP-MS (ERG)                 | EQL-0512-201    | 201         |
| PM <sub>10</sub> /ICP-MS (ERG)      | EQL-0512-202    | 202         |

**Sources or Contacts for Designated Reference and Equivalent Methods****2B Technologies, Inc.**

2100 Central Ave., Suite 105  
Boulder, CO 80301  
(303) 273-0559  
<http://www.twobtech.com>

**ABB Process Analytics**

P.O. Box 831  
Lewisburg, WV 24901  
(304) 647-4358

**Advanced Pollution Instrumentation, Inc.**

[Refer to Teledyne - Advanced Pollution  
Instrumentation, Inc.]

**Andersen Instruments**

[Refer to Thermo Fisher Scientific, Inc.]

**ASARCO Incorporated**

3422 South 700 West  
Salt Lake City, UT 84119  
(801) 262-2459

**Beckman Instruments, Inc.**

Process Instruments Division  
2500 Harbor Blvd.  
Fullerton, CA 92634  
(714) 871-4848

**Bendix**

[Refer to ABB Process Analytics]

**BGI Incorporated**

58 Guinan Street  
Waltham, MA 02451  
(781) 891-9380  
[www.bgiusa.com](http://www.bgiusa.com) (bgiinc@attglobal.net)

**Casella Monitor**

Regent House, Wolseley Road, Kempston  
Bedford, United Kingdom MK42 7JY  
[www.casellameasurement.com](http://www.casellameasurement.com)

**Columbia Scientific Industries**

11950 Jollyville Road  
Austin, TX 78759  
(800) 531-5003

**Combustion Engineering**

[Refer to ABB Process Analytics]

**Dasibi Environmental Corp.**

[Formerly, 506 Paula Avenue  
Glendale, CA 91201  
(818) 247-7601  
[www.dasibi.com](http://www.dasibi.com)

**DKK-TOA Corporation**

29-10, 1-Chome, Takadanobaba,  
Shinjuku-ku  
Tokyo 169-8648, Japan  
+81-3-3202-0211  
[www.toadkk.co.jp](http://www.toadkk.co.jp)

**Eastern Research Group, Inc.**

601 Keystone Park Drive, Suite 700,  
Morrisville, NC 27560  
919- 468-7800  
[www.erg.com](http://www.erg.com)

**Ecotech Pty. Ltd.**

1492 Ferntree Gully Rd.  
Knoxfield, Victoria, 3180, Australia  
+61 1300 364 946  
[www.ecotech.com.au](http://www.ecotech.com.au)

**Environnement S.A**

111, bd Robespierre  
78300 Poissy, France  
[www.environnement-sa.com](http://www.environnement-sa.com)  
Instruments also available from:  
**Altech/Environnement U.S.A.**  
2623 Kaneville Court  
Geneva, IL 60134  
(630) 262- 4400

**Environics, Inc.**

69 Industrial Park Rd. E.  
Tolland, CT 06084-2805  
(203) 429-0077  
[www.environics.com](http://www.environics.com)

**FAI Instruments S.R.L**

Via Aurora, 15 -00013 Fonte Nuova  
Rome, Italy  
(+39) 06.9050248  
[www.fai-instruments.it](http://www.fai-instruments.it)

**Graseby GMW**

[Refer to Thermo Fisher Scientific, Inc.]

**GRIMM Technologies, Inc.**

5833 Stewart Parkway, Suite 203  
Douglasville, GA 30153  
(877) 474-6872  
[www.dustmonitor.com](http://www.dustmonitor.com)

**Horiba Instruments Incorporated**

17671 Armstrong Avenue  
Irvine, CA 92714  
(800) 446-7422  
[www.horiba.com](http://www.horiba.com)

**Inter-Mountain Labs, Inc.**

1673 Terra Avenue  
Sheridan, WY 82801  
(307) 672-8945  
[www.intermountainlabs.com](http://www.intermountainlabs.com)

**Lear Siegler**

[Refer to Teledyne Monitor Labs, Inc.]

**Commonwealth of Massachusetts**

Department of Environmental  
Quality Engineering  
Tewksbury, MA 01876

**Met One Instruments, Inc.**

1600 Washington Blvd.  
Grants Pass, OR 97526  
(541) 471-7111  
[www.metone.com](http://www.metone.com) (metone@metone.com)

**McMillan**

[Refer to Columbia Scientific Industries]

**Mine Safety Appliances**

600 Penn Center Blvd.  
Pittsburgh, PA 15235-5810  
(412) 273-5101

**Monitor Labs, Inc.**

[Refer to Teledyne Monitor Labs, Inc.]

**New Star Environmental, LLC**

3293 Ashburton Chase NE  
Roswell, GA 30075  
(770) 998-2590

**Opsis AB**

Box 244  
SE-244 02 Furulund  
Sweden  
+46 46 72 25 00  
[www.opsis.se](http://www.opsis.se)

Instruments also available from

**Opsis, Inc.**

150 N. Michigan Ave., Suite 1950  
Chicago, IL 60601  
(312) 447-7733

**State of Oregon**

Department of Environmental Quality  
Air Quality Division  
811 S.W. Sixth Avenue  
Portland, OR 97204

**PCI Ozone Corp.**

One Fairfield Crescent  
West Caldwell, NJ 07006  
(201) 575-7052  
[www.pci-wedeco.com](http://www.pci-wedeco.com)

**Phillips Electronic Instruments, Inc.**

85 McKee Drive  
Mahwah, NJ 07430

**Recordum Messtechnik GmbH**

Jasomirgottgasse 5  
2340 Mödling  
Austria  
[www.recordum.com](http://www.recordum.com)

**Rupprecht & Patashnick Co., Inc.**

[Refer to Thermo Fisher Scientific, Inc.]  
[26 Tech Valley Drive]  
[East Greenbush, NY 12061]  
[(518) 452-0065]  
[ [www.rpco.com](http://www.rpco.com) ]

**Sibata Scientific Technology, Ltd.**

1-25, 3-chome  
Ikenohata, Taito-ku  
Tokyo 110, Japan  
81-3(3822)2272  
TTani@email.msn.com

**Seres**

360 Rue Louis de Broglie  
La Duranne - BP 87000  
13793 Aix en Provence - Cedex 3  
France  
+33 (0)4 42 97 37 37  
[www.seres-france.com](http://www.seres-france.com)

**SIR S.A. (Sistemas Instalaciones y Redes, S.A.)**

Avenida de la Industria, 3  
Tres Cantos 28760 Madrid, SPAIN  
(+34) 91 803 66 02  
[www.sirsa.es](http://www.sirsa.es)

**SIR USA**

826 West Braddock Road  
Alexandria, VA 22302-3605  
(703) 837-1883

**Tanabyte Engineering, Inc.**

1210 West Burbank Blvd., Suite B,  
Burbank, CA 91506  
(818) 842-4022  
[www.tanabyte.com](http://www.tanabyte.com)

**Teledyne - Advanced Pollution Instrumentation, Inc.**

9480 Carroll Park Drive  
San Diego, CA 92121-5201  
(858) 657-9800  
[www.teledyne-api.com](http://www.teledyne-api.com)

**Teledyne Analytical Instruments**

16830 Chestnut Street  
City of Industry, CA 91748  
(626) 934-1622

**Teledyne Monitor Labs, Inc.**

74 Inverness Drive East  
Englewood, CO 80112-5189  
(303) 792-3300  
[www.teledyne-ml.com](http://www.teledyne-ml.com)

**Thermo Electron Corporation**

[Refer to Thermo Fisher Scientific, Inc.]  
[27 Forge Parkway]  
[Franklin, MA 02038]  
[(508) 520-0430 / (866) 282-0430]  
[www.thermo.com](http://www.thermo.com)

**Thermo Environmental Instruments, Inc.**

[Refer to Thermo Fisher Scientific, Inc.]  
[www.thermoei.com](http://www.thermoei.com)

**Thermo Fisher Scientific, Inc.**

81 Wyman Street  
Waltham, MA 02454  
(781) 622-1000 / (800) 678-5599  
[www.thermo.com](http://www.thermo.com)

**Tisch Environmental, Inc.**

145 S. Miami Avenue  
Village of Cleves, OH 45002  
(513) 467-9000  
[www.tisch-env.com](http://www.tisch-env.com)

**URG Corporation**

116 Merritt Mill Road  
Chapel Hill, NC 27516  
(919) 942-2753

**U.S. EPA**

National Exposure Research Laboratory  
Human Exposure & Atmospheric  
Sciences Division  
Process Modeling Research Branch  
(MD D205-03)  
Research Triangle Park, NC 27711  
(919) 541- 5691  
[www.epa.gov/heasd](http://www.epa.gov/heasd)

**U.S. EPA**

Office of Air and Radiation  
Air Quality Planning and Standards  
Air Quality Assessment Division  
(MD C404-03)  
Research Triangle Park, NC 27711  
(919) 541- 3372  
[www.epa.gov/air](http://www.epa.gov/air)

**U.S. EPA**

Region 9 Laboratory  
1337 South 46<sup>th</sup> Street, Bldg 201  
Richmond, CA 94804

**Wedding and Associates, Inc.**

[Refer to Thermo Fisher Scientific, Inc.]